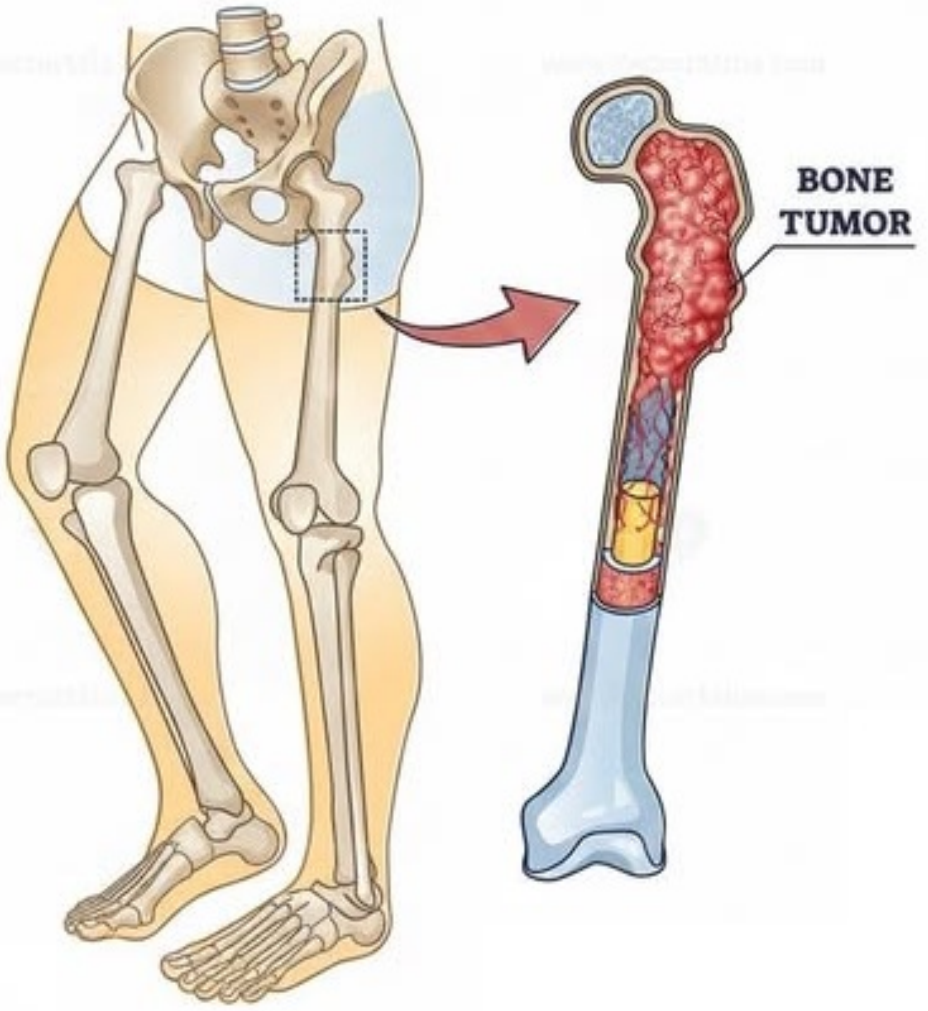
COURSE

Bone Tumors

Nasr City Ortho Tube

9 lectures



Prof Dr Wissam Abousenna

Watch course

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Developed by

Lecturers

Prof Dr Wisam Abosena

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This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

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How to use this book

- Click any line in the Contents to jump to that lecture.
- The time at the bottom right of each page (» 04:57) opens the original lecture on YouTube at that exact moment.
- "↑ Chapter" returns to the lecture's contents; "↑ Book" returns to the book's contents.
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- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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Introduction to Bone Tumors

(Clinical Presentation, Diagnosis and Staging)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 14) “both positive increases sensitivity” → “both positive gives higher specificity”

C2 (p. 16) “T = tumour grade” → “T = primary tumour”

C3 (p. 17) “reaches the epiphysis” → “reaches the physis”

C4 (p. 17) “1B: with regional metastasis” → “1B: M0, no metastasis”

C5 (p. 17) “metastasis = Grade 3” → “metastasis = Stage 3”

C6 (p. 18) “≤ 5 cm or ≥ 5 cm” → “≤ 5 cm or > 5 cm”

Queried - kept as spoken, but looks wrong

Q1 (p. 11) As spoken; standard definition is >5-10% body weight over months, not 10 kg in 10 days.

Q2 (p. 12) As spoken; simple bone cyst is central medullary; cortical metaphyseal lesion in child suggests NOF/fibrous cortical defect.

Clinical Presentation of Bone Tumours Ask AI

- Lecture aim: clinical diagnosis and **staging** of bone tumours
  - How to diagnose, deal with and understand the tumour patient
- Four presentation **scenarios**
  - Incidental finding
    - Small child falls on his leg; perfectly fine before
    - Example: lesion in the **distal tibia**
      - Site: distal tibia, **metaphysis** → not next to the joint
      - **Well-defined**: white sclerotic line around it; internal changes
    - Or older adult imaged for another reason → lesion appears
    - No complaint at all → **discovered accidentally** (incidental finding)
  - Swelling: "I have a swelling"
    - Found on X-ray or MRI; example shown: proximal femur
  - **Pathological fractures**
    - No known lesion, or mild pain for a while → sudden fracture
    - Trivial trauma, e.g. going into or out of the bathroom
    - Examples: **well-defined lesion** with fracture through it
    - Examples: **ill-defined lesion** with fracture through it
  - **Symptomatic cases**: persistent pain that does not go away

History Ask AI

- Fixed questions → always asked
- **Pain**
  - Duration
  - Aggravating factors
  - Relieving factors
  - Onset: **acute or gradual**
  - **Night pain**?
  - **Rest pain**?
  - **Mechanical pain**? → increases with movement
- **Red flags** (worrying features)
  - Night pain
  - Wakes patient from sleep
  - Prevents patient from sleeping
  - Increases over a short period
  - No relation to exertion
- **Swelling and fever**
  - Swelling appeared with the pain?
  - Swelling enlarging?
  - Fever?
- Main concern in tumours → **pain and swelling**
- **Past history**: previous operations → ask again, carefully

  - First answer "no" → re-asked: abdominal operation → later found to be **cancer**
  - "Gland operation, simple" / "small cancer, removed"
  - Simple patients often unaware → **precise history** essential

High Yield
- Night pain
- Pain waking the patient from sleep or preventing sleep
- Pain increasing over a short period
- Pain unrelated to exertion
- Essentials: pain, swelling, past history

Examination Ask AI

- **General examination**: particular attention by tumour specialists
  - May give a **clue to the diagnosis**
- **Local examination**
  - **Swelling**
  - Muscle **wasting**
  - **Deformity**
  - **Neurovascular status**

Why the General Examination Matters Ask AI

- Tumours → usually **systemic diseases**, especially malignant tumours
- Differential diagnoses → **metabolic diseases**, also generalized diseases
- Takes no more than 3–4 minutes, top to bottom

Head-to-Toe Screen Ask AI

- Neck → **thyroid swelling**?
- **Lymph nodes**: cervical, submandibular; ask about swellings
- Quick look at other areas
- Female → **masses in the breast**?
- **Tenderness**; quick abdominal examination
- Swellings in the legs

Diagnosis by General Examination Ask AI

- Elderly + thyroid swelling + **history of thyroid cancer** → diagnosis made
- Woman with breast lump, or known cancer → diagnosis made
- Prostatic symptoms → usually (not always) refer to the **urologist**
- **Mass**, or mass in the loin → suspect tumour
- Case: female with bone lesion; diagnosis not established
  - Work-up: **mammography** → bone scan → core biopsy → open biopsy in theatre
  - After operation: **mass in the breast discharging** for 3 years
  - Examination → **malignant ulcer**
  - Breast examination (own or surgeon's) → would have made things much easier
- One minute of examination → easy diagnosis

Clinical Pearl
- Full work-up (mammography, bone scan, core and open biopsy) failed to diagnose
- Breast lump discharging for 3 years → malignant ulcer
- General examination → extremely important

Weight Loss Ask AI

- Omitted earlier from present illness and examination
- **Significant loss of weight**: more than 10 kg in a week to 10 days *[Q1 · As spoken; standard definition is >5-10% body weight over months, not 10 kg in 10 days.]*
  - ≈ 1 kg per day
- Indicates **catabolic status**

Laboratory Investigations Ask AI

- **Routine laboratory investigations** in suspected tumour
  - **Rule out infection**
  - Tumour vs something else
- **Tumour markers** for primary tumours → no role
  - **No tumour markers in sarcoma**
    - None for **osteosarcoma**
    - None for **Ewing sarcoma**
  - Diagnosis → **biopsy**, not laboratory tests
- Routine tests detect
  - **Infection**
  - **Metabolic disease**
  - **Hyperparathyroidism**
  - **Serum calcium** level
- General rules only → details with individual topics

Must-Know
- No tumour markers in sarcoma (osteosarcoma, Ewing sarcoma)
- Diagnosis → biopsy, not laboratory tests

Plain X-ray Ask AI

- First investigation; **mainstay** of diagnosis
- Determines lesion **extent**, aggressiveness, biological behaviour
- **Crucial step in diagnosis**
- X-ray first → **not MRI first**
- Four questions
  - **Where is the lesion?**
  - What is the lesion?
  - What is the lesion doing to the bone?
  - How is the bone reacting to the lesion?

Site of Lesions Ask AI

X-ray

- Classic X-ray chart of lesions → divided by age: less than 30 / more than 30 years
- Position along the bone: metaphyseal, diaphyseal, epiphyseal

Site	Lesions
Epiphyseal	Chondroblastoma, infection, giant cell tumour
Diaphyseal	Fibrous dysplasia, Ewing sarcoma, metastases, myeloma, lymphoma (adults)
Metaphyseal	Everything else

- **Most tumours metaphyseal**; few diaphyseal or epiphyseal

Spot Diagnosis vs Differential Ask AI

X-ray

- X-ray can diagnose → yes
- Opening lesion: surrounded by **sclerosis**, eccentric
  - Metaphyseal → more precisely **metaphyseal cortical**
- Three items of site
  - Which bone
  - Which part of the bone
  - **Cortical or medullary**
- Cortical + metaphyseal + young patient → most probably **simple bone cyst** *[Q2 · As spoken; simple bone cyst is central medullary; cortical metaphyseal lesion in child suggests NOF/fibrous cortical defect.]*
- Another lesion: typical shape, internal changes → spot diagnosis
- Examples → **X-ray can make the diagnosis**
- Larger lesion, described
  - **Epiphyseal**
  - Well-defined in one part, not in another
  - **Lytic** → no attempt at matrix formation
  - **Eccentric**
  - DD: cyst, giant cell tumour, infection, sarcoma → **not a spot diagnosis**
- **Expansile** lesion → most probably ABC (aneurysmal bone cyst) or giant cell tumour
- Lesion without **surrounding sclerosis**, with ill-defined part
  - vs well-defined lesion with surrounding sclerosis
  - DD: ABC, **chondromyxoid fibroma**
  - Giant cell tumour less likely → does not occur at this age
- **Ill-defined** lesion in a young patient → many possibilities
- First row → X-ray **makes the diagnosis**
- Second row → X-ray gives a **differential diagnosis**
- X-ray alone → not the end

MRI Ask AI

- Second step: **MRI with contrast** → always with contrast
- X-ray → spot diagnosis or differential diagnosis → MRI for the latter

Lesions Needing MRI Ask AI

MRI

- **Epiphyseo-metaphyseal lesion**, partly well/ill-defined, adult → wide DD
  - **Giant cell tumour** → first at this age
  - Cyst
  - Chondromyxoid fibroma
  - Something malignant
- **Distal radius**: expansile lytic, blown-out bone, thinning of the cortex
  - Most probably ABC; DD:
    - Giant cell tumour
    - ABC
    - **Telangiectatic osteosarcoma**
- Large lesion: no surrounding sclerosis, somewhat expansile, cortical thinning → many possibilities
- **Lytic, well-defined lesion** with breakdown → cystic or malignant
- **Osteosclerotic lesion** with new bone formation
- X-ray undecided → add MRI with contrast

Indications Ask AI

MRI

- Diagnosis not possible by X-ray alone
- **Normal plain X-ray** in a symptomatic patient
  - Lesion must destroy 30–50% of the bone → before apparent on plain X-rays

Requirements Ask AI

MRI

- **With contrast**
- **Whole bone**: joint above → joint below → detects skip lesion
- **T1, T2**, STIR studies / fat suppression

What MRI Shows Ask AI

MRI

- **Nature** of the lesion
- **Extent**: inside and outside the bone
- **Relation to the surrounding structures**
- Example: **destruction of the epiphysis**
- **Skip lesion**: another lesion at a distance from the primary; no continuity
- **Soft-tissue component**
- Signal: **low on T1**, high on T2
- Example: lesion away from adjacent structure → no erosion

High Yield

- MRI always with contrast
- Whole bone, joint above to joint below → skip lesion
- T1, T2, STIR / fat suppression
- Normal X-ray until 30–50% of bone destroyed

CT Ask AI

- Useful in **morphology** and affection of cortical bone
- **Osteoid osteoma**: tibia, axial cut → cortical lesion

- Cortical lesion with **destruction**
- CT → **investigation of choice in osteoid osteoma**
- **Staging**: search for secondaries in primary sarcomas
  - **CT chest** → pulmonary metastases
- Other investigations also ordered

Bone Scan, PET and Ancillary Imaging Ask AI

- **Routine**: X-ray + MRI; CT when indicated

Bone Scan Ask AI

- **Sensitive but not specific**
- Not always ordered → case-dependent
- Depends on **osteoblastic activity** → useless in non-osteoblastic lesions
- Extreme value in **metastatic disease** and primary sarcoma
- **Not routine staging**
- **Cortical lesion** → support with CT
- Suspected **sarcoma** or metastasis → bone scan

PET Scan Ask AI

- **PET-CT** / PET scan → same idea as bone scan
  - Bone scan: **radionuclide material** → image
  - PET: **metabolite** → image
- Use → **detecting residual and recurrent lesions**
  - Operated sarcoma → **recurrence**?
  - Metastases → unknown **origin** (primary)
  - After chemotherapy → **effect of chemotherapy** on the lesion
- **Increases specificity** with other radiologic studies
  - PET + CT together; both **positive** → higher specificity *[C1 · transcript: "both positive increases sensitivity"]*
- Tracer: **fluorodeoxyglucose (FDG)** instead of technetium
- Both imaged by gamma camera

	Bone scan	PET scan
Agent	Technetium (radionuclide)	FDG (metabolite)
Image	Black-and-white	Coloured
Main value	Metastatic disease, primary sarcoma	Residual/recurrent lesions, unknown primary, chemotherapy effect
Routine staging	No	No

Mammography and Ultrasound Ask AI

- **Mammography** → suspected breast lesion
- **Ultrasonography** → suspected liver lesion
- Staging not complete → but **diagnosis** may emerge here

  - Female: breast mass on mammography + bone metastases → **breast cancer**

Biopsy Ask AI

- **Tissue pathology** → through biopsy
- Biopsy → **last step, not the first step**
  - Don't hurry; complete proper work-up first
- Purpose: tissue for **histopathological diagnosis**

Types of Biopsy Ask AI

- **Cytology** → not used in bone
  - Bone diagnosis needs **cells and stroma** (architecture), not cells alone
- **Fine needle aspiration** → breast and thyroid; not bone
- **Core** → required in bone
  - **Open biopsy**
  - **Percutaneous**: CT-guided or ultrasound (sonar)-guided

Type	Use in bone	Note
Cytology	No	Cells alone; useful in breast
Fine needle aspiration	No	Useful in breast, thyroid
Core — open	Yes	Operating room
Core — percutaneous	Yes	CT- or ultrasound-guided

Histopathological Diagnosis Ask AI

- **Frozen section** → rarely used in bone
  - Pathologist in theatre with microscope → malignant cells or not
  - Rare → needs large hospitals, special specimen preparation
- **Paraffin section**: haematoxylin and eosin (H&E)
- **Immunophenotyping**
  - Paraffin: **spindle-shaped cells**
    - Atypia
    - Pleomorphism of the nuclei
    - Malignant appearance; many mitotic figures
  - → suspect **spindle cell sarcoma**
  - About ten types → which type of **spindle cell sarcoma**?
  - **Antibody** against specific antigen / protein in spindle cells
  - Positive **antigen–antibody reaction** → tumour type diagnosed

Pitfall

- Biopsy is the last step, not the first
- Cytology and FNA → not diagnostic in bone
- Bone needs cells and stroma → core biopsy

Principles of Staging Ask AI

- Diagnosis sequence: history → general exam → local exam → labs → X-ray → MRI → **biopsy** (definitive)
- Patient asks: **benign or malignant**? Which stage — first, second, third?
- Purposes of staging
  - **Prognosis**
  - **Planning of treatment**
- Malignant tumours: first reason → level of the tumour
- **American Joint Committee**, 1976 → first to introduce TNM
  - T: primary **tumour** *[C2 · transcript: "T = tumour grade"]*
  - N: **lymph nodes**
  - M: **metastasis**
- Second: **biologic behaviour** of the tumour
  - **Origin**
  - **Grade**: low or high
  - **Local spread**
  - **Metastatic spread**
- Staging = biologic behaviour + **extent** → advanced or not

Prognostic Factors Ask AI

- **Grade** (aggressiveness)
  - Low grade or high grade
- **Extent** → signs of aggressiveness
  - Greater extra-compartmental extent → more aggressive
  - Involvement of **neurovascular bundles** → more aggressive
  - **Multifocal** → more aggressive
- **Size**: risk of local recurrence and spread ↑ with size, or size more than 8 cm
- **Metastases**: local or regional
- Bad prognosis
  - Larger size
  - Vascular invasion
  - **Soft-tissue extent**
  - Local or systemic spread

Staging of Benign Bone Tumours Ask AI

- Previous staging → malignant tumours
- **Benign bone tumours** → three stages: latent, active, aggressive
- **Latent**
  - Discovered accidentally, like the incidental lesion earlier
  - **"Leave me alone" lesion**: asymptomatic
    - Leave it alone → it leaves you alone
  - Inactive → does nothing to the patient
- **Benign active**

  - Example: **simple bone cyst** enlarging until it reaches the physis *[C3 · transcript: "reaches the epiphysis"]*
  - Examples: ABC, simple bone cyst
- **Benign aggressive**
  - Progressive → **destroys the bone**
  - Example: **giant cell tumour** → eats bone → articular cartilage → joint

Stage	Activity	Behaviour	Examples
Latent	Inactive, asymptomatic	Discovered accidentally	—
Active	Progressive	Within the bone; no breakout	Simple bone cyst, ABC
Aggressive	Progressive	Invades and breaks out of the bone	Giant cell tumour

Staging of Malignant Musculoskeletal Tumours Ask AI

Enneking System Ask AI

Enneking

- Primary malignant musculoskeletal tumours → **Enneking**
- Stages 1, 2, 3 → by T, M, G
- A / B → by **extent**

Parameter	Category /	Category /
T (extent)	T1 intracompartmental	T2 extracompartmental
M	M0 no regional or distant metastasis	M1 regional or distant metastasis
G (grade)	Grade 1 low grade	Grade 2 high grade

Stage	T	M	G
1A	T1 intracompartmental	M0	G1 low grade
1B	T2 extracompartmental	M0	G1 low grade *[C4 · transcript: "1B: with regional metastasis"]*
2A	T1 intracompartmental	M0	G2 high grade
2B	T2 extracompartmental	M0	G2 high grade
3	Any	M1 regional or distant	Any *[C5 · transcript: "metastasis = Grade 3"]*

- Not metastatic → **Stage 1 or 2**
  - A or B → by T1 / T2
  - 1 or 2 → by low / high grade
- Metastasis → **Stage 3**

- Examples
  - T1 M0 G1 → Stage 1A
  - T1 M1 G1 → Stage 3

AJCC 2003 Bone Staging Ask AI

AJCC bone

- **American Joint Committee (AJCC)** → modified version of 2003, not that of 1976
- Similar to Enneking
  - Adds **lymph nodes**
  - Still intracompartmental vs extracompartmental
  - Adds **size** → 8 cm or not

Category	Definition
T1	8 cm or less in greatest dimension
T2	More than 8 cm
T3	Discontinuous tumour → skip lesions
N0	No lymph nodes
Any lymph node	Immediately Stage 4

- M0, M1 → as in Enneking
- Grade 1, Grade 2 → more detailed, **according to the pathology**
  - Well differentiated
  - Moderately differentiated
  - Poorly differentiated
  - Undifferentiated
- Added to Enneking classification: lymph nodes, size, differentiation grade

Paediatric Soft-Tissue Sarcoma Ask AI

IRSG

- **Intergroup Rhabdomyosarcoma Study Group** → stages 1, 2, 3, 4

Stage	Definition
1	Localized disease
2	Total gross resection with evidence of regional spread
3	Incomplete resection or biopsy with gross residual disease
4	Any size primary tumour, with or without regional lymph nodes, with metastasis

AJCC Soft-Tissue Sarcoma Ask AI

AJCC STS

- Separate **American Joint Committee** system for soft-tissue sarcomas
- Based on size, regional lymph nodes, metastasis
- Heavily size-dependent → **≤ 5 cm or > 5 cm** *[C6 · transcript: "≤ 5 cm or ≥ 5 cm"]*

Must-Know

- Enneking: T (compartment), M, G → Stages 1A–3
- Any metastasis → Enneking Stage 3
- AJCC bone: size cut-off 8 cm; any lymph node → Stage 4
- AJCC soft-tissue sarcoma: size cut-off 5 cm

Summary: Diagnostic and Staging Pathway Ask AI

- Aim → recognise the tumour patient
- Presentation: nothing at all, increasing pain, or swelling
- History: increasing or not
  - Wakes from sleep? Mechanical? Fever? **Loss of weight**?
- **General examination** in 3–4 minutes
  - Any lesions, **skin lesions**, abdomen, thyroid
- Routine laboratory investigations
- **X-ray** first → may diagnose alone
  - Not diagnostic → MRI with contrast
- Suspected malignant lesion → **CT chest**
- Suspected cortical lesion → **CT** on the lesion
- Suspected sarcoma or metastatic disease → **bone scan** and PET scan
- Final answer: the patient's stage
  - **Benign** → latent, active or aggressive
  - **Malignant** → Stage 1, 2 or 3
    - By grade, intracompartmental or not, metastasis or not

Imaging of Bone Tumors

(A Systematic Approach to the Plain Radiograph)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 29) As spoken; interrupted reaction is called aggressive/sarcoma later (a051), contradicting 'less aggressive'.

Q2 (p. 31) As spoken; sunburst is classically osteosarcoma (lecture itself says so at a106); Ewing typically onion-skin.

Q3 (p. 37) As spoken; rugger-jersey spine is horizontal sclerotic bands of renal osteodystrophy, not hemangioma vertical striations.

Work-up Recap and Goals of X-ray Reading Ask AI

Clinical Work-up Recap Ask AI

- Presentation: **pain with swelling**
- Examination → general + local
  - General examination points
    - Young patient → **deformities**
    - Older patient → swellings elsewhere
      - **Neck swelling**
      - Lymph node swelling
- Investigations
  - Routine labs
  - **X-ray**
  - MRI of whole bone, with contrast
- Further tests per X-ray findings → **bone scan** or PET-CT
- Lecture focus: **Radiology of Bone Tumors** → how to read the X-ray
  - Very important in practice and in the exam
- Bone tumors → reading, follow-up, **fixed scheme**

Goals of Reading the X-ray Ask AI

- First goal → **benign or malignant**
  - Questions to answer
    - 1 → **location** of the lesion
    - 2 → what the lesion is doing to the bone
    - 3 → what the bone is doing to the lesion → **reaction**
    - 4 → **matrix**
    - Last, with experience → tissue type
- X-ray → basis of the **patient plan**
  - Full work-up (MRI, bone scan …) vs **MRI only**
  - Plan not fixed → changes with history and X-ray
- No bone tumor diagnosis **without X-ray**
  - X-ray = the **map**
  - X-ray + MRI → **complementary**

Location of the Lesion Ask AI

Four Questions of Location Ask AI

- First point on the X-ray → **location**
- Image recalled from previous lecture
- Location → **four questions**
  - Which bone? → easiest, e.g. **femur**
  - Epiphyseal, metaphyseal, or diaphyseal?
    - Below physis / between physis and diaphysis → **metaphyseal**

        - No fixed measurement for this boundary
      - Beyond that region → **diaphyseal**
    - Cortical or medullary?
- Each location → own **differential diagnosis**
  - Metaphyseal, cortical, medullary lesions → separate differentials
- Depth options
  - **Corticomedullary**
  - Purely **cortical**
- Central or eccentric?
  - Epicenter included in **anatomical axis** of bone → **central**
  - Epicenter away from anatomical axis → **eccentric**
  - Some lesions typically **eccentric**, others typically central
- Location → first element of lesion description

Question	Options
Which bone	e.g. femur, tibia, fibula
Longitudinal plane	Epiphyseal / metaphyseal / diaphyseal
Depth	Cortical / corticomedullary / medullary / surface
Transverse plane	Central / eccentric

Longitudinal Plane Ask AI

- **Most lesions** → metaphysis (previous lecture)
- **Epiphyseal** and diaphyseal lesions → least common
  - Least common in children and adults alike
- Of the four regions → most lesions **metaphyseal**
- Recognition → practice and repetition

Transverse Plane Ask AI

- Two types → **central** and eccentric
  - Central → lesion in **center of the bone**
    - Lesion center in anatomical axis of bone → **central lesion**
  - Most famous central lesion → **simple bone cyst**
  - Central examples → simple bone cyst, **fibrous dysplasia**

Surface Lesions Ask AI

- Lesions outside the bone → **surface lesions**
  - Arising from outside bone, neither medullary nor cortical → think **surface lesion**
- Features of each lesion → with repetition
- Current aim → **describe** a lesion and how to think
- Typical sites → in individual lesion lectures

Bone Cysts Ask AI

- Bone cysts → **not true tumors**
  - Classified under **tumor-like conditions**
- Two most famous cysts
  - **Simple bone cyst**
  - **Aneurysmal bone cyst** (ABC)
- Some cysts → **secondary** to other bone lesions
  - e.g. fibrous dysplasia → **secondary ABC**
- Commonest lesions with **secondary ABC**
  - **Fibrous dysplasia**
  - **Enchondroma**
  - **Chondromyxoid fibroma**
  - **Chondroblastoma**
  - **Giant cell tumor**
- Each → covered in detail later
- Main primary cysts → simple cyst, ABC
  - Many other lesions → **secondary ABC changes**
    - Fibrous lesions
    - Cartilage lesions
    - **Giant cell tumor** → mechanism explained in GCT lecture

Must-Know

- Cystic lesions of bone → **primary or secondary**
- Cystic lesions of bone → **tumor-like conditions**, not true tumors
- Secondary ABC → fibrous dysplasia, enchondroma, chondromyxoid fibroma, chondroblastoma, GCT

Lesion Effect on Bone: Lysis and Margins Ask AI

Applying Location: Femoral Lesion Ask AI

Case

- Description within limits of what is learned so far
- **Femur**, skeletally immature patient
  - Longitudinal → **metaphyseal**
  - Center outside **anatomic axis** → eccentric (more or less)
  - Cortical or medullary → needs **lateral view**
    - Appears **corticomedullary** → part cortex, part medulla
- Four location questions → answered

Lysis and Margin Definition Ask AI

- Lesion effect on bone → **lysis** (lytic)
  - Appears as **black dots**
- Pattern → well-defined or ill-defined?
  - Margin traceable with finger or pen → **well-defined**
  - Not traceable, beginning and end unknown → **ill-defined**

        - Examples: diaphyseal lesion, metaphyseal lesion
- Well-defined pattern → **geographic** (has subtypes)
- Ill-defined pattern → **permeative** or moth-eaten
- Example: well-defined lesion + surrounding **sclerosis**
  - Sclerosis around lesion → **very good sign**
  - Mother bone surrounds lesion → **sclerotic margin**
- Each pattern → own **differential diagnosis**, learned with practice

Margin and Aggressiveness Ask AI

- **Margin** → zone between tumor and native bone
  - Slowly progressive → **distinct margin** (clear, visible)
  - Rapidly progressive → **indistinct margin** (unclear, not visible)
- More aggressive lesion → **less clear margins**
- Aggressive lesions → **destroy the bone**
- Aggressive lesions → **benign aggressive** or malignant

Pitfall

- X-ray → **no pathological diagnosis**, only differential diagnosis
- List 3–4 possibilities → **pathology** decides
- "Looks like ..." → from viewing many radiographs

Geographic Destruction Subtypes Ask AI

- Traceable + surrounded → **geographic destruction** (lytic)
  - With surrounding sclerosis → **well-defined with sclerosis**
  - Traceable, no surrounding sclerosis → **geographic, well-defined, without sclerosis**
  - Partly traceable, partly ill-defined → **more or less well-defined**
    - = geographic destruction with **ill-defined margin**
- Aggressiveness increases → from **sclerotic margin** to ill-defined margin and beyond

Pattern	Margin	Aggressiveness
Geographic, well-defined with sclerosis	Traceable, sclerotic rim	Least
Geographic, well-defined without sclerosis	Traceable, no rim	↑
Geographic with ill-defined margin	Partly traceable	↑↑
Moth-eaten / permeative	Not traceable	Most

Case: Distal Tibial Lesion with Sclerotic Margin Ask AI

Case

- Traceable margin + **sclerotic bone** from mother bone
- Skeletally immature, **distal tibia**
  - Metaphyseal, **eccentric**, corticomedullary
  - Well defined → **well-defined sclerotic margin**
  - Margin **distinct**; transition from mother bone visible

- Appearance → well known with a specific lesion

Case: Proximal Fibular Lesion without Sclerosis Ask AI

Case

- **Proximal fibula**, metaphyseal
  - Traceable → **well-defined**
  - **Thinning out** of the cortex
  - Metaphyseal
  - Central, with bone **expansion**
- Example → **well-defined without surrounding sclerosis**
- Both cases → **benign-looking**
  - Diagnosis → pathology; X-ray → differential

Case: Ill-defined Lesion of Proximal Tibia Ask AI

Case

- **Osteolytic lesion**, proximal tibia, adult
  - Location → **epiphyseo-metaphyseal**
  - Definitely **lytic**
  - Center with center of tibia → **central**
  - **Medullary** → takes whole medulla
  - Nothing traceable → **ill-defined margin**
- Central, in an adult → could be anything
  - Resolved by → history, examination, MRI, **biopsy**
  - Plus **staging** (previous lecture)

Ill-defined Destruction: Moth-eaten and Permeative Ask AI

Moth-eaten Pattern Ask AI

- **Moth-eaten** → punched-out lesions
  - Multiple scattered holes, **varying size**
  - Normal bone between holes
  - Seen in → metastasis, **myeloma**
  - Bone gnawed from inside, like moth eating fabric → **very characteristic**
- Example → **thinning out** of cortex + destruction inside bone
  - Whole bone involved, beginning and end unknown → **ill-defined destruction**

Permeative Pattern Ask AI

- Famous ill-defined pattern of **sarcomas** → **permeative**
  - Beginning and end unknown
  - Extension → **cortical destruction** + soft tissue mass
  - **Soft tissue mass** → characteristic feature
  - Any **aggressive lesion** → most famous: sarcoma
- Lesion takes whole bone → cortex and bone affected
  - These features → evidence of **aggressiveness**
  - Lucency also present
- Sound cut out during moth-eaten part → repeated

Recap After Audio Interruption Ask AI

- Moth-eaten → starts at one point, takes **whole bone**
  - Bone **gnawed from inside**
  - Well known in → **myeloma**, metastases
  - Part here, part there → beginning and end unknown
- Not traceable → think **aggressive** immediately
- Permeative → beyond gnawing
  - Takes **whole medulla**
  - Very violent, with **soft tissue mass**

Feature	Moth-eaten	Permeative
Appearance	Multiple punched-out holes, varying size	Whole medulla, beginning and end unknown
Between lesions	Normal bone	—
Cortex / soft tissue	Thinning out of cortex	Cortical destruction + soft tissue mass
Typical lesions	Myeloma, metastases	Sarcomas, any aggressive lesion

Case: Sclerotic Metaphyseo-diaphyseal Lesion Ask AI

Case

- **Metaphyseo-diaphyseal**, skeletally immature
  - Not lytic → **sclerotic lesion**
  - **Central** → takes whole medulla
  - Very violent periosteal reaction
  - Very large **soft tissue mass**
- Very well known in **bone sarcomas** → e.g. osteosarcoma

Bone Reaction: Periosteal Reaction Ask AI

Destruction versus Reaction Ask AI

- So far → tumor location + **bone destruction**
- **Destruction** → lucency or lysis
- **Reaction** → sclerosis, remodeling, periosteal reaction
- Periosteal reaction → part of **remodeling of bone**; has types
  - Formation time → **10 days to 3 weeks**
  - Configuration → depends on **nature of the process**
    - Plus intensity and aggressiveness of lesion

Types of Periosteal Reaction Ask AI

- Lesion + periosteal reaction + large soft tissue mass → **aggressive**
- **Less aggressive** reaction → thick, interrupted, or laminated *[Q1 · As spoken; interrupted reaction is called aggressive/sarcoma later (a051), contradicting 'less aggressive'.]*
- Destruction with periosteal reaction → possible
- Names and details of types → **not very important**
  - Important → recognizing appearance

- Examples
  - Very mild reaction around **fibular lesion**
  - Reaction extending top to bottom
  - Reaction above and below, **none in the middle**
  - **No periosteal reaction** vs continuous reaction

Continuous versus Interrupted Ask AI

- More **continuous** reaction → less aggressive pathology
- **Interrupted** → broken by bone destruction in the middle
  - Bone formation not completed → more in **sarcoma**
- **Thin lamellated** → usually benign
  - Benign lesions
  - Stress fractures
  - Simple healing processes
- Periosteal reaction → bone's **attempt of remodeling**
- Exception: continuous type with **sunburst appearance**
- Not continuous → most likely **malignant**
- = reaction of the mother bone

Periosteal reaction	Implication	Typical setting
Continuous	Less aggressive, reassuring	—
Thin lamellated	Benign	Benign lesions, stress fractures, healing
Interrupted	Aggressive, most likely malignant	Sarcoma
Continuous sunburst	Very aggressive (exception)	Ewing sarcoma

Codman Triangle Ask AI

- Sign strongly incriminating a lesion → **triangle**
  - Base → the lesion
  - One side → **periosteal reaction**
  - Other side → cortex (mother bone)
- Ill-defined lesion + periosteal reaction → **Codman triangle**

Exam Pearl

- Codman triangle → evidence of **aggressive lesions**
- Not necessarily sarcomas
- **Not diagnostic** → radiologic finding only
- Can be part of the periosteal reaction

Sunburst Appearance Ask AI

- Long periosteal reaction, **long segment** of bone → sunburst appearance
  - Very long segment, **continuous on both sides** of bone
  - Areas of **new bone formation** in the middle → most aggressive
- Interrupted → disaster; uninterrupted → usually reassuring

  - Exception → continuous + **soft tissue mass** in middle
    - → very aggressive
- **Destruction** + very rapid new bone formation → sunburst
  - Especially in **Ewing sarcoma** → classic look *[Q2 · As spoken; sunburst is classically osteosarcoma (lecture itself says so at a106); Ewing typically onion-skin.]*

Tumor Matrix Ask AI

- Last item, comes with experience → **tumor matrix**
- Ill-defined lesion → matrix often not seen, really difficult
- **Calcification** within lesion → lesion has matrix
- Matrix most in **cartilage tumors**
  - Also osseous and **fibrous tumors**
- Matrix → very important, most of all in cartilage tumors
  - Cartilage tumors → visible **calcification**

Benign versus Malignant Features Ask AI

- Aim → **benign or malignant**, not pathology
- Benign lesion
  - **Well-defined**
  - Sclerotic margin
  - Periosteal reaction mild to continuous → **solid**
  - Geographic → traceable with pen or finger
  - Central or eccentric → **irrelevant**
- Benign aggressive or malignant
  - **Destruction** of mother bone
  - Not traceable
  - Areas of destruction in middle → **moth-eaten** or permeative
  - **Interrupted** periosteal reaction
  - Perforates cortex → extends outside
- Classification → **benign latent** (inactive), benign aggressive, malignant
  - Based on → **pattern of destruction**, soft tissue mass, periosteal reaction

Feature	Benign	Benign aggressive / malignant
Definition	Well-defined, geographic	Not traceable, moth-eaten or permeative
Margin	Sclerotic	No sclerosis
Periosteal reaction	Mild to continuous (solid)	Interrupted
Cortex	Intact	Perforated → extends outside
Soft tissue mass	Absent	Present

Surface Lesions Ask AI

Case: Distal Femoral Surface Lesion Ask AI

Case

- Applied without lesion names
  - **Distal femur**, adult
  - **Metaphysis**
  - Slightly unclear → more or less **well-defined**
  - On bone from outside → **surface lesion**
- Differential of **surface lesions**
  - **Osteochondroma**
  - **Parosteal osteosarcoma**
  - **Myositis ossificans**
- Differentials → come with practice
  - Many tumors → differentials added progressively

Surface versus Cortical Lesion Ask AI

- Objection → cortex involved, eccentric, expansion → cortical?
  - **CT** → no continuity with cortex → **surface lesion**
- Routine → MRI or CT for diagnosis
  - This case → **characteristic surface lesion**
- **Cortical lesions** → do not produce this expansion
- **Expansile lesions** → must be continuous with region of origin (medulla)
  - Expansion → must arise from **medulla**
  - Lesion → from medulla, from bone, or extending out of bone
- Lesion name → not important; **description** is

Worked Cases: Description and Differential Ask AI

Case 1: Cortical Eccentric Metaphyseal Lesion Ask AI

Case

- Aim → **differential diagnosis**, not final name
  - Skeletally immature, **lytic lesion** in metaphysis
  - Partly traceable → **geographic destruction**
    - Ill-defined margin in places
  - **No surrounding sclerosis** (or mild)
  - **Eccentric**
  - Appears **cortical**
- Possibilities → **infection** or tumor
- Cortical eccentric lesion in young patient → apply the rule
  - More or less **well-defined**
  - **No periosteal reaction**
  - **No soft tissue mass**
  - → **benign-looking**
    - Fits infection, benign, or benign aggressive, far more likely

Benign Lesion Families Ask AI

- Benign tumors → four families

Family	Key feature
Osseous	e.g. osteoma, **osteoid osteoma** → cortical lesion
Cartilage	Contain calcification
Fibrous	Fibrous matrix
Cysts	Expansion; not in cortical region

- Differential for Case 1

Candidate	Fit	Reason
Cyst	No	—
Giant cell tumor	No	Not the age, not the site
Cartilage lesion	No	No calcification
Osseous → osteoid osteoma	Yes	Cortical
Infection	Yes	—
Chondroblastoma	No	Chondroblastoma → **epiphysis**

- Way of thinking → read all tumor types, know distinguishing features
  - Clearer in **benign tumors** lectures (next two)
- Goal → **pattern of thinking** + fixed descriptive scheme
  - Any lesion → own differential diagnosis

Case 2: Central Metaphyseal Lesion with Sclerosis Ask AI

Case

- **Metaphyseal**, skeletally immature, lytic
  - **Geographic destruction**, partly ill-defined
  - **Central**
  - Some surrounding **sclerosis**
  - **Medullary**
  - No periosteal reaction
  - **No thinning** of cortex; no soft tissue mass
- Complete description → **scheme** fully applied
- Diagnosis → after reading all tumor types

Case 3: Lytic Central Expansile Metaphyseal Lesion Ask AI

Case

- **Metaphyseal** lesion
- Mostly traceable → **geographic destruction**
  - Small part not traceable
- **Central**
- Cortical **expansion** + crack in cortex
- Lytic, central, metaphyseal, young patient

Candidate	Fit	Reason
Cyst	Yes	—
Giant cell tumor	No	Not its age
Osteoid osteoma	No	Cortical
Enchondroma	No	Enchondroma → calcification
Chondroblastoma	No	Epiphyseal

- Current stage → **describe well**

Case 4: Inactive Cyst Ask AI

Case

- Child
- **Metaphyseo-diaphyseal** → metaphyseal, extending into diaphysis
- Clearly **medullary** → takes whole medulla
- Center with axis of bone → **central**
- Traceable
- Drawn white line → **sclerotic border**
  - Previous case → no sclerosis, **cavitations**
  - → well-defined sclerotic margin, **geographic destruction**
- No **cortical breakdown**
- No **pathological fracture**
- No periosteal reaction
- No soft tissue mass
- No periosteal reaction, mass, cortical destruction → most probably **benign tumor**
  - Appearance of **inactive cyst**
  - Active cyst vs inactive cyst → compared images
- Details → after all tumor types
- Lesion names → not important; **description** is

Case 5: Eccentric Epiphyseo-metaphyseal Femoral Lesion Ask AI

Case

- Bone → **femur**
- Region → **epiphyseo-metaphyseal**
- Center not in anatomical axis of bone
  - → **eccentric**
- Medulla + cortex → **cortical expansion**
  - Term "eccentric expansion" → not important
  - Key → eccentric lesion expanding **mother bone**
- No new bone formation → **purely lytic**
- **Thinning out** of cortex + extension outside bone
- Periosteal reaction → mild, not very clear
- Definite **soft tissue mass**
- → **signs of aggressiveness**

Candidate	Fit
Infection	Possible
Simple bone cyst	No → not this aggressive
Aneurysmal bone cyst	Fits
Giant cell tumor	Fits in adult, not in child

Candidate	Fit
Sarcoma	Fits → lytic, aggressive-looking

- Differential varies → **description fixed**

Case 6: Central Tibial Lesion with Eccentric Expansion Ask AI

Case

- **Metaphyseal**, tibia, skeletally immature child
- Center central + bone expansion → **central with eccentric expansion**
- **Medullary**, definitely
- Four location questions → tibia, central, medullary, expansion
- Definitely **lytic**
- **Cortical breakdown** → expanded, starting to break
- Soft tissue mass → not much
- Beginning **lamellated periosteal reaction** → aggressive-looking
- Fits → **ABC**, sarcoma, others
- Key → description and thinking, same way each time
- Similar lesion → same questions

Case 7: Central Lytic Lesion of Distal Femur Ask AI

Case

- **Distal femur**, skeletally immature
  - **Metaphyseal**
  - **Central**
  - Mainly **medullary** → MRI or CT confirms
  - Some cortical expansion → **ballooning** of mother bone
    - Shown on lateral view
  - **Lytic**, certainly
  - Traceable, no line around → **more or less well-defined**
    - **No sclerotic margin** from mother bone
  - Center → approximately with the axis
  - Periosteal reaction → very mild
  - No soft tissue mass
- Summary description
  - **Metaphyseal**, skeletally immature
  - **Lytic** → partly well-defined, partly ill-defined
  - No cortical expansion
  - **Central**, medullary
  - No periosteal reaction

Candidate	Fit	Reason
Cyst	Yes	—
Giant cell tumor	No	Not its site, not its age
Infection	Unlikely here	Infection → great simulator
Fibrous dysplasia	Yes	Can occur at this site

High Yield
- Benign-looking bone tumor → think four families only
- **Cyst**, fibrous, cartilage, osseous
- **Infection** → great simulator
- GCT excluded in child → not its age

Characteristic Appearances on Plain X-ray Ask AI

Calcified Matrix: Enchondroma Ask AI

- **Calcified matrix** → immediately enchondroma
  - Multiple spotted calcification → characteristic of **enchondroma**
  - **Enchondroma in the hand** → different, lytic lesion
    - Commonest **bone tumor of the hand**

Case: Eccentric Expansile Proximal Tibial Lesion Ask AI

Case

- **Proximal tibia**, skeletally immature
  - **Metaphyseal**
  - **Eccentric**
  - Cortex involved → **expansion and thinning** of cortex
- Differential → **ABC**, chondromyxoid fibroma

Differential Diagnosis Scheme to Come Ask AI

- End of **benign bone tumors** → scheme for differential diagnosis
  - **Central** benign-looking lesion → what to think of
  - **Surface lesion** → what to think of
  - **Calcified lesion** → what to think of
- Prerequisite → bone tumors read well
- Today's aim → **scheme**, not diagnosis
  - Correct X-ray description + **fixed way of thinking**

Synovial Chondromatosis Ask AI

- Classic appearance → **synovial chondromatosis**
  - Sometimes visible on X-ray

Case: Central Metaphyseo-diaphyseal Lesion (Brown Tumor) Ask AI

Brown tumor

- **Metaphyseo-diaphyseal**, adult
  - **Destruction**, central, medullary

  - Cortical thinning/expansion
- 100% description → differential ABC or **fibrous dysplasia** acceptable
  - At level of **systematic thinking**
- Central lesions taking most of bone → may be **metabolic**
  - This lesion → **brown tumor**
- Diagnosis not the point → **description** is
- Description → lytic, central, **medullary**
  - No aggressive features → benign-looking
  - **Multiple**

Vertebral Striations Ask AI

- Plain X-ray alone → diagnoses **a few lesions**, not all
  - Striations in vertebral body → e.g. **hemangioma**
  - **Corduroy** or rugger-jersey → vertical striations (not very important) *[Q3 · As spoken; rugger-jersey spine is horizontal sclerotic bands of renal osteodystrophy, not hemangioma vertical striations.]*

Punched-out Lesions Ask AI

- **Punched-out lesion** → like moth-eaten
  - Piece taken out of bone
  - Very characteristic → **eosinophilic granuloma**, myeloma

Malignant Bone Tumors: Radiographic Patterns Ask AI

Case: Permeative Diaphyseal Lesion in a Child Ask AI

Case

- Skeletally immature
  - Diaphyseal
  - **Medullary**
  - **Central**
  - Lesion indistinguishable from bone → **ill-defined**
  - Long segment + considerable periosteal reaction + **soft tissue mass**
- **Aggressive-looking** in young patient → same reasoning

Tumor Bone: Osteosarcoma Ask AI

- Any lesion with destruction + **new bone formation** → same approach
- Bone formation = tumor bone → characteristic of **osteosarcoma**
  - Also **sunburst appearance** → but not Ewing sarcoma
- Description
  - Skeletally immature (approximately)
  - **Metaphyseo-diaphyseal**
  - Beginning and end unknown
  - Destruction, periosteal reaction, **soft tissue mass**
  - → definitely **malignant tumor**

Chondrosarcoma Ask AI

- Beginning and end unknown
  - Reaction unclear

  - Very violent **calcification**
- → **chondrosarcoma**
  - Calcification vs **ossification** → chondrosarcoma lecture

Fibrous Matrix: Fibrosarcoma Ask AI

- Purpose → show **fibrous matrix**
- Fibrous matrix → causes **lysis** → fibrosarcoma
  - Adult
  - **Ill-defined destruction** of bone
  - Mild reaction (image not clear)
- Own distinctive appearance

Metastases and Myeloma Ask AI

Case: Pathological Fracture with Renal Cancer Ask AI

Case

- Adult
  - **Destruction**, no sclerosis
  - Aggressive-looking
  - **Central** in bone
- History → **kidney cancer** + fracture without trauma
  - → most likely **metastasis**
- X-ray → **linking ring** between history, examination, diagnosis
- Not pathology → appearance: **aggressive**
  - Destruction
  - Cortex involved
  - Partly traceable only
  - **Soft tissue mass**
- Elderly + kidney cancer → **serious problem**

Osteolytic Metastases Ask AI

- Frank appearance → **osteolytic metastasis** from breast cancer
- Same appearance → **myeloma** and metastasis
  - No pathological diagnosis from X-ray
- Also → **multiple ill-defined osteolytic lesions**
- Bone metastases → may be **osteolytic**

> **Clinical Pearl**
> - **Multiple lesions**, one or both sides → think systemic
> - Systemic → **metabolic** or metastatic

Sclerotic Metastases Ask AI

- Sclerotic, slightly destructive, **new bone formation** → metastatic bone disease
  - Some **punched-out lesions** also
- Bone scan of the spine

Myeloma Ask AI

- **Myeloma** → punched-out lesions (myeloma lecture)
  - Lesions all over the bone
  - **Multiple punched-out lesions** → serious problem
  - Small holes → **moth-eaten** pattern
  - Lesion boundaries indistinguishable

Summary of the Reading Scheme Ask AI

Case: Mixed Lytic-Sclerotic Lesion in an Adult Ask AI

Case

- Lecture → continuation of **previous lecture**
- Goal → describe X-ray + **differential diagnosis**
- Description
  - **Mixed lytic-sclerotic** → destruction + sclerosis
  - Adult, not traceable → **ill-defined destruction**
  - No periosteal reaction, but takes **whole medulla**
- → **malignant-looking**
  - Affects **staging** and diagnosis
- X-ray → differential diagnosis, **not pathology**

The Scheme for Long-bone Lesions Ask AI

- First → **where is the lesion**: four questions
  - Which bone
  - Part of bone → **metaphyseal** or diaphyseal
  - Cortical or medullary
  - Central or eccentric
- Second → **lysis** or new bone formation
- Third → **bone reaction**
  - Lysis → ill-defined or **well-defined**
  - **Periosteal reaction**? soft tissue mass?
  - Bone reaction to lesion → appearance
- Fourth → **calcification** or matrix
- Last → signs of benignity vs malignancy
  - Well-defined, sclerosis, no cortical thinning, no periosteal reaction, no mass → **benign**
  - Destruction, cortical breach, mass, interrupted periosteal reaction → **malignant**
  - Big difference → **differential diagnosis**
- Tumor types → after the tumor-type lectures

High Yield

- Location → **four questions**
- Lysis vs new bone; margin; periosteal reaction; soft tissue mass
- **Matrix** → calcification
- Benign vs malignant → pattern of destruction, mass, periosteal reaction

New Bone Formation versus Calcification Ask AI

- **New bone formation** → continuous, bland sheet
  - Takes bone segment beginning to end
  - **No gap**, no normal bone in middle
  - Not as spots
- New bone formation → accompanied by **attempt of remodeling**
  - → must have **periosteal reaction**
- **Calcification** → completely different
  - Described as benign process
  - **Intramedullary**, very small adjacent spots
  - Surrounded by **lysis**
  - **Far away** from mother bone → not a segment
  - Spots with **bone between them**
  - **Small dots** → like scattered seeds in middle of medulla
- New bone → reaches mother bone, **periosteal reaction**, continuous, bland sheet
- Calcification → normal bone in between, away from cortex → **classic calcification**

Feature	New bone formation	Calcification
Distribution	Continuous bland sheet, beginning to end	Very small spots / dots
Normal bone in between	None, no gap	Present
Relation to mother bone	Reaches mother bone	Far away, away from cortex
Location	Bone segment	Intramedullary
Surroundings	Periosteal reaction (remodeling)	Lysis

Benign Bone Tumors

(Describing the Lesion: Surface, Cortical and Medullary Tumors)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 51) “slightly older than osteoblastoma” → “slightly older than osteoid osteoma”

C2 (p. 54) “osteoblastoma mainly cortical” → “osteoid osteoma mainly cortical”

C3 (p. 63) “really causes toxic manifestations” → “rarely causes toxic manifestations”

Queried - kept as spoken, but looks wrong

Q1 (p. 54) As spoken; osteoblastoma typically presents in second-third decades, mostly under 30, not thirties.

Q2 (p. 66) As spoken; hamartoma is disorganised native tissue at normal site; altered location is choristoma, not dysplasia.

Systematic Description of Bone Lesions (Review) Ask AI

Descriptive Checklist Ask AI

- **Site** — which bone; number → solitary vs multiple
- Part of bone → **metaphyseal**, epiphyseal, diaphyseal
- Position → **central** vs eccentric
- Compartment → **cortical**, medullary, corticomedullary
- **Lesion → bone**: lytic, cavitation, cystic cavities
- **Bone → lesion**: periosteal reaction or sclerosis
- **Tumor matrix** — clarified on the radiographs below

Surface, Cortical and Medullary Lesions Ask AI

- **Surface lesion** — exophytic, grows entirely outside bone, has a stalk
- Neck continuous with **mother bone** → surface; not cortical, not medullary
- Surface lesions
  - **Osteochondroma**
  - **Myositis ossificans**
  - **Parosteal osteosarcoma** — later lectures
- Typical surface appearance → exophytic, outside bone
- Alternative → **cortical** lesion
- **Osteoid osteoma** → typical cortical lesion, inside cortex
- Cortical lesion may be surrounded by
  - sclerosis → **non-ossifying fibroma**
- **Geographic** → inner line + surrounding outer line
- Example → **ill-defined**, epiphyseal–metaphyseal, medullary
  - Edges untraceable → not **geographic**
  - → aggressive-looking
- Contrast
  - Previous → well-defined, **cortical**
  - This one → ill-defined, **medullary**

Matrix: Osseous versus Cartilaginous Ask AI

- New bone formation → **osseous matrix** (osteoid)
- Example lesion
  - Ill-defined → end not traceable
  - **Periosteal reaction** → interrupted
  - Large **soft-tissue mass**
- New bone inside lesion → "**bone inside bone**" → typical osseous lesion
- Alternative → **calcified matrix**
- Typical **enchondroma** → dots clustered together

Matrix	Appearance	Example
Osseous (osteoid)	New bone formation; "bone inside bone"	Osseous lesion

Matrix	Appearance	Example
Cartilaginous (calcified)	Dots clustered together	Enchondroma

From Description to Diagnosis Ask AI

- Images → review of previous lecture
- Example → no periosteal reaction; **cortical breakdown** present
- So far → descriptive features determined and recorded
- Diagnosis → study each tumor type and its characteristics
- Description = background → characteristics **fill in the spaces**

Benign Bone Tumors — Classification and Study Framework Ask AI

Tissue-Based Classification Ask AI

- Classification follows bone composition
  - Undifferentiated **mesenchymal** cells
  - Fibrous tissue
  - Cartilage
  - Osseous tissue
- → mesenchymal, fibrous, cartilaginous, osseous tumors
- **Hemangiomas**, lymphangiomas — cited as the commonest in this group (attribution uncertain)
- Bone = **living tissue** → blood vessels, nerves, marrow
  - → vascular, neural, lipogenic, notochordal tumors
- All above → benign bone tumors

WHO Classification Ask AI

- Older modified classification → not important
- **WHO classification** → followed; 2013, 2015, 2017 versions similar
- Basis → **histological sub-classification**

How to Study Any Tumor Ask AI

- Checklist
  - **Definition**
  - Age; sex
  - Bones → **male-to-female ratio**; long vs flat bones; metaphyseal vs epiphyseal
  - Clinical picture
  - Diagnosis
  - Differential diagnosis
- Same layout as a theoretical exam answer

Osteoid Osteoma Ask AI

Definition and Epidemiology Ask AI

- **Benign tumor** in vascular osteoblastic mesenchyme
  - Stroma → vascular

  - Osteoblasts → produce the tumor
- Mesenchyme → osteoblasts → **osteoid**; very vascular
- Age → 10–35 years
- 65% → long bones, mainly **femur** and tibia
- Feet, hands → possible
- Vertebrae → **rare**
- Mainly **cortical**
  - May be medullary
  - Subperiosteal
  - Intra-articular
- Spine → mainly **lumbar segment**
- **Males** > females
- Sites → long bones, flat bones, spine

Theories and Pathology Ask AI

- Theories
  - Vascular
  - Ischemic
  - Traumatic
  - Congenital
  - Infection
- Vascular osteoblastic tissue + **nerve endings** acted on by prostaglandins
  - → marked response to **NSAIDs**
- **Night pain** → ~1.5× morning level
  - Very vascular tumor → nocturnal vasodilatation → ↑ vascularity
- Theories → know explanation only

Clinical Picture Ask AI

- **Nocturnal localized pain** → main feature
- History → particular time? wakes from sleep?
- Night pain → **red flag** (alarming sign and symptom)
- Intra-articular → **synovitis**, arthritis, joint effusion — especially knee
- Spine → **painful scoliosis** — important cause
  - Analgesic → straightens in morning
  - Night → pain returns → leans forward or sideways
  - Deformity comes and goes → may become **permanent** until ablation
- **Point tenderness** → patient points with one finger
- **Referred pain** → somewhat rare, confusing
  - Hip lesion → knee pain
  - Knee lesion → tibia or ankle pain
- Mechanism → vascular tumor + nerve endings + **prostaglandins**

High Yield
- Nocturnal localized pain → alarming symptom
- Point tenderness → one finger
- Prostaglandin-mediated → marked NSAID response
- Spine → painful scoliosis

Imaging Ask AI

- X-ray → **cortical lesion** surrounded by active sclerosis
- **Bone scan** (skeletal scintigraphy) → confusing or intramedullary lesion
  - Shows **double density sign**
    - High-activity focus → **nidus**
    - Surrounding lower activity → reactive bone sclerosis
- **CT** → hallmark of diagnosis
- Cortical lesions → **CT better than MRI**
  - MRI → surrounding oedema obscures
- Classic → **intracortical lytic lesion** + reactive sclerosis
- CT → **nidus** + excessive surrounding sclerosis
- Appearances vary; CT very obvious
  - Nidus inside cortex
- **MRI** → surrounding oedema → not clear
  - Nidus may be visible
  - Not hallmark of diagnosis
- Suspected but not localised → **bone scan**
  - Intramedullary
  - Subperiosteal

Reading a Bone Scan Ask AI

- Bone scan → separate specialist topic
- Symmetric uptake in all **physes**, on anterior and posterior views → physiological
- Not every hot spot significant
  - Rib spots on both sides → usually **traumatic** or hyper-uptake
- Hot spot with **no counterpart** on other side → significant
  - E.g. hyperactive physis vs non-active opposite physis
- **PET-CT** → very rarely used
- Routine → bone scan
- Bone scan uses
  - Confusing cases
  - **Response to treatment**
    - Fading → correct treatment given
- Size → **2 cm maximum**
- Spine, femur → same principle
  - **Cortical lesion** + surrounding sclerosis

Modality	Role in osteoid osteoma
X-ray	Cortical lesion + active sclerosis
CT	Hallmark of diagnosis; nidus + sclerosis
MRI	Oedema obscures; nidus may show; not hallmark
Bone scan	Double density sign; intramedullary or subperiosteal lesion; response to treatment
PET-CT	Very rarely

Differential Diagnosis Ask AI

- **Osteoblastoma**
- Other sclerosing conditions
  - **Stress fracture**
  - Sclerosing osteomyelitis
- Cortical lesion → confused with **sclerosing bone lesions**

Histopathology Ask AI

- **Benign osteoblastic mesenchyme** → osteoid surrounded by
  - osteoblasts → **osteoblastic rimming**
- Rimming → **benign**; abundant osteoid; very vascular
- **Spindle osteoblasts** + osteoid
- Osteoid surrounded by its osteoblasts → benign lesion
  - Osteoblastoma or osteoid osteoma
- **Osteosarcoma** → osteoid in one place, osteoblasts elsewhere
  - Osteoblasts produce osteoid wildly → osteoid far from mother cells
  - → **lack of osteoblastic rimming**

	Benign (osteoid osteoma, osteoblastoma)	Osteosarcoma
Osteoid–osteoblast relation	Osteoid rimmed by osteoblasts	Osteoid far from mother cells
Rimming	Present	Absent

Treatment Ask AI

- **Radiofrequency ablation** → unequivocal
- Recurrent cases → needle under **CT guidance** → ablation
- RFA → **hallmark of treatment**

Summary Ask AI

- **Uncommon tumor**
- Males more
- Long bones
- Hands, feet → everywhere
- Clinical hallmark → **nocturnal pain** + point tenderness

- X-ray → **sclerotic lesion**
- Treatment → radiofrequency ablation
- Learned → **differential diagnosis of sclerotic lesions**

Osteoblastoma Ask AI

Definition and Epidemiology Ask AI OB

- Second **bone-forming tumor**
- 1% of all primary bone tumors
- Slightly **older age** than osteoid osteoma *[C1 · transcript: "slightly older than osteoblastoma"]*
- **Big sister** → almost same pathology, larger size
- **Unknown etiology** → no prostaglandin mechanism established, unlike osteoid osteoma
- Sites
  - Long tubular bones
  - Appendicular skeleton
  - Vertebral
    - **Vertebral body** and column → ~40% of osteoblastomas
- → more common in spine than hands, feet, long bones
- **Scoliosis** → as in osteoid osteoma

Toxic (Aggressive) Osteoblastoma Ask AI

- **Rare**; lecturer saw one case
- **Toxic manifestations** → fever, weight loss
  - → fear of malignant change
- Low exam priority

Clinical Picture Ask AI OB

- **Painful scoliosis**, pain
- Pain → **not nocturnal**

Imaging Spectrum Ask AI OB

- Difficult to diagnose by imaging
- Various appearances
  - Neat small **cortical lesion** → like little sister
  - Large **corticomedullary** lesion
  - Longitudinal
  - Well-defined or ill-defined
  - **Periosteal reaction**
  - Eccentric
- → **spectrum of diagnosis** on X-ray
- **Lytic** or ossified — even excessive ossification
- X-ray → differential, not diagnosis
- Diagnosis → X-ray + history + **MRI**
- **Biopsy** → 95% of cases

Role of Imaging Modalities Ask AI OB

- **Bone scan** → as in bone-forming tumors
- **CT** → size, location — as in osteoid osteoma
- **MRI** → extent in bone and soft tissue
  - Better → osteoblastoma not always cortical
- Osteoid osteoma → **always cortical** unless proven otherwise
- Osteoblastoma → cortical or non-cortical

X-ray Patterns and Differential Reasoning Ask AI OB

- Neat lesion + sclerosis → misleading; not cortical → shown by **CT**
- **Pattern 1** — diaphyseal lesion
  - Ill-defined-looking; some sclerosis
  - No periosteal reaction; no **cortical thinning**
  - Diaphyseal, central
  - Fairly well-defined + surrounding sclerosis → probably **benign**
  - Acceptable at X-ray level
    - **Infection**
    - Lymphoma → patient young, still acceptable
    - Fibrous dysplasia
  - **ABC** → less likely; usually metaphyseal, more expansion
  - Resembles larger **osteoid osteoma** + sclerosis
- **Pattern 2** — metaphyseal lesion
  - **Cortical** or corticomedullary
  - Fairly well-defined → can be drawn
  - Mild sclerosis
  - Larger than osteoid osteoma → **benign-looking**
  - Osteoid osteoma → too large
  - **Infection** → possible
  - **Giant cell tumor** → impossible; must be epiphyseal
  - **Chondroblastoma** → must be epiphyseal
  - **Simple bone cyst** → no; cortical lesion
  - Large osteoid osteoma → **giant osteoid osteoma** = osteoblastoma
- Strange lesion → systematic description + **narrow differential diagnosis**
- Other appearances → CT
- **Pattern 3** — CT
  - No **nidus** (vs sclerosis + nidus in osteoid osteoma)
  - Large cortical lesion → **expansile destruction**
  - Non-ossifying fibroma → X-ray needed to decide
- Not always cortical
- Osteoblastoma → **confusing**; difficult on X-ray; several patterns
- **Pattern 4** — superior pubic ramus
  - **Ill-defined**, lytic
  - Very faint bone formation
  - Pelvis, adult
  - Pelvic tumors → usually **cartilaginous**

        - Acceptable even without calcification
        - Usually calcified
        - Possible at this age
    - Not osteoid osteoma → **lytic osteoblastoma**
    - **Fibrous tumors** (non-ossifying fibroma) → not at this site or size
    - Malignant or aggressive impression → acceptable
    - Diagnosis → **MRI**, biopsy, history
    - **Metastasis** → correct at X-ray level
        - Depends on known primary
            - E.g. **cancer prostate**, cancer breast
        - No known primary → undecidable
    - Needs **MRI with contrast** + biopsy
- **Pattern 5** — excessive ossification
    - **Corticomedullary** → enters cortex
    - Aggressive → areas lose **blood supply** → black on MRI
    - **Ill-defined** metaphyseal lesion, adult
    - Calcification vs ossification → undeterminable
    - No **cortical breakdown**; no periosteal reaction
    - Not osteoma; not **bone cyst**
    - Osteosarcoma → acceptable; **osteoblastoma** → excellent
- **Always include osteoblastoma** in the differential of unclassifiable lesions
- **Pattern 6** — aggressive-like periosteal reaction + soft-tissue mass
    - → **osteosarcoma-like osteoblastoma**
    - Mimics osteosarcoma → especially **skeletally immature**
    - Metaphyseal; lysis + solid **new bone formation**, almost continuous top to bottom
    - No cortical breakdown; aggressive-looking
    - **Osteoid osteoma** → impossible
- Reasonable differential → accepted
- Same principle throughout

Exam Pearl

- Unclassifiable lesion → always include osteoblastoma
- Giant cell tumor → must be epiphyseal
- Chondroblastoma → must be epiphyseal
- Pelvic tumors → usually cartilaginous

Histopathology Ask AI

OB

- Like osteoid osteoma
    - Greater **osteoid production**
    - More vascular
- **Epithelioid** subtype → must know
    - Epithelioid osteoblasts → epithelium-like, **large nuclei**
- Pink osteoid between cells → probably **benign**
- Epithelioid type → larger cells, still **osteoblasts**
    - **Giant cells** possible → later grouped with giant-cell-containing tumors

Differential Diagnosis Ask AI OB

- Little sister → **osteoid osteoma**
- Other cortical lesions
- **Osteosarcoma** → very important; same origin and lineage
- Osteoblastoma → abundant osteoid produced around osteoblasts
- Osteosarcoma → osteoid and osteoblasts apart → **no osteoblastic rimming**
- Confusion greatest → **aggressive types**
- Overlap entities → **osteoblastoma-like osteosarcoma** vs osteosarcoma-like osteoblastoma
  - Osteosarcoma resembling osteoblastoma
  - Osteoblastoma resembling osteosarcoma
- Extensive literature; very difficult **differential diagnosis**
- Cases treated as osteosarcoma → lecturer twice

Treatment Ask AI OB

- Same lineage as osteosarcoma; both produce osteoid
- **Curettage** or resection → aim at complete clearance

Summary Ask AI OB

- **Always keep osteoblastoma in mind** → especially thirties
- **Rare** → seldom seen
- Diagnosis → difficult
- Lesion fitting nothing → think osteoblastoma
- Sites → long bones, flat bones, **spine**
- Age → **thirties** *[Q1 · As spoken; osteoblastoma typically presents in second-third decades, mostly under 30, not thirties.]*
- Causes **scoliosis**
- Forms → central, eccentric, ossified, **cortical**, medullary
- Treatment → **curettage** or resection

Osteoid Osteoma versus Osteoblastoma Ask AI

- Benign osseous tumors → two near-identical
  - **Osteoid osteoma** → known mechanism (nerve endings + vascular tissue); small
  - **Osteoblastoma** → giant osteoid osteoma; larger
- Both → all bones + spine; osteoblastoma **more in spine**
- Both → **painful scoliosis**
- Both → cortical or medullary
  - Osteoid osteoma → **mainly cortical** *[C2 · transcript: "osteoblastoma mainly cortical"]*
  - Osteoblastoma → not necessarily cortical; **variable sclerosis**
- Osteoblastoma → very difficult; **always in differential**
- Osteoid osteoma → confused with **sclerosing lesions**
  - Sclerosing osteomyelitis; **stress fracture**
- Osteoblastoma → same + infection + **osteosarcoma** — especially aggressive osteoblastoma

Feature	Osteoid osteoma	Osteoblastoma
Size	Small; up to 2 cm	Larger — "giant osteoid osteoma"

Feature	Osteoid osteoma	Osteoblastoma
Frequency	Uncommon	1% of primary bone tumors; rare
Age	10–35 years	Slightly older; thirties
Sites	Long bones 65% (femur, tibia); hands, feet; vertebrae rare (lumbar)	Spine ~40%; long and flat bones
Position	Mainly cortical	Cortical or not; X-ray spectrum
Etiology	Prostaglandins + nerve endings	Unknown
Pain	Nocturnal; point tenderness; NSAID-responsive	Pain; not nocturnal
Scoliosis	Painful	Painful
Key differential	Osteoblastoma, stress fracture, sclerosing osteomyelitis	Same + infection + osteosarcoma
Malignant change	Never	Never — but mimics osteosarcoma
Treatment	Radiofrequency ablation	Curettage or resection

Benign Cartilaginous Tumors — General Features Ask AI

- Today → most cartilage tumors; two simple ones → next lecture
- Distinguishing features
  - Origin → metaplastic or unclear
    - **Metaplasia** → cell changes nature into another cell
    - Others → origin and mechanism unknown; signals under study; know names only
  - Location → **intramedullary** or juxtacortical
  - Some → **malignant transformation**
    - Contrast: **osteoid osteoma** → never malignant
    - Osteoblastoma → never malignant; mimics osteosarcoma
  - Cause → **IDH gene** mutations — IDH1, IDH2
- Mixed group with **known causes**
- All, most or some → malignant potential; intramedullary or juxtacortical

Must-Know

- Cartilage tumors → intramedullary or juxtacortical
- Some → malignant transformation
- Osseous tumors → never malignant
- Cause → IDH1, IDH2 mutations

Enchondroma Ask AI

Definition and Pathogenesis Ask AI

Enchondr.

- **Second most prevalent** after osteochondroma
- Most common tumor in **phalanges** and hand
- Predilection → hand
- Etiology → **gene mutations** → inhibit osteoblastic differentiation
- Normal → **endochondral ossification**: cartilage cells produce bone cells
- Enchondroma → cells deposited within **metaphysis** → displaced into diaphysis
- Natural maturation blocked → mainly gene mutation
  - Related to **oncometabolite**
  - Research aim → medical treatment for uncontrolled lesions
    - Not yet successful
- Cells → trapped instead of maturing

Presentation and Sites Ask AI

Enchondr.

- Usually **asymptomatic** → discovered accidentally
  - Fall, trauma, elderly knee-pain X-ray
- Sites → **short tubular bones of hand** → feet (less common) → long bones
- Refers to **solitary enchondroma**

The Alarming Sign Ask AI

Enchondr.

- **Pain in a previously painless region** → ominous
  - X-ray 3 months ago fine
  - Now severe pain → very dangerous
- → **red flag** in cartilage tumors and tumors generally
- Egyptian patients → often misleading
- Ask → **mechanical** pain? wakes from sleep? related to walking?
- Enchondroma starts to **hurt** or enlarge → worry immediately

Imaging Ask AI

Enchondr.

- **Hallmark of diagnosis** → dots
- Long bone → **typical cartilaginous matrix** (dots), very clear
- **Short tubular bones** → lytic lesion
  - Metaphyseal; well-defined, ill-defined in parts
  - **Purely lytic**
  - Cortical expansion
  - No periosteal reaction
- Long bones → **calcified matrix**
- MRI → separate short lecture later
- MRI → **alignment of cells** in magnetic field → image
  - Low signal → dark
  - High signal → bright
- Cartilaginous tumors → **low T1**, high T2
  - Most tumors → same

- **Calcifications** → always black on MRI
- Example → **metaphyseal**; fairly well-defined; ill-defined areas; no reaction
  - **Characteristic** appearance

Site	Enchondroma appearance
Short tubular bones	Purely lytic; expansile; no periosteal reaction
Long bones	Calcified matrix — dots
MRI	Low T1, high T2; calcifications black

Histopathology Ask AI

Enchondr.

- Cartilage cells in **lacuna**; little collagen
- Short bones → some **cellularity** in lacunae
- Lacunae → chondroblasts or **chondrocytes** — typical chondrocytes

Differential Diagnosis — Low-Grade Chondrosarcoma Ask AI

Enchondr.

- Innocent → **medullary infarct**
- Very difficult → **low-grade chondrosarcoma**
  - Recent **biomarkers** to differentiate
- Among the most difficult problems to date
  - **Tumor markers** → last 2–3 years
- Differentiation → compare X-rays
- Hallmark → **spots** side by side
- Experience → appearance recognised reliably
- Differences → length; relation to **cortex**
- **Endosteal scalloping** → cortex eaten from inside

Feature	Enchondroma	Low-grade chondrosarcoma
Length	Short segment	Long segment
Cortex	Medullary; no contact	Approaches, rubs cortex
Endosteal scalloping	Absent	Present
Periosteal reaction	None	Faint; visible on MRI
Margins	Fairly well-defined	Ill-defined in places

- ↑ length + cortical approach + scalloping → ↑ concern for
  - **Low-grade chondrosarcoma**
- Pathology → few cells **malignant**
- Formerly → **grade ½** or grade 0 — between enchondroma and chondrosarcoma
- Treatment of both → **curettage**
- Low-grade → may progress to **high-grade chondrosarcoma**
- **Core biopsy** → does not solve problem
  - Low-grade tumors → core may sample inactive cartilage, miss active area
- Solution → **curette it all** → send to pathologists

  - Pathologist → chondrosarcoma areas or active chondroblasts?
- **Main differential in long bones** → low-grade chondrosarcoma

Pitfall
- Core biopsy → may miss low-grade area
- Solution → curette all, send for pathology
- Long, cortex-hugging, scalloping lesion → suspect low-grade chondrosarcoma

Treatment Ask AI

Enchondr.

- **Asymptomatic** → no treatment
- **Pathological fracture** → may heal → follow-up
- Painful, problematic, or low-grade chondrosarcoma suspected
  - → **curettage and graft** or curettage and cement

Summary Ask AI

Enchondr.

- **Most common tumor in the hand**
- Short and long bones
- Children
- Usually **asymptomatic**
- Intramedullary or **juxtacortical**
- Hallmark of diagnosis → **calcification**
- Treatment → curettage
- Long bones → beware **low-grade chondrosarcoma**

Juxtacortical (Periosteal) Chondroma Ask AI

- Enchondroma variant → **periosteal chondroma**
- Appearance → bite taken out of bone
- Periosteal, beside cortex → mimics **surface lesion**
- **Rare**
- Also termed **juxtacortical** → sits beside cortex

Enchondromatosis — Ollier Disease and Maffucci Syndrome Ask AI

Significance of Multiplicity Ask AI

- Cartilage tumors → intramedullary, juxtacortical, or **multiple**
- More multiple → higher **malignant transformation** incidence

Ollier Disease Ask AI

- Multiple enchondromas → **enchondromatosis**; metaphysis and diaphysis
- **No hereditary tendency** → bone dysplasia
- Like enchondroma, several sites
- Sites → femur, tibia, ilium → phalanges, **metacarpals**
- Eponym → **Ollier disease**

Maffucci Syndrome Ask AI

- Ollier + lymphangiomas or hemangiomas → **Maffucci syndrome**
- **Congenital, non-hereditary**
- Enchondromatosis + soft-tissue lesion
- One side of body

Malignant Transformation and Follow-up Ask AI

Ollier

- Malignant transformation → Ollier 30%; **Maffucci** 50%
- Never discharge
  - Clinic → every 3 months
  - **X-rays** → every 6 months at least
  - **Follow-up card**
- Ollier chondrosarcoma → quite **aggressive**
- Maffucci → **mortality**; lecturer saw one case

Imaging and Histology Ask AI

Ollier

- Many enchondromas; larger; more **aggressive in hand**
- Sometimes → **deformities**
- **Phleboliths** (angiomas) → Maffucci
- Histology → **myxoid cartilage**; small cells; abundant myxoid
- Long bones → **longitudinal streaks**
  - Lytic lesions running lengthwise
- **Railway tracks** → characteristic of Ollier
- Metaphyseal lesions → deformities, **shortening**
  - Corrected by **paediatric orthopaedics**

Enchondroma Recap Ask AI

- Cause → not fully known
- **Juxtacortical** or medullary
- Hand; no distinctive clinical picture
- X-ray → long bone **calcification**; short bone lytic
- Long bone differential → **low-grade chondrosarcoma**
- May be **multiple**; may be juxtacortical

Osteochondroma Ask AI

Definition and Epidemiology Ask AI

OC

- **Most common benign tumor**
- Second cartilage tumor
- Older books → separate; **WHO classification** → cartilage tumor
- 45% of benign tumors; 12% of all **bone tumors**
- Before **third decade**
- **Metaphysis**
- Long bones; **spine** rare
- Stops growing at **skeletal maturity**
- Presentation → **swelling** noticed after fall or while bathing child

Clinical Picture and Alarming Signs Ask AI OC

- Usually **asymptomatic**
  - Unless traumatic fracture or pressure manifestations
- Alarm → **pain** without fracture or pressure
  - E.g. lateral humerus lesion, small 2 months ago → enlarging, painful, no nerve compression
    - Bone not fractured → **suspicious**
- **Osteochondroma variants** → lesions resembling it, e.g. under the nail

Imaging Ask AI OC

- **Characteristic lesion**
- **Surface lesion**
- Continuous with **mother bone**
  - Cortex → continuous with cortex
  - Medulla → continuous with medulla
- Points **away from the joint**
  - Proposed causes → rapid growth; **muscle pull**
- Today's lesion groups
  - **Cortical lesions**
  - Surface lesions → osteochondroma, myositis ossificans
  - Differential diagnosis of cortical lesions; **differential of surface lesions**
- Varied appearances; key → **corticomedullary continuity**
- Same → cortex with cortex, medulla with medulla
- Denser → more **aggressive**
- Large surrounding mass → possible **pseudoaneurysm**
  - → careful imaging; sometimes **angiography**

Histopathology Ask AI OC

- Zone of **provisional calcification** of
  - Physis-like structure → spherical, rounded
  - Corresponds to calcification area of its **stalk**
- **Cap of hyaline cartilage** → physis grown outward
- **Gene mutations**
- **No medical treatment**; nothing reduces growth
- Cartilage cap → **hyaline cartilage** related to growth plate

Multiple Hereditary Exostoses Ask AI MHE

- Cartilage tumors → juxtacortical or medullary
  - Osteochondroma → not medullary; **surface**
- Multiple → **multiple hereditary exostosis** (osteochondromas)
- **Bone dysplasia**
  - **Dysplasia** = abnormal configuration of cells
- Aka **diaphyseal aclasis**
- **Autosomal dominant**
- Contrast → Ollier not hereditary; **Maffucci** congenital, not hereditary
- **Hereditary** → father often affected
- Positive family history → 66–70%

- Onset → 2–3 years: **multiple knobs**, swellings → multiple deformities
- Importance → like **Ollier**
- **Malignant transformation**
  - Solitary osteochondroma → <1%
  - Multiple hereditary exostosis → 5–10%
- Follow-up → exam every 3 months; **X-rays** every 6 months at least
- **Flat bones** → higher transformation
- Child with osteochondroma → full **skeletal exposure**
- **Never forget scapula and spine**
- Check → leg **deformities**; other swellings in leg or back
- **Spine** osteochondroma → rare (1–3%) → still examine
- **Pelvis**, scapula → very high chondrosarcoma incidence
- Famous **deformity** → e.g. knee
- Appearance → benign or more aggressive

Feature	Ollier	Maffucci	Multiple hereditary exostoses
Lesions	Multiple enchondromas	Enchondromatosis + he mangiomas/lymphangio mas	Multiple osteochondromas
Heredity	Not hereditary	Congenital, not hereditary	Autosomal dominant; family history 66–70%
Malignant transformation	30%	50%	5–10% (solitary <1%)
Key feature	Railway tracks; longitudinal streaks	Phleboliths; mortality	Diaphyseal aclasis; deformities
Follow-up	Every 3 months; X-ray every 6 months	Same	Same

> **High Yield**
> • Ollier → 30% malignant; Maffucci → 50%
> • Multiple hereditary exostoses → 5–10%; solitary <1%
> • Flat bones → higher transformation
> • Never forget scapula, pelvis, spine

Malignant Transformation — Exostotic Chondrosarcoma Ask AI OC

- Differential → **exostotic chondrosarcoma**
- Suspicion
  - Solitary or multiple exostosis → **starts to enlarge**
  - Growth after **skeletal maturity** (e.g. age 30)
  - Night pain; not mechanical; no fracture or pressure
- Key → **cartilage cap** thickness
  - Normal → ~0.5 cm; 1 cm already a lot
  - 2 cm → **impossible** in benign lesion

  - Example → 4 cm
  - >0.5–1 cm → **active** or aggressive, or low-grade chondrosarcoma
- Microscopy → same cartilage appearance as
  - the ordinary type

Osteochondroma Variants Ask AI

- **Osteochondroma variants** → paediatric entities
  - **Trevor–Fairbank disease** (dysplasia epiphysealis hemimelica) → intra-articular osteochondroma
  - **BPOP** (bizarre parosteal osteochondromatous proliferation) → arises from short tubular bones
    - Hand or foot
  - Under the nail → **subungual exostosis**
- Resemble osteochondroma in **pathology** and appearance; not osteochondroma
- Importance
  - Trevor–Fairbank → **deformities**, especially distal femur, proximal tibia
    - Treatment troublesome; **excision** very difficult
    - Ongoing debate with paediatric orthopaedics → remove or not
  - BPOP → confuses surgeons; may mimic **malignancy** — especially subungual
    - Severe pain
    - **Biopsy** → more confusing; reports calcified; looks like osteochondroma

Variant	Description	Problem
Trevor–Fairbank (dysplasia epiphysealis hemimelica)	Intra-articular osteochondroma	Deformities — distal femur, proximal tibia; difficult excision
BPOP	Osteochondroma-like; short tubular bones of hand/foot	Mimics malignancy; severe pain; confusing biopsy
Subungual exostosis	Under the nail	Confusing; mimics malignancy

Osteochondroma — Treatment and Surgical Indications Ask AI

Indications for Surgery Ask AI

OC

- Treatment → **follow-up**
- No routine operation → difficult in Egyptian mindset
- **Absolute** → complicated osteochondroma
  - Fractured
  - Pressing on a **nerve**
  - Surrounding **bursitis**
  - **Pseudoaneurysm**
  - Suspicion of **malignancy**
- **Cosmesis** → relative indication
  - Disfigurement (e.g. a girl) → removal acceptable
  - Explain + signed **consent**

- **Site** may force surgery
  - **Axillary**
  - Pelvis extending into **groin**
  - Pressing on vastus medialis obliquus (as recorded) → cannot move
  - Directly anterior or posterior → **cannot sit**
- Hidden, lateral humerus or lateral distal femur, silent, not growing → **never operate**
- Absolute → **complication** or malignant transformation; all else relative
- Malignant transformation → **history**
  - E.g. scapula lesion grown since exam 6 months ago → suspect immediately
- Important sites → pelvis, **scapula**, spine
  - **Flat bones** → higher transformation

Summary Ask AI

OC

- **Commonest primary benign bone tumor**
- Any age
- Usually **around the knee**
- Usually asymptomatic
- **Surface lesion**
- Treatment → **observation**
- Worry → **pain in previously painless lesion** or progressive enlargement

Lecture Summary Ask AI

Benign Osseous Tumors Ask AI

- Covered → review; **benign osseous** tumors; half of benign cartilaginous
- Two similar innocent tumors → **osteoid osteoma**, osteoblastoma
  - Osteoid osteoma → cortical; **nocturnal pain**; point tenderness; surrounding sclerosis
    - Confused with **sclerotic lesions**, osteoblastoma, infection
    - **Differential of sclerotic lesions**
  - Osteoblastoma → **osteoid osteoma**, somewhat larger
    - Mainly **spine**; long and flat bones
    - **Pain**, swelling; rarely toxic manifestations *[C3 · transcript: "really causes toxic manifestations"]*
    - **Spectrum of diagnosis** → eccentric, cortical, medullary, ossified, lytic
      - **Aggressive**, vascular → many patterns

Cartilaginous Tumors — Enchondroma Ask AI

- Cartilage tumors → changed differentiation or **configuration** of cells
  - Cause → **gene mutations**
- **Solitary** or multiple; medullary or juxtacortical
- Short tubular bones → **lytic**; long bones → calcification
- Calcified lesion → think **cartilaginous lesion**
- Multiple → **enchondromatosis** (Ollier disease)
- Long bone differential
  - Innocent → **bone infarct**
  - Aggressive → **low-grade chondrosarcoma**

- Long lesion + **endosteal erosion** → suspicious
- Any lesion
  - Painful when painless, or enlarging → **alarming signs** and symptoms
- **Enchondromatosis** → malignant potential → regular follow-up

Osteochondroma Ask AI

- **Surface lesion** → very easy
- Differential of surface lesions
  - **Parosteal osteosarcoma**
  - Myositis ossificans
- Innocent → unless enlarging or **painful**
- Feared → **chondrosarcoma** → judged by cartilage cap
- Multiple → **osteochondromatosis** → deformities; malignant potential

Questions and Answers Ask AI

Ablation in Hand, Foot and Spine Ask AI

OO

- Q: RFA role in **osteoid osteoma** of hand and foot?
  - Yes — also **spine**
  - Not all spinal cases → **next to root** → serious problem
  - Hand, foot → also risky
  - → radiologist performs **marking** with thin
    - **K-wire** → surgeon ablates if radiologist cannot
- Wrist ablation case → **avascular necrosis of lunate** + arthritis
- Caution sites → hand, foot, spine, **tibia**
  - Tibia → subcutaneous → **skin necrosis**
- Always **refer to the radiologist** → at least marking
  - **CT-guided thin K-wire** into lesion
  - Surgeon (e.g. talus, hand) → ablates or burrs down

En Bloc Excision and Recurrence Ask AI

OO

- Q: **en bloc excision** of osteoid osteoma?
  - Not done; lecturer twice, last time **inaccurate**
  - Only in special cases
  - Radiologists publish more on this
  - **Recurrent osteoid osteoma** → ablation
  - Surgery → only after **2 recurrences** at least
- **Biopsy**, MRI → light explanatory slides later

Osteochondroma Recurrence Ask AI

OC

- Q: avoiding recurrence; does recurrence raise malignancy suspicion?
  - Structure → **cartilage cap** + base continuous with mother bone
  - **Cartilage cap** → remove completely
  - Base → curette down to **normal cancellous bone**
    - No cartilage cell buried in healthy **metaphysis**
  - Base → curetted **concave**

  - **Recurrence** → surgical skill, not tumor
  - Recurrence ≠ **malignant transformation**
  - Cause → **incomplete excision**: cap or cap–mother bone junction left

Frozen Section — Enchondroma versus Low-Grade Chondrosarcoma Ask AI

- Q: frozen section to differentiate?
  - No → **open biopsy** does not tell
  - **Frozen section** → no answer
  - Daily problem: observed cases → low-grade; operated cases → enchondroma
  - Insufficient cells → **low-grade chondrosarcoma**: few malignant among 10,000; none among a million
  - → **pathologist** must cut many sections

Skin Necrosis After Ablation Ask AI

- Q: risk of **skin necrosis** with RFA in tibia?
  - Very low → ~2–3%; tibia = **subcutaneous bone**
  - Hand, foot, spine → **refer to radiologist** for opinion
    - Risk of **arthritis** or damage → marking → surgeon removes or burrs down

Calcification versus Ossification Ask AI

- Previous-lecture question → **calcification** vs ossification
  - Detailed in later lectures
  - **Calcification** → central, medullary → endosteal scalloping
  - **Ossification** → usually grows outwards

	Calcification	Ossification
Location	Central, medullary	Grows outwards
Cortex	Endosteal scalloping — eats from inside	—

Femoral Neck Osteoid Osteoma Ask AI

- Q: ablation in **femoral neck** — femoral artery risk?
  - None → radiologist approaches **away from blood supply**
  - Posterior lesion
    - Approach → **from opposite side**
    - Posterior entry → blood supply unaffected
- **Radiofrequency ablation** → hallmark of osteoid osteoma treatment
  - **Kasr Al-Ainy** radiologists → excellent; report problems
- Femoral neck → among most difficult
  - 0.5–1 cm lesion inside **cancellous mass**

Osteoid Osteoma with Scoliosis Ask AI

- Q: does **scoliosis** improve after RFA?
  - Improves → takes **long time**
  - Next to **root**, pedicular, or not feasible → marking + spine surgeons
  - Lecturer's two cases

        - One → improved in ~3–4 weeks
        - Other → took longer
    - **Spine surgeons** see more

Hand Enchondroma with Fracture Ask AI

Enchondr.

- Q: fractured hand **enchondroma** → curette or wait?
    - Wait → cases **united** and stopped complaining
    - **Displaced** or malunited → treat as fracture (hand surgeons or self)
    - United + not complaining → nothing
    - Hand → **conservative** management
- Ablation → better understood by **radiologists**

Adjuvants Ask AI

- Q: use of an **adjuvant**?
    - Curettage slides → later in course
    - **Kasr Al-Ainy** → not used
    - Adjuvant → **high effectiveness**
    - Used by **Ain Shams**; likely Al-Azhar
    - Not available everywhere; **proper adjuvant**

Cartilage Cap Size Ask AI

OC

- Q: cap size threshold for removal?
    - **Cartilage cap** >0.5–1 cm → do not wait
    - Remove **whole osteochondroma** → treat as chondrosarcoma
- Transformation (osteochondroma, enchondroma) → usually **low-grade chondrosarcoma**
- Cap 0.5–1 cm or more (2, 3, 4 cm) → **remove immediately**; mandatory

Scattered Medullary Enchondroma Ask AI

- Q: **enchondroma** as scattered medullary lesion in long bone?
    - Yes → possible
- Remaining questions → repeated

Dysplasia versus Hamartoma Ask AI

- Q: dysplasia vs **multiple hereditary exostoses**?
    - Multiple hereditary exostoses → **is a dysplasia**
    - Tissue → cells arranged in particular way
    - Change in shape, arrangement, appearance, location
        - → **hamartoma** *[Q2 · As spoken; hamartoma is disorganised native tissue at normal site; altered location is choristoma, not dysplasia.]*
    - Change in arrangement or location only → **dysplasia**

Prophylactic Removal at Maturity Ask AI

OC

- Q: remove asymptomatic osteochondroma at **skeletal maturity**?
    - No
    - **Solitary osteochondroma** → transformation <1% → not needed

When to Request a Bone Scan Ask AI

- Q: **bone scan** in benign bone tumors?
  - Possibly **multiple** lesion → enchondroma, osteochondroma
  - Formerly routine → now **not routine**
  - Possibly multiple (**fibrous dysplasia**, enchondroma, osteochondroma) → bone scan or skeletal survey
    - Bone scan → **superior**; more sensitive
  - Diagnostic difference → **osteoid osteoma**
  - Treatment → **response** assessment
    - Persistent pain after ablation
      - **Faded out** → expected to heal
- Session scope → benign tumors only

Non-Ossifying Fibroma and Osteoblastoma Ask AI OB

- Q: **non-ossifying fibroma** in osteoblastoma differential?
  - No → NOF = well-defined, eccentric, **geographic**, sclerotic
  - Not in differential
  - **Osteoblastoma** → eccentric, ossified, lytic or cortical; X-ray spectrum
  - Some books include NOF → lecturer **never**

Prophylactic Fixation After Curettage Ask AI

- Q: when to fix after **curettage**?
  - Indications
    - **Window** > two-thirds of bone
    - Only 1 cortex intact
    - Lesion length > 5–6 cm
  - **Kasr Al-Ainy** → cement preferred
  - Femur, tibia → fixation with **two or three screws** above and below
    - → easy plate removal later
- One cortex intact, or > two-thirds removed, + **aggressive cavity** → prophylactic fixation

Cement, Grafts and Masquelet Ask AI

- Q: **Masquelet** for bone defect, or always cement?
  - **Bone cement** → also used as adjuvant at Kasr Al-Ainy in curettage
  - **Extended curettage** → curettage aided by other tools → adjuvant + filler
  - **Adjuvant** → painted or placed in lesion (e.g. phenol) → reduces recurrence
  - **Filler** → fills cavity: cement, bone graft, others
  - **Cement** → filler and adjuvant at the same time (Kasr Al-Ainy; much literature)
  - Some sites → cement unsuitable, e.g. **enchondroma in hand**
    - → cement not good there
  - **Masquelet technique** → almost never; long defects only
  - Kasr Al-Ainy → mainly cement; sometimes **bone graft**
  - Al-Azhar, Ain Shams → **bone graft**
  - Each school → own approach

Component	Role	Examples
Adjuvant	Placed in lesion → reduces recurrence	Phenol; cement
Filler	Fills cavity	Cement, bone graft, others

Osteochondroma Excision Depth Ask AI

OC

- Q: remove lesion
  - Remove **cortex** also? No
  - Medulla continuous with medulla → excise to **mother medulla**
  - Curette to normal bone → no cartilage escaping into **metaphysis**
  - May level it down
- Closing

Tumor-like Lesions of Bone

(Bone Cysts and Other Non-neoplastic Mimics)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 84) As spoken; only osteopetrosis is osteoclast failure. Osteopoikilosis and melorheostosis are not osteoclast defects.

Q2 (p. 88) As spoken; elsewhere the lecture places eosinophilic granuloma in a child's diaphyseal lesion, not an epiphyseal one.

Q3 (p. 91) As spoken; a lesion stuck to the bone is parosteal osteosarcoma. Myositis ossificans is classically separated from the cortex.

Tumor-like Lesions of Bone Ask AI

Concept Ask AI

- Continuation of **benign bone tumors**
- Benign-looking or tumor-looking lesion → keep **tumor-like conditions** in mind
  - Lesions simulating bone tumors
  - Wide spectrum of lesions
  - **Non-neoplastic**
  - Simulate tumors clinically, radiologically, or both

Classification Ask AI

- Two groups
  - **Tumor-like lesions of bone** → this lecture
  - Tumor-like lesions of soft tissue
- Commonest tumor-like lesions → **bone cysts**
- **WHO classification** → tumors + tumor-like conditions listed together
  - "Tumors and tumor-like conditions of bone" → osseous, cartilaginous, etc.
- **Dysplasias** and non-true bone tumors → grouped as tumor-like lesions
  - Osteochondroma
  - Fibrous dysplasia
  - Eosinophilic granuloma
- Rarities → **metabolic bone diseases** simulating tumors → tumor-like conditions
  - Osteopoikilosis
  - Osteopetrosis

Bone Cysts: Definition and Types Ask AI

- Cyst → **fluid-filled cavity**
- Bone cysts → **pseudo-cysts**, not true cysts
  - No lining endothelium
- Most important → **unicameral bone cyst** + aneurysmal bone cyst
  - Remaining types → awareness only
- Other cystic bone lesions
  - **Primary bone cysts** → UBC, ABC
  - Simple bone cyst or ABC changes **secondary to bone tumors**
    - **Secondary ABC changes** → ABC changes developing on a bone tumor
  - Other bone cysts (not important)
    - Osteitis fibrosa cystica
    - Hydatid cyst
    - Brodie's abscess
    - Brown tumor
    - Intraosseous ganglion
    - Osteoarthritic (degenerative) cysts
    - Infective cysts
    - Mucous cyst

- UBC + ABC → no lining endothelium → **pseudo-cysts**
  - Reason → etiology

Unicameral (Simple) Bone Cyst Ask AI

Radiological Features Ask AI

UBC

- Uncomplicated cyst
  - **Metaphyseal**, skeletally immature
  - Well-defined; clear surrounding sclerosis
  - Purely lytic
  - Central, not eccentric
  - Cortex not thinned; no reaction
- Complicated cyst
  - Cortical thinning
  - Pathological fracture
  - **Fallen fragment sign** (fallen leaf sign) → characteristic of UBC
    - Broken fragment → falls into fluid-filled cavity → gravity → bottom

High Yield

- **Fallen fragment sign** → characteristic of UBC
- Proximal femur + proximal humerus → 90%
- UBC → no biopsy; X-ray usually enough

Epidemiology Ask AI

UBC

- First described 1876 (attributed to Virchow; name uncertain)
- Males > females
- First decades of life
- Sites
  - **Proximal femur** + proximal humerus → 90%
  - Calcaneus

Etiology Ask AI

UBC

- Several theories
  - **Traumatic theory** → most famous
    - Child → trauma → mechanical rupture of physeal plate
    - No time for reparative process → no endothelial lining
  - Fluid build-up theory
  - Growth anomaly theory
  - **Genetic translocations** → chromosomes 4, 6, 8, 12
  - Venous stasis
- Theories not important in themselves
  - Most important → growth anomaly / traumatic + fluid build-up

Pathogenesis: Fluid Build-up Theory Ask AI

UBC

- Most famous theory
  - Rapid growth curve, young children, metaphyseal region
  - Breakdown → **prostaglandins**, interleukin 1 and 2, proteolytic enzymes
  - ↑ Bone-resorbing activity, especially after trauma
  - Complements traumatic theory
- Rapid process → no **reparative process** → no lining → pseudo-cyst
- Theories only

Histopathology Ask AI

UBC

- **Thin fibrous membrane**
  - Giant cells
  - Prostaglandins

Active versus Inactive Cyst Ask AI

UBC

- Key question → **active** or inactive?
- Worrying features → active cyst

Feature	Active (worrying)	Inactive
Age	Younger child	Older child
Position	Close to physis	Far from physis; diaphyseal
Bone expansion	Present	—
Cortex	Thinned out	Not thinned
Pathological fracture	Present	—

- UBC → eventually heals
- Problem → **proximal femur** → very high-stress area
  - Child may fracture before cyst heals
- Proximal humerus + proximal femur → 90% → must know their management

Complications and Presentation Ask AI

UBC

- **Recurrence rate** → 10–20%
- Femoral problems
  - **Growth arrest**, especially after fracture
    - Femoral neck fracture; displaced femoral neck fracture
  - AVN of head
- Presentation
  - Usually asymptomatic
  - **Traumatic determinism** → healthy child falls → lesion discovered
  - Pathological fracture

Outcome Classification Ask AI

UBC

- **Self-limited** cyst → heals
- Outcome after treatment

Outcome	Definition
Healed fully	Whole cyst → new bone formation
Healed with defect	Defect < 50% of bone diameter
Persistent cyst	> 50% of bone diameter; thin cortical rim
Recurrent cyst	Reappeared at original size

Follow-up Philosophy Ask AI

UBC

- Classic urge (Dr. Walid Ebeid) → hole completely filled with bone → repeated **bone graft** packing → not quite correct
- Judge the **cyst wall** + activity, not the cavity
  - Asymptomatic metaphyseal cyst, sound walls → leave; review every 3 months
  - Contents persist for a period
  - Big cyst → wait for fracture → union → reassess
- Persistent cyst appearance → no concern
- Two things to watch
  - Patient **symptoms**
  - Cyst wall → sclerotic (good) vs thinned out (likely to fracture)
- Hence → outcome classification very important
- Proximal femur management → see treatment

Clinical Pearl

- Follow the **cyst wall** and symptoms, not the cavity
- Sclerotic wall → reassure; thinned wall → fracture risk
- Asymptomatic → review every 3 months

Aneurysmal Bone Cyst Ask AI

Definition and Epidemiology Ask AI

ABC

- Completely different entity from UBC
- Described by **Jaffe and Lichtenstein**
- Etiology → many theories, not traumatic
  - Varicosities
  - Arteriovenous fistula
  - Vascular anomaly
- Any age; no gender predilection
- Any bone → flat bones, spine
- Solid form → **reparative giant cell granuloma**
- Incidence percentages → not needed
- **Blood-filled spaces**

Radiology Ask AI

ABC

- X-ray
  - Metaphyseal
  - Definite expansion
  - Central, but **eccentric expansion** → bulges outward
  - Cortical thinning
  - Reaches epiphysis
  - Some periosteal reaction → periosteum involved
- Eccentric expansile lesion → differential
  - **ABC** → number one
  - Chondromyxoid fibroma → acceptable; age a bit young
  - Osteosarcoma → acceptable at X-ray level
- Second example → expansion, cortical thinning, some periosteal reaction

MRI Ask AI

ABC

- **Fluid–fluid levels** → characteristic of ABC
  - Cavity with fluid + blood → gravity → heavier settles below, lighter rises

Work-up and Biopsy Ask AI

ABC

- Work-up → X-ray, MRI, biopsy
- UBC → **no biopsy**
  - Unilocular cyst → characteristic
- ABC → **biopsy mandatory**
  - Benign aggressive
  - Confused with **telangiectatic osteosarcoma**
- Previously not all ABCs biopsied → now routine biopsy
- Same **staging** as before

Must-Know

- ABC → **biopsy mandatory**; UBC → no biopsy
- ABC confused with telangiectatic osteosarcoma
- MRI fluid–fluid levels → characteristic of ABC

Clinical Scenarios of Bone Cysts Ask AI

Fibrous Dysplasia (Recap) Ask AI

FD

- Covered previous lecture
  - Fibrous dysplasia
  - Albright syndrome; **McCune–Albright syndrome**
- All → tumor-like conditions
- Fibrous dysplasia → gene mutation (GNAS, as heard) → considered tumor-like condition by some

Scenario 1: Accidental Discovery Ask AI

- Applies to UBC, ABC, fibrous dysplasia
- **Commonest** scenario

  - Healthy child → minor knock → X-ray → lesion found
- Imaging
  - X-ray may suffice
  - **MRI** → unilocular or not; confirm fluid
- Management
  - Humerus → observation, **conservative**
  - Femur → never leave
    - Usually asymptomatic; symptomatic femoral lesion → very rare
    - **High-stress area** → many microfractures → intervene
- Accidental → humerus: leave; femur: intervene

Scenario 2: Pain or Swelling Ask AI

- → **MRI**
- + **Serum parathyroid hormone**
  - Hyperparathyroidism → in ABC differential
- Overlapping triad → diagnose together
  - ABC
  - Giant cell tumor
  - **Hyperparathyroidism**
- ABC + systemic disease or weak-looking bone → serum PTH
  - **Brown tumor** → in differential
  - e.g. child, weak-looking bone, cystic lesions
- Generalized bony pains or **osteopenia** → serum PTH + vitamin D
- Neither → treat as ordinary UBC or ABC

Treatment of Bone Cysts Ask AI

Prophylactic Fixation of the Femur Ask AI

- Femur → intervene
- Kasr Al-Ainy method
  - **Proximal femoral locked plate**
  - Screws into lesion → depends on physeal reach; usually not done
  - Nancy (elastic) nails, or intramedullary nail (rigid type uncertain)
  - Cystic fibrous dysplasia or fibrous lesion at fracture risk → **prophylactic fixation**

Indications and Options Ask AI

- Accidental discovery → leave
- Pain or swelling → intervention mandatory

Lesion	Intervention
UBC-like	Injection; simple curettage; simple fixation (per stage, proximal femur vs proximal humerus)
ABC	Extended curettage
Recurrent	En bloc resection

Curettage Types Ask AI

- **Extended curettage** = curettage + burr
- Types

Type	Technique
Simple curettage	Ordinary curette (spoon)
Extended curettage	Curette + high-speed burr → 50,000–60,000 or 120,000 rpm; not ordinary power drill

Injection and Adjuncts Ask AI

ABC

- Most famous injection now → **polidocanol**; unavailable in Egypt
- Studies → cortisone, bone morphogenetic proteins, bone marrow injection

	UBC	ABC
Injection	Mainly cortisone; rarely used now	Polidocanol
Mainstay of treatment	Conservative	Surgery

- Before surgery → **selective embolization**
- **Denosumab** → targeted therapy; proven very beneficial
  - ↓ Osteoclastic resorption
  - ↓ Bone destruction
- Radiotherapy → rare; nobody uses it now
- All → options in ABC treatment; each with pros and cons

Summary: Asymptomatic Patient Ask AI

- Unilocular or **multilocular**?
  - Unilocular → simple bone cyst
    - Proximal humerus → conservative
    - Proximal femur → not left
- UBC → no biopsy; X-ray or MRI; **X-ray** usually enough
- ABC → **biopsy mandatory**
  - Confused with brown tumor → consider if generalized bony pains or osteopenia
  - Telangiectatic osteosarcoma
- ABC treatment → mainly **surgical**
  - Polidocanol → unavailable; similar agent in Egypt
  - Embolization; denosumab

Pathological Fracture Through a Bone Cyst Ask AI

Scenario 3: Pathological Fracture Ask AI

- Third scenario → after asymptomatic, pain/swelling
- Humerus → **sling**; review after three weeks → reassess after union
- Proximal femur → must be **fixed**
- Kasr Al-Ainy → fibula in proximal femur → controversial

- Fibula + graft in fibrous dysplasia → controversial
  - Some still use **cortical strut grafts**
  - Some abandoned it
  - Some still convinced

Grafting: Classic Teaching versus Current Literature Ask AI

- Neither answer wrong

Source	Position
Classic teaching	Any benign bone tumor → extended curettage + graft (fibula or cancellous bone graft)
Current literature	Fibrous dysplasia → mainly no fibula; many cases do very well

- ABC → Alexandria + Ain Shams favour **fibula**; many papers
- Femoral lesion, per some papers
  - Bone graft substitute + **cast**
  - Fibula + cast
  - Cast only
- Kasr Al-Ainy experience → grafts **resorbed** (many cases)
  - → Fixation, with or without grafting
- Acceptable answers → graft, fibula, cancellous graft
  - Essential point → **curettage**

Curettage after Fracture Ask AI

- Proximal femur, pathological fracture → curettage? Not necessarily → controversial
  - Fixation alone → fracture already opens and decompresses cyst
  - **CuroBiopsy** → small window + curette → curettage + biopsy together → usual practice
  - Some surgeons → fixation only
- Schools of thought → curette or not, open or not
  - All correct; all practiced

> **Must-Know**
> • Three clinical scenarios of bone cysts
> • Accidentally discovered
> • Pain and swelling
> • **Pathological fracture**

Biopsy Ask AI

ABC

- Pathological fracture → routine staging + biopsy
  - Aggressive, expansile → ABC-like
  - ABC → **CT-guided core biopsy**

Fixation of the Proximal Femur Ask AI

- Depends on → **maturity** + cyst position

	Immature	Mature
Physis	Limits implant choice	No problem
Screws alone	Room below physis → possible; but insufficient → collapse into varus	Acceptable
Side plate	Very important; much better	Much better
Other options	Smooth K-wires; Wagner-type wires + cerclage (side-plate-like)	—

- Lesion reaching physis
  - Screws → no hold; very weak bone
  - Must cross physis → **smooth** implant only
- Mature, or proximal lesion with room for screws → no problem

Pitfall

- Screws alone in immature proximal femur → **collapse into varus**
- Crossing the physis → smooth implants only

UBC versus ABC: Summary Ask AI

- Bone cysts → most important **tumor-like lesions**

Feature	UBC (simple, unicameral)	ABC
Nature	Unilocular cyst	Aggressive, expansile
Diagnosis	X-ray + history	Biopsy required
Associations	—	Metabolic (hyperparathyroidism); aggressive lesions (giant cell tumor)
Mainstay of treatment	Conservative, except proximal femur	Surgery
Injection	Cortisone (rarely)	Polidocanol

Lesions Usually Not Needing Biopsy Ask AI

- Osteochondroma
- **Enchondroma** of the hand
- UBC
- Everything else → biopsy usually needed or mandatory

Exam Pearl

- No biopsy usually → osteochondroma, enchondroma of hand, **UBC**

ABC Treatment after Biopsy Ask AI

- Options
  - Extended curettage
  - **En bloc resection** → large + recurrent
  - Cryotherapy → then extended curettage
- After curettage
  - Nothing
  - Fibula
  - **Cement** → mature only; dangerous to physis; ABC usually immature

Shared Rules Ask AI

- Any cystic lesion, **proximal femur** → never leave → fracture, AVN
- Proximal humerus → conservative
- Fracture through UBC or ABC → wait for **union**
  - Then manage → ABC treated; UBC heals

Bone Infection: The Great Simulator Ask AI

Epiphyseal Lesion Ask AI

Infection

- **Infection** → mimics any tumor
- X-ray → epiphyseal, skeletally immature, well-defined, lytic
  - Differential → infection vs **chondroblastoma**

Infection versus Ewing Sarcoma Ask AI

- Children → infection confused with **Ewing sarcoma**
- MRI

MRI finding	Diagnosis
Central breakdown + peripheral enhancement (peripheral thickening)	Infection (most likely)
Whole marrow affected, no peripheral enhancement	Ewing sarcoma (usually)

- Distinction → extremely difficult

CT-guided Biopsy Ask AI

Infection

- Routine → **CT-guided biopsy**; non-diagnostic → exploration biopsy
- Child, femoral lesion; X-ray + MRI inconclusive
  - CT-guided biopsy + culture
  - Rule → **culture what you biopsy, biopsy what you culture**
  - Send → culture + pathology
  - Difficult → needs experienced radiologist

Exploration Biopsy Ask AI

Infection

- Indication → CT-guided biopsy unavailable; ill, **febrile** child; infection vs sarcoma undetermined
- Steps
  - Lateral approach (not necessarily medial)
  - **Transmuscular** incision down to bone
  - Bone window; incision 5–6 cm; retractors or self-retainers
  - Pus → infection → drain → culture + pathology
  - Unknown tissue → culture + pathology
  - Splint in hospital → await **pathology**
    - Ewing sarcoma → chemotherapy; biopsy done according to the rules → nothing lost
    - Infection → return 2–3 days → widen incision → **debride**
- Not for obvious infection → fever 40, sinuses discharging pus
- Unclear picture + inconclusive MRI → immediate biopsy or **open biopsy**
- Always keep bone infection in mind

Clinical Pearl

- **Culture what you biopsy**, biopsy what you culture
- Exploration biopsy → pus or tissue → culture + pathology

Rare Bone Diseases Simulating Tumors Ask AI

- Names only; frequent on X-ray
- All → **metabolic diseases**; tumor-like conditions

Osteopoikilosis Ask AI

- **Spotted bone disease**; children

Osteopetrosis Ask AI

- **Failure of osteoclastic resorption** → bone formation intact
- **Erlenmeyer flask deformity**

Trevor Disease Ask AI

- Trevor–Fairbank → **intra-articular osteochondroma** (previous lecture)
- Two problems
  - Osteochondroma → may limit movement → knee, ankle
  - Growth disturbance (uncertain)
- Treatment → shared with **pediatric orthopedics**

Melorheostosis Ask AI

- Metabolic bone disease
- **Dripping candle** (candle wax) appearance

Common Features Ask AI

- All → **failure of osteoclastic resorption** → osteoclast defect *[Q1 · As spoken; only osteopetrosis is osteoclast failure. Osteopoikilosis and melorheostosis are not osteoclast defects.]*
- Bone marrow injection → helps a lot
- Often untreated; often no treatment at all

- Recognize appearance → metabolic diseases simulating bone tumors

Gorham–Stout Disease Ask AI

- **Vanishing bone disease** (disappearing bone disease)
- Early films → sarcoma-like → then bone disappears
- Many treatments; largely experimental (uncertain)
- This appearance → think sarcoma; keep Gorham–Stout in mind

Eosinophilic Granuloma Ask AI

Definition Ask AI EG

- Tumor-like condition
  - Most famous tumor-like condition in **children**
- Group of multiple **systemic diseases** → bone marrow, skin, mucosa
- **Pathogenesis unknown** → viral, bacterial, genetic, immunological?
- Simulates tumors in children
- Aim → many lesions resemble tumors → keep them in mind
- End-of-lecture scheme → when to add tumor-like lesions to **differential diagnosis**
  - Not considered → osteopoikilosis, melorheostosis (rarities)
  - Considered → bone cysts, infection, eosinophilic granuloma (children)
- Sites
  - Children
  - **Flat and short bones**; skull
  - Long bones → femur usually; tibia

Clinical Presentation Ask AI EG

- Presentation
  - Asymptomatic
  - Possibly non-tender (uncertain)
  - Pain or swelling
  - Rare → pathological fracture
  - Rare → **spinal cord compression** → spine
  - Rare → ↑ ESR → confusion with Ewing sarcoma / infection

Imaging Ask AI EG

- Variable appearances
- Example → **diaphyseal**; some reaction
- MRI → very high signal; **severe edema**
- Fluid or not → check **T1**
  - Calling it a cyst → acceptable → enters differential of cysts
- Aggressive appearance possible
- Skull → **punched-out lesion**
  - Differential of punched-out lesions → myeloma + eosinophilic granuloma
- Any appearance → history, examination, **biopsy**

Diaphyseal Lesion in a Child Ask AI EG

- Diaphyseal lesion → age + site exclude other lesions → **way of thinking**
  - Sarcoma → moth-eaten; childhood sarcoma → Ewing sarcoma
  - Resembles it → eosinophilic granuloma
- Reasoning by child, diaphyseal, epiphyseal → **scheme** at end

Pathology Ask AI EG

- Inflammatory cell background, eosinophils, **Langerhans giant cells**
- Langerhans giant cells → cell **conglomeration** → giant cells
  - Well-defined membrane
  - Pale nucleus
  - Also called eosinophilic granuloma cells
- Inflammatory cells; lymphocytes

Course and Treatment Ask AI EG

- Course → depends on child's **genetic background**
  - Rarely evolves → multiple types (e.g. Letterer–Siwe)
  - Usually localized → responds to treatment
- After needle (CT-guided) biopsy
  - Responds very well; biopsy alone → may **heal spontaneously**
  - Bone marrow examination
  - Examine other sites
    - < 1% → systemic, **lethal** (rare)
  - Mainstay → **curettage** + steroid injection
- Child, diaphyseal lesion → infection, Ewing sarcoma, eosinophilic granuloma

Key Points Ask AI EG

- Pain and swelling
- May be well-defined
- Eosinophils, **histiocytes**, lymphocytes
- Differential → osteomyelitis vs Ewing sarcoma
- Diagnosis → **immunohistochemical stains** (not very important)

High Yield

- Child + diaphyseal lesion → infection, Ewing sarcoma, **eosinophilic granuloma**
- Punched-out skull lesion → myeloma vs eosinophilic granuloma
- Systemic, lethal form → < 1%

Paget's Disease of Bone Ask AI

Features Ask AI Paget

- Tumor-like condition of bone
- **Etiology unknown** → viral or bacterial?
- Old males
- Severe pain; deformity

- Loose connective tissue replacement
- Osteoclastic resorption + **new bone formation**
- Responds very well → **bisphosphonates**
- Sarcoma → < 1%
- Bone bigger, thicker than normal; cortex widened (uncertain) → always think **Paget's disease**

Sclerotic Lesion in Old Age Ask AI

Paget

- Old patient; **arthritic hip**; separate lesion
  - One option → fits neither appearance nor site; still acceptable in differential
  - Comparison to follow
- Cartilaginous lesion → excluded
  - Enchondroma → wrong age, wrong site
  - Chondroblastoma → epiphyseal
  - **Chondromyxoid fibroma** → expansile
- **Ill-defined** → possibly not benign
  - Old age, non-benign → metastases, myeloma, lymphoma
- Osteosclerotic → **osteosclerotic metastasis** possible → correct answer
- Old age + osteosclerotic lesion → osteosclerotic metastasis or Paget's disease

Exam Pearl

- Bone bigger + thicker than normal → **Paget's disease**
- Old age + sclerotic lesion → osteosclerotic metastasis vs Paget's

Mucoid Cyst (Intraosseous Ganglion) Ask AI

- **Mucoid cyst** = intraosseous ganglion = mucous cyst
- Theory exists → not needed
- Characteristic appearance
- Recognized on sight
- Usually **epiphyseal**
- Finger → instantly recognizable; **characteristic**

Hyperparathyroidism Ask AI

Features Ask AI

HPT

- **Metabolic disease** → primary or secondary
- In ABC differential; also with giant cell tumor
- Female + giant cell tumor or ABC → **parathormone**, especially with osteopenia
  - Three → one unit
- Usually **middle-aged females**; appears with giant cell tumor
- **Medical treatment** first

Radiology Ask AI

HPT

- **Osteopenic** bone
- Cystic lesions → **brown tumors** (osteitis fibrosa cystica)

- Multiple enchondromatosis → acceptable differential
  - Clue → **quality of bone** → osteopenia; abnormal
- Example
  - **Epiphyseo-metaphyseal**
  - Central
  - More or less well-defined; partly ill-defined
  - Distal ulna
  - Bone expansion + cortical thinning
  - No (or mild) periosteal reaction
  - No soft tissue mass
- Expansile lesion differential → ABC, chondromyxoid fibroma
  - + **Giant cell tumor** → epiphyseal lesion
  - All correct → systematic thinking

Reasoning by Age and Site Ask AI

Lesion	Patient	Add to differential
Epiphyseal	Adult	Hyperparathyroidism; infection; intraosseous ganglion
Epiphyseal	Child	Eosinophilic granuloma *[Q2 · As spoken; elsewhere the lecture places eosinophilic granuloma in a child's diaphyseal lesion, not an epiphyseal one.]*

- Every site → own tumors + tumor-like conditions
- Diagnosis → **MRI + biopsy + staging**; follow the system

> **Clinical Pearl**
> • ABC, giant cell tumor, hyperparathyroidism → **one unit**
> • Female + giant cell tumor or ABC + osteopenia → parathormone

Giant Cell Reparative Granuloma Ask AI

Features Ask AI GCRG

- **Giant cell reparative granuloma** → clustering of giant cells
- Confused with bone cysts, especially ABC
- Causes expansion
- Usually **hands and feet**

Expansile Lesion of the Hand Ask AI GCRG

- Expansile, cortical thinning, **purely lytic**
- Differential (tumors)
  - Enchondroma
  - ABC
  - Chondromyxoid fibroma → very difficult in hand; list last

  - Giant cell tumor → very difficult in hand; list last
- Tumor-like condition → reparative giant cell granuloma = **solid ABC**
  - ABC without the cyst → giant cells instead
  - All differentials above → acceptable at X-ray level
  - Diagnosis → MRI + biopsy; same staging
- Expansile lesion, **hand or foot** → add reparative giant cell granuloma
- Foot → enchondroma, ABC → all fit

Gaucher's Disease Ask AI

Features Ask AI

Gaucher

- Included because in books; lecturer saw it 2–3 times
- **Lysosomal storage disease**
- Autosomal recessive
- Neuropathic + non-neuropathic types
- Adults → **adult Gaucher's** → lytic lesions → confusing
- Similar → **Niemann–Pick**; different metabolite from glucocerebroside of Gaucher's
- Two encounters
  - Child unable to walk → **non-neuropathic** Gaucher's → management question
  - Adult → aggressive lesion

Radiological Reasoning Ask AI

Gaucher

- Ill-defined, **metaphyseal**, skeletally immature → many fit
- Resembles **infection** → correct answer
- Systematic exclusion
  - Osteoid osteoma → wrong age, wrong appearance
  - Cartilaginous → no **calcification**; lytic
  - Fibrous dysplasia → not found here
  - Non-ossifying fibroma → has sclerotic rim
  - Metabolic → fits; bone looks weak
  - Cystic → not really
  - Sarcoma → possible
  - Eosinophilic granuloma, Ewing
    - Possible
- Confusing cases → **MRI + biopsy**
- May involve **whole femur**
- Case → core biopsy → fixation → result lymphoma → final diagnosis adult Gaucher's
- Gaucher's in differential → **adults** + children
  - Children → neuropathic or non-neuropathic by stage; usually non-neuropathic
  - No intervention unless fracture (uncertain) or other need → treated and fixed normally

Simulators Diagnosed by History Ask AI

Charcot (Neuropathic) Joint Ask AI

Charcot

- Causes → **diabetes**; leprosy (uncertain)
- Seen very often
- **Destruction of articular surface**
- Swelling + edema
  - X-ray → mistaken for **osteolytic lesion**
- History → Charcot joint immediately
- May present with pathological fracture
- **Painless** range of motion
- Deep-sensation loss; **syringomyelia** → neurologist
- Key → history + general examination
- Diabetic + swollen joint + instability + painless + destruction on X-ray → **Charcot arthropathy**
  - No biopsy; no MRI
- Treatment → usually arthrodesis (uncertain); very complex

Hemophilic Pseudotumor Ask AI

Hemophilia

- **Factor VIII deficiency**
- Huge swelling; encircling dilated vessels (uncertain)
- History → hemophilia + swollen leg → easy diagnosis
- **Terrifying appearances**
- Sometimes **amputation**; resection impossible
- Hence name → pseudotumor

Avascular Necrosis Ask AI

AVN

- Common around **hip**
- Causes → cortisone; hematologic malignancy
- Every clinic → leukemia/lymphoma patient on **cortisone** → referred by chemotherapy doctor as "lesion" → not a lesion
- Sclerosis + joint destruction → **stage 3–4** AVN
- MRI → **marrow edema** + marrow displacement
- Confuses many

Hydatid Cyst Ask AI

- **Echinococcus granulosus** infection
- Middle age
- Pelvis + femur
- Cystic lesions
- X-ray characteristic; microscopy → **scolices**
- Example → **metaphyseo-epiphyseal** lesion
- Any reasonable differential acceptable; resembles familiar lesions; rarely suspected
- Diagnosis → MRI + **biopsy**
  - Specific MRI appearance → diagnosis before biopsy possible

  - **Serum testing** before biopsy → depends on history

Myositis Ossificans Ask AI

- Very important; causes many problems
- Any age
- Long bones; around hips; **elbows**
- Accompanies **severe trauma**
- **Surface lesion** → first in this lecture
- Surface lesion differential (previous lectures) → osteochondroma, **parosteal osteosarcoma**, myositis ossificans
- **Zonal stratification** (zonal phenomenon)
  - Periphery mineralized; centre not mineralized
  - Microscopy → many **fibroblasts** centrally, few peripherally
  - Characteristic appearance
- Versus parosteal lesion
  - **Termite mound** stuck to the bone → like a ball stuck onto bone from the front → called myositis ossificans *[Q3 · As spoken; a lesion stuck to the bone is parosteal osteosarcoma. Myositis ossificans is classically separated from the cortex.]*

Thinking Scheme for Tumor-like Conditions Ask AI

By Site Ask AI

- Site → think of

Site	Think of
Diaphyseal	Infection; fibrous dysplasia; previous diaphyseal lesions; eosinophilic granuloma (especially child); Gaucher's disease (rare)
Surface	Myositis ossificans
Epiphyseal	Chondroblastoma; giant cell tumor (next lecture); intraosseous ganglion; hyperparathyroidism

By Pattern Ask AI

- Pattern → think of

Pattern	Think of
Sclerotic	Paget's disease → old age; bone bigger, thicker than normal
Lytic	Hemophilia; Gaucher's disease
Joint destruction	Charcot joint
Expansile	Hydatid cyst
Hand	Enchondroma; reparative giant cell granuloma

- **Infection** → great simulator → include in any differential
- **Synovial lesions** → tumor-like lesions of soft tissue → covered with soft tissue tumors
- Large part of tumor-like conditions → diagnosed by **history** + clinical examination

> **High Yield**
> • **Infection** → great simulator → in every differential
> • Diaphyseal, child → eosinophilic granuloma
> • Hand, expansile → reparative giant cell granuloma
> • Joint destruction → Charcot joint

Closing Advice Ask AI

- Topic completed before **giant cell tumor** → last benign tumor topic
- Read; follow the fixed **systematic scheme**

Questions and Answers Ask AI

- **Symptomatic UBC** in humerus?
  - CT → no microfractures + symptomatic → treat accordingly
  - Injection, curoBiopsy (small hole + curette), or simple curettage
  - Still painful after 3 months → intervene
- Injection agent?
  - Lecturer → no injections
  - **Steroid** ≈ bone marrow; steroid easiest; literature → roughly equal
- **Tumoral calcinosis**?
  - Tumor-like condition; more in soft tissue
  - WHO → soft tissue tumor-like conditions
- Secondary bone cyst diagnosis → **pathology**
- ABC → pathology mandatory; two reasons
  - Confusion with **telangiectatic osteosarcoma**
  - Possible secondary ABC
    - Lesions with secondary ABC changes → next lecture (giant cell tumor, rules of curettage); very important
- **Pathological fracture** in ABC → biopsy first or fix first?

  - Biopsy → wait → fix
  - Not same setting → CT-guided core biopsy
  - Non-diagnostic → open biopsy + **frozen section**
- ABC → very **deceptive**
  - Sarcoma-like lesions → turned out ABC
  - Biopsy delaying treatment → open biopsy + frozen section, pathologist present
  - Rare; only if delay would harm the child
- Simple bone cyst, **femoral neck**, adult → when to fix?
  - Never left → always prophylactic fixation
  - Open cyst or not → controversial
    - Lecturer → window + curettage
    - Dr. Walid, senior professors → no opening
    - Ain Shams, Alexandria → open + graft
- Adolescents → wires?
  - Yes → **3 thick wires** + screws short of physis + side plate (locked plate)
  - Wagner-type construct with wires on plate instead of cerclage → possible
- Acute vs **recurrent cyst** on X-ray?
  - **Serial X-rays** → healed, healed with defect, persistent, recurrent
  - Recurrence → follow-up child; cyst healed (e.g. 6 months ago) → returns + expansion + reaches physis
  - Only for previously seen cysts; otherwise impossible → known by **follow-up**
- Asymptomatic **enchondroma** of finger → do nothing
  - Accidentally discovered → nothing
- Case → multiple **bone infarctions**, distal femur + distal tibia; normal labs; no cortisone history
  - Question unclear → no answer
- **Heterotopic ossification** vs myositis ossificans?
  - Heterotopic ossification → more scientific term
  - Reason → myositis ossificans includes myositis ossificans progressiva → completely different entity

	Heterotopic ossification	Myositis ossificans
Term	More scientific; describes pathology	Describes multiple lesions more
Trauma	After trauma (required)	Traumatic or non-traumatic

- Role of **calcitonin injection**?
  - Not heard of; lecturer looked into it
- Gaucher's images → osteosarcoma in differential?
  - Yes → **osteosarcoma**, fibrosarcoma, spindle cell sarcoma
  - Gaucher's included because in books; systematic approach → no errors
  - Aggressive appearance → sarcoma possible
- **Melorheostosis** → tumor-like lesion?
  - Yes; more a metabolic bone disease
  - WHO classification → in children, under tumor-like lesions
- Immature femoral neck, smooth K-wires → cross physis?

- Yes → **three smooth wires**; no problem
- Wagner-type construct + cerclage around shaft
- Smooth wires → head vascularity never affected by 3 smooth wires
- Physeal zone spared → **reparative zone**

Giant Cell Tumor of Bone

(Giant Cell-Containing Lesions: Diagnosis and Management)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 113) “extensor carpi radialis (volar approach)” → “flexor carpi radialis”

Queried - kept as spoken, but looks wrong

Q1 (p. 104) As spoken; fat stays bright on standard fast spin-echo T2, it is dark only with fat suppression.

Q2 (p. 104) As spoken; cartilage lesions are typically very high on T2, so they are not an exception to low T1/high T2.

Q3 (p. 105) As spoken; RANK is activated by RANKL (the denosumab target), not by matrix metalloproteinases.

Q4 (p. 107) As spoken; reported malignant transformation after radiotherapy for GCT is far lower (roughly 5–15%).

Q5 (p. 112) As spoken; contradicts the same lecture: curettage recurrence 25–30% vs en bloc 0–12%, function at expense of recurrence.

Q6 (p. 112) As spoken; standard GCT denosumab is 120 mg every 4 weeks with day 8 and 15 loading doses; not 3-monthly.

Lesions with Secondary ABC Changes Ask AI

- One benign member per tumor family → **secondary ABC** changes
  - Osseous → osteoblastoma
  - Cartilage → **chondroblastoma**; also chondromyxoid fibroma
  - Fibrous → non-ossifying fibroma, fibrous dysplasia
  - Rare: **hemangioendothelioma** — not important
- Malignant: osteosarcoma; rarely chondrosarcoma
  - Osteosarcoma + secondary ABC changes = **telangiectatic osteosarcoma**
- Diagnosis → **biopsy**
  - Every ABC must be biopsied

Giant Cell Lesions Ask AI

- Giant cell tumor ≠ only giant cell–containing tumor — written-exam question
- Other giant cell–containing lesions
  - ABC
  - **Brown tumor**
  - Non-ossifying fibroma
  - Benign fibrous histiocytoma
  - **Giant cell osteosarcoma**
- Mnemonic **"A-B-O"** — not A-B-C
  - A → ABC
  - B → brown tumor
  - B → benign fibrous histiocytoma
  - O → non-ossifying fibroma
  - Plus giant cell osteosarcoma
- All = **giant cell lesions** → differential diagnosis of giant cell tumor
- Most important member → giant cell tumor itself

Exam Pearl

- Giant cell tumor is not the only tumor containing giant cells
- Others: ABC, brown tumor, non-ossifying fibroma, benign fibrous histiocytoma, giant cell osteosarcoma
- Mnemonic: A-B-O

Giant Cell Tumor of Bone: Features and Etiology Ask AI

Distinctive Features Ask AI

GCT

- **Giant cell formation**
- **Osteolytic destruction** of bone
- Many controversial therapeutic options
- Relatively high **local recurrence**: 5–30%
- Occasionally **metastasizes**

- Sometimes turns **malignant** → 2% of cases

Etiology — Three Theories Ask AI

GCT

- Theories
  - **Vascular theory**
  - **Telomerase** enzyme (chain) dysfunction
  - **TIMP failure** — tissue inhibitor of matrix metalloproteinase
- Telomerase mechanism — picture the chromatid
  - End of chromosome sequence → **telomere**
  - Telomere contains enzyme telomerase
  - Function → inhibits fusion with adjacent cell's telomere
  - Inhibition fails → telomere fuses with neighbour
  - Fusion → cell behaves differently from its own sequence → cascade begins
- Telomerase failure → not specific to GCT; involved in many cancers
- **MMPs** (matrix metalloproteinases) — same idea as in osteoarthritis
  - Actions
    - Cell migration
    - Mobilization of growth factors
    - Extracellular matrix degradation
    - Angiogenesis
  - Types 2, 3, 4 … 9; most destructive → **type 9**
- Normal inhibition → fibroblast → **TIMP**
  - Defective TIMP → MMPs act → cascade: growth factors, migration, angiogenesis, matrix degradation
  - Same mechanism → osteoarthritis to some cancers
- GCT arises from **telomerase** reaction or TIMP dysfunction
- Etiology theories somewhat peculiar to GCT → must know

Sites and Clinical Presentation Ask AI

Epidemiology and Sites Ask AI

GCT

- **Female predominance**
- Not only around the knee — other sites
  - Proximal humerus
  - Distal radius
  - Hands and feet
  - Sacrum, sternum
  - Vertebral body
  - Calcaneus

Foot and Ankle Grouping Ask AI

- First tumor peculiar to the **foot** → tumors grouped by site
- Classic pairs by bone

Site	Lesions
Calcaneus	Giant cell tumor, ABC
Talus	Giant cell tumor, chondroblastoma

- Ankle and foot → **giant cell tumor, ABC, chondroblastoma**
  - Osteoblastoma → far behind
- GCT in the foot → usually **calcaneus**
- Sites on slide
  - Proximal humerus
  - Pelvis → rare; mainly **sacrum**
  - Proximal femur, distal femur, proximal tibia
  - Hands and feet

Symptoms and Signs Ask AI

GCT

- **Pain** → most common presenting symptom
- Swelling
- Deformity
- **Pathological fracture** → 10–30%
  - Clinical problem → management below

Radiological Features Ask AI

Describing the Lesion Ask AI

GCT

- Description by the standard scheme
  - **Epiphyseal** to subchondral area, or epiphyseo-metaphyseal
  - **Eccentric**
  - Margin not drawable → not well defined
  - No **sclerotic rim**
- Differential → one chondrosarcoma type, covered later
  - **Clear cell chondrosarcoma**

Epiphyseal Lesion — Differential Ask AI

- Epiphyseal lesion → one of two
  - Skeletally immature → **chondroblastoma**
  - Skeletally mature → **giant cell tumor**
- Tumor-like conditions in the epiphysis
  - **Infection** → theoretically included; does not fit
  - **Intraosseous ganglion** → does not fit
  - Neither fits a lesion this size
- Definite fit → giant cell tumor
- **ABC** → hesitantly; not typically epiphyseal
- Rare epiphyseal chondrosarcoma → clear cell chondrosarcoma

X-ray Features Ask AI

GCT

- Subchondral epiphyseal or epiphyseo-metaphyseal
- **Radiolucent**, eccentric
- Local bone destruction
- **Cortical breakdown**
- Different forms
- Lytic epiphyseal lesion → **giant cell tumor until proved otherwise**

Feature	Chondroblastoma	Giant cell tumor
Skeletal maturity	Immature	Mature
Position	Always subchondral	Subchondral epiphyseal or epiphyseo-metaphyseal
Shape	Rounded, with sclerosis	Large, lytic, no sclerotic rim
Location	Usually posterior, or medial / lateral	Eccentric

- Work-up sequence
  - **Staging**
  - History + local clinical examination
  - **MRI with contrast**
  - Biopsy

Distal Radius Ask AI

GCT

- Different forms in the **distal radius**
  - **Soap bubble** appearance — like soap foam
  - Lytic but **cavitary**
  - Other form → purely lytic; cavitation only slight
- Epiphyseal lesion in distal radius → **giant cell tumor until proved otherwise**

Tibia Ask AI

GCT

- **Tibia** → margin faint in part
  - Reaches subchondral bone
  - Epicentre somewhat eccentric
  - More or less well defined
  - Epiphyseal + adult → GCT until proved otherwise
- Same pattern in the **distal tibia**
  - Distal tibia → GCT, ABC, rarely chondroblastoma
  - Adult → not chondroblastoma
  - Eccentric, lytic, reaching epiphysis → GCT until proved otherwise

Hand Ask AI

GCT

- Hand lesion differential
  - **Enchondroma** → most common tumor in the hand; acceptable answer
  - Expansile → ABC — correct, occurs there
  - Tumor-like → **reparative giant cell granuloma** = solid ABC
- All → differential diagnosis of hand tumors

- Other hand lesions → covered in their own lectures

Fibula Ask AI

GCT

- Possible site → **fibula**
  - Lesion partly not visible; appears epiphyseal
  - Alternative diagnosis acceptable — esp. if MRI matches last lecture's findings
  - Some expansion → possible but rare

Sacrum Ask AI

- Sacral tumors
  - Giant cell tumor
  - ABC
  - **Chordoma**
  - Metastasis
  - Hematological malignancies, myeloma
- Sacral tumor → think GCT, chordoma, metastasis
- GCT-like appearance ≠ treat as GCT → full **work-up** (staging, biopsy) — fixed rules
- Follow the scheme → no errors

Calcaneus Ask AI

GCT

- Ankle lesions
  - Distal tibia or talus → chondroblastoma, GCT, ABC
  - **Calcaneus** → GCT, ABC
- Commonest foot tumors → talus, calcaneus, distal tibia

> **High Yield**
> - Lytic epiphyseal lesion in an adult → giant cell tumor until proved otherwise
> - Distal radius epiphyseal lesion → GCT; soap bubble or cavitary
> - Calcaneus → GCT, ABC; talus → GCT, chondroblastoma
> - Sacrum → GCT, chordoma, metastasis

MRI and CT Ask AI

MRI Features Ask AI

GCT

- **Well-defined**, well-circumscribed intramedullary lesion
- Homogeneous, low signal on **T1**
- T2 → high (possibly intermediate)
- **Gadolinium** → enhancement — tumor cells take it up

Reading T1 vs T2 Ask AI

- Signal reference → fat
  - Darker than fat → **low intensity**
- Sequence identification → TR / TE numbers or fat signal

Clue	T1	T2
TR / TE numbers	< 1000	≥ 2000
Fat	Bright	Dark *[Q1 · As spoken; fat stays bright on standard fast spin-echo T2, it is dark only with fat suppression.]*

- Fat check → quick alternative; no shame looking at the numbers
- Rule → almost all lesions **low T1, high T2**
  - Stated exceptions → fibrous lesions, cartilage lesions *[Q2 · As spoken; cartilage lesions are typically very high on T2, so they are not an exception to low T1/high T2.]*
  - Or intermediate T1, high T2
- Slide example → fat dark → T2
- Always check the sequence before reading signal

Role of MRI and CT Ask AI

- **MRI** → the map for planning
  - Vessels — e.g. posterior
  - Oedema
  - Cortical breakdown
  - Relation to the canal
  - Foramina; state of the canal
  - Lesion itself and its nature
- **CT** → bone detail; used selectively
  - Intra-articular lesions
  - Intralesional **mineralization**
  - Margins
  - Pathological fractures, cracks
  - Not the planning map
  - Very useful → pelvis, sacrum
- Routine → **MRI**, not CT

Pathology of Giant Cell Tumor Ask AI

Gross Ask AI GCT

- **Chocolate brown** or brown brain tissue — gelatinous, like brain
- Yellowish or orange discoloration

Microscopy and Grading Ask AI GCT

- **Multinucleated giant cells** dispersed
  - Within a **stroma**
  - Stroma → **mononuclear cells**
- Giant cell : mononuclear cell ratio → **grading**
  - Low-grade
  - Intermediate-grade
  - High-grade

- **Rule of 30** → 30 cells per high-power field, 30 nuclei
  - → giant cell tumor until proved otherwise
- Higher grade with
  - More giant cells
  - More nuclei
  - Giant cell **conglomeration**
- Giant cells → very large, with giant nuclei

Treatment: Medical and Surgical Ask AI

Medical — Denosumab Ask AI GCT

- **Denosumab** → FDA-approved targeted therapy
  - Acts on **RANK receptors**
  - RANK receptors → acted on by the MMPs (as stated) *[Q3 · As spoken; RANK is activated by RANKL (the denosumab target), not by matrix metalloproteinases.]*
  - Inhibited through denosumab
- RANK receptors → role in **metastatic bone disease** (next lecture)
  - Receptors on cells → enhance osteoclastic resorption
  - Denosumab → inhibits **osteoclastic resorption**

Surgical Options Ask AI GCT

- Surgery → **main treatment**
  - **Wide resection** → whole bone removed → autograft, allograft or endoprosthesis
  - **Joint salvage** → curettage + bone graft; curettage + cement; curettage + cement + fixation

Types of Curettage Ask AI

- **Curettage** = scraping; four types
  - Simple
  - Extended
  - Excisional
  - Curopsy (curettage-biopsy)
- **Simple curettage** → open cavity (rules below) → ordinary curette → scrape
- Plus burr → mechanical or **extended curettage**
- Soft tissue extension outside bone
  - Eccentric lesion + soft tissue extension
  - Scrape inside + remove outside part
  - → **excisional curettage** = excision of soft tissue extension
- Summary

Type	Technique
Simple	Open cavity → ordinary curette → scrape
Extended (mechanical)	Simple + burr
Excisional	Curettage + excision of soft tissue / muscle outside

Type	Technique
Curopsy	Small opening, decompression, blind scraping

Curopsy Ask AI

Curopsy

- **Curopsy** (curo-biopsy) → curettage + biopsy at once
  - Small incision, 2 cm opening — not a proper window
  - Under image or not
  - Decompression of the cyst; blind scraping
  - Indications
    - Simple bone cyst
    - Small intraosseous ganglion
    - Suspected infection
  - Not acceptable → **ABC**, giant cell tumor

> **Pitfall**
> • Curopsy is not acceptable in ABC or giant cell tumor

Adjuvants and Fillers Ask AI

- Plain curettage → must add **adjuvant** + filler
- Adjuvant → helps curettage, **reduces recurrence**
- Curettage and burr alone → insufficient
- **Fillers** available → bone graft, bone graft substitutes, cement itself
- Adjuvants compared

Adjuvant	Advantages	Disadvantages	Use
Cement	Comfortable; mechanical support; easy, reliable; reduces recurrence; thermal killing of residual cells	—	Kasr Al Ainy — cement school
Phenol	—	Possible toxicity to the patient; complications at times	Ain Shams
Liquid nitrogen	Much higher efficacy	Careless application → burns, fractures	—
Argon beam (gas)	Very effective	Not available in Egypt; much debate	Sometimes used

- Fillers at Kasr Al Ainy → **cement** preferred
  - Ordinary **bone graft** → very slow incorporation; masks recurrence
  - Bone graft substitutes → **induce infection** in many cases
  - Substitutes → only in children, reluctantly
- Curettage → **mainstay of treatment** of benign tumors and GCT
  - Whole cavity opened → curettage; small part (decompression) → curopsy

Variants of Giant Cell Tumor Ask AI

- **Metastatic giant cell tumor**
  - Benign tumor that metastasizes
  - Other benign metastasizing tumor → **chondroblastoma**
  - Both epiphyseal; both metastasize
- **Multicentric** → very rare, < 1%
- **Malignant**
  - Primary entity → malignant giant cell tumor
  - After irradiation → **radiation-induced giant cell tumor** — radiotherapy widely used in the past
- **Radiotherapy** → still used by some; not worldwide
  - About 30–40 (percent, as heard) → radiation-induced sarcoma or radiation-induced giant cell osteosarcoma *[Q4 · As spoken; reported malignant transformation after radiotherapy for GCT is far lower (roughly 5–15%).]*

Exam Pearl
- Chondroblastoma and giant cell tumor → both epiphyseal, both benign tumors that metastasize
- Multicentric GCT → less than 1%

Surgical Technique of Curettage Ask AI

Pathological Fracture — Options Ask AI

GCT

- **Pathological fracture** → solved case by case
  - Cement only
  - Bone graft inside
  - Plate and screws
  - Choice depends on the case

Biopsy Before Surgery Ask AI

- **No surgery without biopsy**
- No "curette it and see"
- Patient objections → all rejected
  - "Biopsy spreads the tumor"
  - "One operation instead of two"
  - "No money for a biopsy"
- Biopsy done by the treating team

Must-Know
- No patient is operated on without a biopsy
- Biopsy done by the treating team

Approach Rules Ask AI

Curettage

- Incision → **through the muscle**, never between muscles
- No intermuscular plane
- Minimize **contamination**
- Drain → in line with the incision
- Goal → no contamination of other planes

Cortical Window Ask AI

Curettage

- Example → lesion in the **proximal tibia**
  - Opening cortex without breaking the bone → difficult
  - Window → directly to bone
- Window dimensions
  - Length = cavity length — 8 cm cavity → window ≥ 8 cm
  - Width ≤ two-thirds or half of bone diameter
  - No small holes → **large cortical window** for view of all walls
  - Purpose → no residual tumor cells on the walls
- Rules of curettage = **rules of biopsy**
  - Longitudinal incision
  - Directly to bone
  - Width → two-thirds or half of diameter
- Large curette first → then smaller curettes
  - Large tumor → sterilized **teaspoon** at times

High-Speed Burr Ask AI

Curettage

- Used in all cases
- Not any power tool
  - Proper speed → 120,000, 130,000 or 150,000 rpm
  - 20–30 thousand rpm → not a true burr
- No **Hohmann retractors**
- Copious lavage → washes out burr debris; burr very powerful
- Confirm **entire cavity walls** curetted → all tumor cells and debris removed

Cavity Filling Ask AI

Curettage

- Preferred filler → **cement**
- Bone graft first, then cement over it, if
  - Very close to the **sub-articular surface**
  - Defect in the surface
- Rationale (some authors) → cement heat may damage; possibly high pressure
  - Graft → protects **articular cartilage**
- Pack the lesion after filling; graft may go first
- Cement → **from inside outward**; outer part last
  - Black on the slide → cement

Fixation in Giant Cell Tumor Ask AI

- Pathological fracture → conventional or **locked plates**

Conventional Plates Ask AI

Fixation

- Sequence
  - Fix to the **mother bone** first
  - Reduce the fracture
  - Measure distal screw lengths into the cavity
  - Insert cement
  - Cement still soft → distal screws → grip plate and cement

Locked Plates Ask AI

Fixation

- **Locked plates** → better; easier configuration
  - For pathological fracture or prophylactic fixation
  - Screws "in the air" lock into plate → then pack with cement

Prophylactic Fixation Ask AI

Fixation

- Indications for **prophylactic fixation**
  - Lesion involves more than 3 cortices, or 2½ cortices
  - Anteromedial window (anterior + medial halves) + crack during curettage → three → fix
  - Cortices weak or likely to break
  - Bone begins to move

Intra-articular Fractures Ask AI

Fixation

- **Articular surface** first
  - Reduce + temporary fixation; bone graft under articular surface possible
  - Plate on articular wall → proximal screws
  - Cement → distal screws
- **Steinmann pins** → support articular surface
  - Strut from inside — like reinforced concrete
  - Reinforcement of the fracture
  - Early range of motion

Clinical Cases Ask AI

Case 1 — Fracture with Crack Ask AI

- 53-year-old male; pain, swelling; **fracture** and crack
- Staging → age 53 raised suspicion of something sinister
- Biopsy → giant cell tumor
- Initial conservative treatment (cast, as heard) → weight bearing → fracture healed
  - Fracture present → no immediate curettage
  - Wait for **fracture union** and cavity closure
  - Then opening by the same rules
- Curettage + burr + plate
  - Screws → only three available positions → construct held

- Fixation reason → crack
  - **Crack** → **fix** in most cases
- Large lesion
  - Anterior cortex opened
  - **Posterior cortex** cracked, already broken
  - Age 53 → fixation mandatory
- Cement first
- Result → cement filling well; plate holding above and below; **union**
- Functional score → 30

Case 2 — Recurrence After Cement Without Fixation Ask AI

- 27-year-old female
- Appearance may mimic **sarcoma**
  - Cortical breakdown
  - Metaphyseal extension; limits unclear
  - Young patient → think before treating
- Needle biopsy non-diagnostic → **open biopsy**
  - Biopsy site → lateral window
- Lesion → **lateral condyle**
  - Burr + cement; phenol possible, not routine at Kasr Al Ainy
  - Fracture → reduced
- Cement, no **fixation** → fixation would have been better, safer
- Weight bearing after six weeks
- Problem at three months
- Recurrence or collapse (as heard) → **resection**
- Why not salvage?
  - Rule → good **bone stock** that can bear load
  - → joint salvage; large surrounding defect → resection
  - Cement extended front and back
  - Cement removed → nothing of the condyle remains
  - Large defect → **resection and replacement**
- Message → **large cavity window**
  - Length = lesion length; width = lesion width
  - Wide lesion → up to two-thirds or half of bone diameter
- **High-speed burr** → main factor reducing local recurrence in the lecturer's study
  - With and without pathological fracture

Differential Diagnosis of Giant Cell Tumor Ask AI

- Other **giant cell lesions** (listed at lecture start)
- Tumor-like → **reparative giant cell granuloma** = solid ABC
  - Rare; hands and feet; include in differential

Outcomes and Take-Home Points Ask AI

Oncologic Outcome Ask AI GCT

- Burr → much better **local recurrence** in all studies
- Not every curettage achieves this
- Local recurrence incidence → 15–30%, 15–25%
- Some reports → 5–35%
- Lecturer's study → burr vs no burr compared

Functional Outcome Ask AI GCT

- Joint preservation vs **en bloc resection**
  - Joint preservation → much higher **MSTS scores**
  - En bloc → less local recurrence, at the expense of function
  - Joint preservation → more function, at the expense of local recurrence

Key Conclusions Ask AI GCT

- Treatment → **joint preservation surgery** — best balance of oncologic and functional outcome
- **High-speed burr** → a must; significant reduction in recurrence rate
- Cement without fixation → better function than with **fixation**
  - Avoid fixation when possible
  - Fix if → weak bone; 3 walls involved; 2–3 walls cracked; crack in the other wall
- **Primary GCT** → better function than recurrent GCT
- Bone graft under cement → **not routine**
  - Cement → arthritis: thoroughly researched (paper by Dr. Waleed)
  - Patients lived years with cement → no arthritis complaints
  - Graft indications → defect, suspected or seen; subchondral bone like **cigarette paper**
  - Reason → cement **thermogenic reaction** may affect articular surface
  - Recurrence → easy perforation → intra-articular
- Goal → optimum function + oncologic outcome

Resection vs Curettage Ask AI GCT

- Risk of **local recurrence**

	En bloc resection	Curettage (joint preservation)
Local recurrence	0–12%	Up to 25–30%
Function (MSTS)	Lower	Much higher
Trade-off	Recurrence ↓ at expense of function	Function ↑ at expense of recurrence

- En bloc not preferred
  - Benign tumor
  - Young patient population
  - Substantial complications of **joint replacement**
  - Need for revision surgery

- Joint preservation → no significant increase in local recurrence (as stated) *[Q5 · As spoken; contradicts the same lecture: curettage recurrence 25–30% vs en bloc 0–12%, function at expense of recurrence.]*

Take-Home Points Ask AI

GCT

- **Benign aggressive** tumor — not "locally malignant" (obsolete term)
- Epiphyseal
- Treatment → curettage = **joint preservation surgery**
- Differentiate from other **giant cell lesions** — GCT one member of the family
- Exam phrase → "**giant cell–containing lesion**"
  - Know its members → includes dangerous giant cell osteosarcoma

High Yield

- GCT → benign aggressive, epiphyseal; "locally malignant" obsolete
- Treatment → curettage = joint preservation surgery
- High-speed burr → significant reduction in recurrence
- Recurrence → en bloc 0–12%, curettage up to 25–30%
- Bone graft under cement → not routine

Questions and Answers Ask AI

Cement Removal and Denosumab Dosing Ask AI

- Q: Remove the cement later? → **No**
  - Old view; literature → not necessary; cement harmless
  - Defect, small hole or cigarette-paper bone → **cancellous bone** under cement
  - No later removal and grafting
- Q: **Denosumab dosing** — like that in osteoporosis (as heard), but
  - Higher dose
  - Monthly first → then every three months; 6–12 months, sometimes 12–24 months
  - Detail: two injections first → every two weeks until end of month 2 → from month 3 every three months for one or two years *[Q6 · As spoken; standard GCT denosumab is 120 mg every 4 weeks with day 8 and 15 loading doses; not 3-monthly.]*
  - Long course → recurrence frequent after stopping (as heard)

Retractors and Approaches Ask AI

- Q: Why no **Hohmann retractors**?
  - Avoid spreading tumor cells to two compartments
  - Example → lesion in **lateral femoral condyle**
    - Open over lateral supracondylar ridge; through vastus lateralis
    - Self-retaining retractors → stay on lateral cortex = **weak cortex**
  - Hohmann anteriorly + washing → cells scatter anteriorly
  - Hohmann posteriorly → cells scatter posteriorly
  - Goal → **one track** → removed with specimen if recurrence or resection
  - Scattered cells → track unremovable → **soft tissue recurrence**
  - Hohmanns → never
- Q: **Distal radius** or spine?

  - Spine → its own rules
  - Distal radius → usually dorsal
  - Weak cortex volar → incisional **volar approach**
    - No direct descent as with the lateral window
    - Separate **flexor carpi radialis** and vessels — as for a radius plate *[C1 · transcript: "extensor carpi radialis (volar approach)"]*
    - Onto pronator quadratus → incise → perforate cortex
    - Self-retainers, no Hohmanns
  - **Dorsal approach** → easy; wall in front; self-retainers

Pitfall
- Hohmann retractors → tumor cells scatter front and back → soft tissue recurrence
- Use self-retainers → one removable curettage track

CT Chest, Steinmann Pins, Multicentricity Ask AI

- Q: Routine **CT chest**? → no, never routine
  - Recurrent cases → CT chest mandatory
  - Reason → benign tumor that metastasizes
- Q: **Steinmann pin** technique
  - Before cement → 2 Steinmann pins, criss-cross
  - Held in bone above; one on lateral cortex, one on medial
  - Adequate length → wedged, supporting articular surface
  - Difficult, tedious at times
  - Not too long → sufficient length only
- Q: **Multicentric** cases? → yes; multicentric GCT exists

Brown Tumor and Metastatic Disease Ask AI

- **Brown tumor** of hyperparathyroidism → in GCT differential
  - Any female, young or old, ABC- or GCT-like lesion → measure **parathormone**
  - Lecturer's cases operated as GCT → turned out brown tumor
  - Patient lost nothing; no cure while **hyperparathyroidism** persists → recurs elsewhere
- Q: **Metastatic GCT**, esp. lungs?
  - Own chemotherapy
  - Interferon → sometimes effective
  - **Denosumab** → excellent in metastatic patients
  - Resistant to denosumab → some drugs of an osteosarcoma-type regimen (as heard); not identical

Clinical Pearl
- Any female with an ABC- or GCT-like lesion → measure parathormone
- Missed brown tumor → recurs elsewhere while hyperparathyroidism persists

Fibular Graft and Liquid Nitrogen Ask AI

- Q: **Vascularized fibular graft** for knee fusion?
  - Resection → autograft, allografts or joint replacement

  - **Fusion** → very rarely relied on; few patients accept
  - Role exists → vascularized fibula = the autograft
  - Advantage → **pedicled**; rotated to distal femur or proximal tibia; no anastomosis
- Q: Concentration of phenol and liquid nitrogen?
  - **Liquid nitrogen** → −196 °C standard; 25–30 minutes
  - Warm saline soaks everywhere → risk of **third-degree burns**
  - Famous for fractures; freezing effect very violent
  - Many precautions; no concentration

Salvage, Fixation Order, Biopsy Ask AI

- Q: When **joint salvage**?
  - Thick, good bone, easily curettable → joint salvage
  - Involvement everywhere (front and back) → nothing left → resection
  - Rule → **joint preservation**
- Q: Fix fully before cementing (as heard)?
  - Very difficult; some insert cement with little finger, curettes or a **McDonald dissector**
  - Lecturer's method → reduce fracture, proximal screws
  - Fracture usually **varus** or valgus → manual reduction; assistant maintains
  - Proximal screws → slide plate → check → measure distal screws in empty cavity
  - Verify under image and direct vision → soft cement → screws
  - Fix then cement → also not wrong
  - Aim of reduction → **weight bearing**, not healing (as heard)
- Q: Radiologically benign → one-stage **excisional biopsy**? (e.g. calcaneus, bad wound healing with two operations)
  - No → **biopsy first**, then treat
  - Biopsy costs patient nothing
  - Needle biopsy non-diagnostic → open biopsy
  - Rarely → **frozen section** — rare in bone; acceptable
    - Specimen to pathologist → benign, atypical, or proceed?

Adequacy of Curettage and Cementing Ask AI

- Q: Burr → when is excision adequate?
  - Curette + burr
  - Walls **sanded** — no elevation, no cavitation
  - Bloody walls → saline → completely white
  - → adequate resection
  - **Corners** → recurrence always comes from corners
  - Curette several times; think finished → curette again
  - Inaccurate curettage → recurrence
  - White, bleached, sanded walls → **adequate curettage**
- Q: Cement → gun or by hand?
  - Example → lateral femoral wall, under image
  - Order → **articular surface** → walls → exterior
  - All packs at once preferred → hardening, time → one piece
  - Or one pack on inner walls, second outside
  - Little or big finger by cavity size; **no cement gun**

  - Outer part → very end
- Q: **Antibiotic cement**?
  - Antibiotics → little relation to recurrence
  - **Methylene blue**– and bismuth (as heard)–impregnated cements → said to reduce local recurrence; not really available in Egypt
  - Antibiotic-impregnated cement → no difference

> **Clinical Pearl**
> • Adequate curettage → walls sanded, white after saline
> • Recurrence always comes from the corners
> • Cement → articular surface first, then walls, exterior last

Burr Availability, Amputation, Bone Scan Ask AI

- Q: No burr → **no operation**; postpone
  - Applies to benign aggressive lesions → GCT, chondroblastoma, low-grade chondrosarcoma
  - Facilities differ
  - GCT operated without burr at Nasser Institute → recurrence → wide resection + **modular prosthesis**
  - Fractured femur → recurrence possible anyway; no burr = defect in the work — confirmed in every series
- Q: **Amputation** ever?
  - Yes → some patients
  - Resistant cases → radiation (very rare) and amputation
- Q: Role of **bone scan**?
  - Less relied on; not routine
  - Role sometimes → pathological fractures (next lecture)
  - Not wrong to do; shows multiplicity and activity (as heard)
  - GCT → 99% hot → little help
  - Suspected metastasis, older patient (e.g. the 53-year-old case) → bone scan reasonable
  - Middle-aged female, epiphyseal GCT-like lesion → not needed

Inoperable Cases, Cautery, Phenol, Follow-up Ask AI

- Q: Inoperable → radiotherapy? → no; **denosumab** → yes
  - Large sacral lesion
  - Large vertebral lesion at risk of hemiplegia
  - Intra-abdominal lesion, e.g. next to the aorta
  - Denosumab → lesions shrink, vanish
  - FDA-approved since 2013; excellent results so far
- Q: **Cauterization** of the cavity after curettage? → no role
  - Old technique; abandoned, including by the lecturer's team
- Q: **Phenol** timing?
  - Curette → burr → phenol, e.g. a second or third time
  - Kasr Al Ainy → after; Ain Shams → straight away
  - Same tools; each school its own way
- Q: Observation without intervention?
  - **Follow-up** → very important; recurrence 15–30%; some say 5–30%

- Observation without surgery → no
- Exception → large sacral lesion, denosumab two years → regression — very, very rare

Osteosarcoma

(Conventional Osteosarcoma: Etiology, Diagnosis and Treatment)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 126) “chromosomes 9, 10, 30 and 17” → “chromosomes 9, 10, 13 and 17”

C2 (p. 135) “CT chest without contrast” → “CT chest with contrast”

Queried - kept as spoken, but looks wrong

Q1 (p. 122) As spoken; osteosarcoma is mainly a childhood/adolescent tumour, so 38% adults vs 2.4% children looks garbled.

Q2 (p. 122) As spoken; third peak given as above 30 here but above 40 or 50 at a011.

Q3 (p. 131) As spoken; most series link pathological fracture in osteosarcoma with worse survival.

Q4 (p. 132) As spoken; standard 5-year survival for non-metastatic osteosarcoma is about 60-70%, not 80-90%.

Q5 (p. 135) As spoken; same fall in survival given as 10-20% at a092.

Definition and Aetiology Ask AI

Definition and History Ask AI

- Osteosarcoma → covered over two lectures
  - This lecture: what it is, how it is diagnosed
  - Next lecture: same rules, minor changes → **subtypes**
- Subtypes = all types other than **conventional**
  - Conventional = usual, most common type
- **Conventional osteosarcoma**: high-grade malignant tumour of osteoblastic mesenchyme
  - Cells: malignant mesenchymal cells producing osteoblasts
- First description: **Abernethy**, 1804
- Abernethy → sarcoma in general
- **Boyer** (French; served in the French campaign) → described origin from bone
- Historical notes only

Risk Factors Ask AI

- Several theories on aetiology
  - **Rapid bone growth** period
  - Factors raising incidence
- **Radiation** → e.g. prior radiotherapy for Ewing sarcoma → osteosarcoma
- **Chemical agents** → fluoride
  - **Fluorinated areas** (some US states, e.g. around Seattle) → ↑ incidence
- **Genetic predisposition** → main current discussion
  - Oncologists' domain; surgeons need an idea of it
- Important associations
  - Gene mutation conditions → more than half of those affected develop it
  - **Paget disease** → osteosarcoma rare, but occurs in some patients
  - **Bone dysplasias**, multiple bone lesions
- Proportion of multiple bone lesions → turn **malignant**
- Syndromes ↑ incidence
  - **Li-Fraumeni syndrome**
  - **Rothmund-Thomson syndrome**
- **Chromosomal abnormalities** → very large, detailed subject
  - Duplication, deletion, translocation
- Risk factors listed for awareness only
- Oncologists → **gene therapy**, treatment of osteosarcoma
- Aetiology unknown → keep in mind
  - **Genetic predisposition**
  - **Rapid bone growth**
  - **Irradiation** exposure
- Four main risk factors
  - **Environmental exposures**
  - **Somatic alterations** (genetic predisposition)
  - **Cancer predisposition syndromes**

  - **Bone growth**

Epidemiology and Site Ask AI

Incidence and Age Ask AI

- **Third most common** bone malignancy → after myeloma, metastases
- Incidence: 38% in adults → 2.4% in children *[Q1 · As spoken; osteosarcoma is mainly a childhood/adolescent tumour, so 38% adults vs 2.4% children looks garbled.]*
- **Three age peaks**
  - First: second decade of life
  - Second: late adulthood
  - Third: above 30, or old age above 50 *[Q2 · As spoken; third peak given as above 30 here but above 40 or 50 at a011.]*
- More common in **African-Americans**, males
- Main peak → **adolescence**
- Main peak → between adolescence and second decade
- Third peak → later, above 40 or above 50

Site Ask AI

- **Long bones**
- **Metaphysis** → conventional osteosarcoma mainly metaphyseal
- Also possible
  - **Diaphyseal**
  - **Epiphyseal** → rare
- Bones: **femur, tibia, humerus**
  - Rare, atypical: **hand**, skull
- Subtypes lecture → which subtypes diaphyseal vs epiphyseal
- Slide: sites with approximate incidence, percentage

Clinical Presentation and Work-up Ask AI

Presentation Ask AI

- **Pain** + swelling (preceding mass)
- Possible **preceding lesion**
  - Multiple bone lesions
  - Previous radiation exposure
- **Pathological fracture** → rare
  - Pain → early medical advice → fracture uncommon at presentation
- Pathological fracture → about 10% of cases
- Swelling examples → **distal femur**
- Swelling also in **forearm**

Laboratory Investigations Ask AI

- **No tumour marker** for sarcoma type
- **Routine labs** → patients will receive chemotherapy

- Relevant labs
  - **Alkaline phosphatase**
    - Tumour from osteoblasts → osteoblasts produce ALP
    - Lower ALP → lower activity → better prognosis
  - **LDH** → prognosis, response to chemotherapy
- ALP + LDH → **prognostic, not diagnostic**
- **No diagnostic labs** for sarcoma

> **Clinical Pearl**
> • No tumour marker for bone sarcoma type
> • ALP, LDH → prognostic, not diagnostic
> • LDH → response to chemotherapy

Clinical Approach Ask AI

- Same rules as before → **history**
  - Swelling
  - Previous cancer
  - Family history
  - **Weight loss**
- **Systemic manifestations** → rare, but possible in osteosarcoma
- Sequence: history → general exam → local exam → labs → **radiology**

Plain Radiography Ask AI

Description Ask AI

- First question from the X-ray lecture
- **Blastic or lytic**?
- Mainly metaphyseal; may be diaphyseal, epiphyseal
- **Blastic** mainly; may be lytic
  - Some authors: lytic more frequent than blastic
- **Codman's triangle**
- **Epiphysis not a barrier** → invades epiphysis
- Describe as in the X-ray lecture

Descriptor	Finding
Site	Metaphyseo-epiphyseal
Margin	Ill-defined — no outline above or below
Position	Central, medullary
Matrix	Mainly blastic → bone formation
Periosteum	Periosteal reaction

Blastic versus Sclerotic Ask AI

- **Blastic** → lesion itself forms new bone
- **Sclerotic** → mother bone's behaviour, surrounding lesion with sclerosis
- No bone formation → **lytic**
- Lesion inside the mother bone
- Famous question: **benign lesion** never looks like this
  - Must show a gap
  - Gap may be small, but must exist

Radiological Appearances Ask AI

- Many appearances, all aggressive
- Same descriptive rules apply
- Example: **metaphyseal**, no defined start or end
- Patient `~15–20 years` → another appearance
- Reaction above → in such a patient → **sarcoma**
- Way of thinking: benign look-alike? No
  - **Osteoblastoma** never looks like this
- Other lesions → never look like this → **malignant**
- Some examples → subtypes, covered later
- Classic lesion: **lytic, ill-defined, adolescent** → osteosarcoma number one
  - Even without new bone formation, even if not blastic

High Yield

- Ill-defined, central, medullary, mainly blastic, metaphyseal
- Codman's triangle; periosteal reaction
- Epiphysis not a barrier → invaded
- Blastic = lesion forms bone; sclerotic = mother bone reaction
- Lytic, ill-defined lesion in adolescent → osteosarcoma first

Staging Investigations Ask AI

Bone Scan Ask AI

- Lesion looks **benign** → MRI with contrast ± bone scan (not routine)
- Lesion looks **malignant** → bone scan mandatory
- **Two types** of bone scan
  - **Technetium** → shows the lesion exactly
  - **Thallium**
    - Rarely used → replaced by **PET CT**
    - Know it exists for rare use

MRI Ask AI

- MRI = **the map**
  - **Soft tissue extent**
  - Relation to vessels
  - Level of the cut

  - Muscle to remove
  - **Skip lesions**
- Example
  - **Skeletally immature**, lesion reaching epiphysis
    - → remove 1 cm of epiphysis, or **trans-epiphyseal** resection
  - Measured on MRI → coronal view
- Axial → map: lesion + **soft tissue mass**
  - Anterior, posterior orientation; **patella**
  - **Vastus medialis**
  - **Vastus lateralis**
  - Far posterior → most likely the vessels
  - Masses taken vs masses left behind
- Sagittal → better
  - **Intra-articular extent**
  - Relation to patellar tendon
  - Posterior extent
- Example: lesion inside bone + large **soft tissue extent** outside (coronal, sagittal)
- Each view shows something

View	Shows
Coronal	Longitudinal extent; resection length, cut level
Sagittal	Extent, especially intra-articular
Axial	Relation to vessels; muscle cut

Other Investigations Ask AI

- **Angiography** → rarely used; replaced by MRI → MR angiography
- **CT** → rarely used
- **PET CT** → response to chemotherapy; suspected local recurrence
- **Audiometry**, echocardiography → because of chemotherapy
  - **Cisplatin**, Adriamycin → hearing, heart (cardiomyopathy)
  - Tests for **treatment complications**; not routine; not requested by surgeons

Biopsy, Pathology and Staging Ask AI

Summary of Approach Ask AI

- **Adolescent** + painful swelling → history → staging
  - Staging = X-ray + bone scan + MRI

Biopsy Ask AI

- **CT-guided core biopsy**
  - Fails → **open biopsy**

Gross Pathology Ask AI

- **Solid hard mass**
- Cut section → fleshy, as in any malignant tumour
- Cut section → **hard white mass**
  - After formalin → yellowish
  - Large size

Microscopy Ask AI

- **Spindled or rounded cells** + high vascularity
- Key finding → **malignant osteoid** (pink material)
  - = osteoid without cells around it

Feature	Normal osteoid	Malignant osteoid
Cells	Surround the osteoid	Separate, unrelated
Osteoblastic rim	Present	Absent — unrimmed

- **Unrimmed by osteoblasts**
- Osteoblasts **abnormal** → many cells, many nuclei
- Osteoblasts in one place, osteoid in another → malignant osteoid
- Sarcomas diagnosed by **cells + stroma**
  - Stroma → osteoid, chondroid or fibrous
- Osteosarcoma → **spindle or round cells**
  - Malignant, with a background → osteoid, chondroid or fibrous
  - Details → **subtypes** lecture
- **Mitotic figures**
- **Pleomorphism**
- **Multinucleated** cells

Enneking Staging Ask AI

- **Enneking** staging (staging lecture)
  - T1, T2, T3
  - Intracompartmental or extracompartmental
  - ± lymph nodes; ± metastases
- Nothing new versus the staging lecture

Cytogenetics Ask AI

- Not surgeons' concern → basics only
  - Journals (e.g. **Cancer**, Sarcoma, Journal of Oncology, Surgical Oncology) → extensive literature
- **Oncologists** → aetiology, treatment
- **Aneuploidy** → chromosome loses normal protein arrangement → abnormal ploidy
- **Non-diploid** chromosomes
- Diploid chromosomes with altered arrangement (adenosine, guanosine)
- Chromosomal **gains and losses** → especially chromosomes 9, 10, 13, 17 *[C1 · transcript: "chromosomes 9, 10, 30 and 17"]*

- **Retinoblastoma gene**
- **MDM2 gene**
  - Diagnostic in some cases of osteosarcoma
- Disturbed arrangement of chromosomal proteins, units
  - → diploid or non-diploid chromosomes → **aneuploidy**
- Oncologists' work; know extensive cytogenetics exists
- **Telomeres** → chromosome ends → prevent fusion between chromosomes
  - Also in aetiology of **giant cell tumour**
- Other genes
  - **Oncogenes**
  - **Tumour suppressor** genes
- Awareness only for surgeons
- Diagram → osteosarcoma cells
  - **Extensive genomic instability**
  - **Gene amplification** → overproduction of certain proteins

Spread and Differential Diagnosis Ask AI

Spread Ask AI

Site	Frequency
Lungs	Main — 20%–80% of cases
Pleura	Possible
Bone	Less common
Lymph nodes	Rare
Viscera	Very, very rare

- Bone sarcomas → **lungs** and bones
- Lecturer: never seen visceral or lymph node metastases

Differential Diagnosis Ask AI

Differential	Reason
ABC — main	Telangiectatic osteosarcoma resembles ABC
Infection	Great similarity; included in everything
Ewing sarcoma	Younger patients; sometimes diaphyseal
Osteoblastoma	As agreed earlier
Osteosclerotic metastases	Rare; sclerotic lesions
Lymphoma, neuroblastoma	Round cell osteosarcoma

- Remember, don't memorise → osteosarcoma = **bone-forming neoplasm**

  - Its own types mimic other lesions → those enter the DDx

Chemotherapy Ask AI

Principles Ask AI

Chemo

- **Chemotherapy → chemotherapy → chemotherapy**
- **Neoadjuvant** → before surgery; introduced by Rosen
- **Adjuvant** → after surgery

Benefits of Neoadjuvant Chemotherapy Ask AI

- 1 → control **micrometastases**
  - Sarcoma = **systemic disease**; micrometastases in circulation
  - Surgery without chemotherapy → recurrence
- 2 → ↓ **tumour size** → easier local control
- 3 → assess chemotherapy potency on resected specimen
  - % dead sarcoma cells = **tumour necrosis index**
    - More than 90% or 95% → chemotherapy successful
    - Less → failed → change drugs, doses, or second-line drugs
- **Adjuvant** chemotherapy → according to neoadjuvant efficacy
- Sequence (as in **Ewing sarcoma**): chemotherapy → surgery → chemotherapy

Must-Know

- Chemotherapy → surgery → chemotherapy
- Neoadjuvant: micrometastases, tumour size, chemo efficacy
- Tumour necrosis index >90–95% → successful
- Below → change drugs, doses, second-line

Recent Trends Ask AI

- Oncologists' domain
  - **Immunotherapy**
  - **Interferon**
  - **Monoclonal antibodies**
  - **Targeted gene therapy** → delivers cytotoxic drug inside tumour cells → ↓ systemic effect
  - **Interleukins**
  - **Gene therapy** → targets aneuploid, diploid, non-diploid chromosomes, telomeres
    - Very large studies; surgeons just aware

Surgical Resection and Reconstruction Ask AI

Resection Ask AI

- Surgeon's concern → type of resection → **wide surgical resection**
- Margins, resection types → dedicated lecture
  - Includes **reconstruction**
- Options: wide resection, radical resection, **amputation** → then reconstruction

Reconstructive Options Ask AI

- **Biological**
  - **Vascularized fibula**
  - Same bone reimplanted → **recycling techniques**
- **Prosthetic** → modular prosthesis
- **Rotationplasty** → counted as biological; modified amputation
- Example: wide resection of distal femur → replaced by fibula, joint, or the bone itself
  - Or amputation, or **modified amputation** (rotationplasty)

Prosthetic Reconstruction Ask AI

- Proximal humerus sarcoma → wide resection → **modular prosthesis**
  - Humerus-shaped; long stem
- **Distal femur** prosthesis
- **Proximal femur** prosthesis
- **Intercalary** prosthesis
  - Not a very good joint; shown for appearance
  - One case very bad; know it exists

Biological: Condyle-Sparing Fibula Ask AI

- Case: lesion above the joint → cut down to **distal femur**
- Wide resection, condyles preserved → **condyle-sparing resection**
- **Fibula** inserted
- Old case → fixed with ordinary **distal femoral plate** (only option then)
- Graft placed in the resected gap
- = **reconstruction**: resect → replace

Recycling Ask AI

- Resect tumour → bone in **container** (e.g. liquid nitrogen) → returned

Technique	Method
Pedicle freezing	Bone left in the leg → leg placed in container
Free freezing	Tumour segment removed completely → placed in container

- **Japanese** → very skilled at it
- Example: recycled bone + **plate** on mother bone
- Appearance of recycled bone; no cement used

Rotationplasty Ask AI

- **Foot**
  - → rotated 180 degrees → used as **knee**
- Example: osteosarcoma of **proximal tibia** or distal femur → resect whole segment
- Foot rotated → attached to **femur** → tibia against femur; fixed
  - X-ray: tibia, fibula below; femur above
- Patient wears a **prosthesis**
- "Knee" = **foot** rotated 180 degrees

Reconstruction Case Examples Ask AI

Free Vascularized Fibula — Humerus Ask AI

- Examples from our cases
- **Free fibula transfer** (not pedicled) → into remaining humerus, young child
  - Fibula hypertrophied → became like the humerus
- **Vascularized fibula** → preserves growth at proximal end
- **Physis** working
- Alternative → fibula against **glenoid**, fixed
- Harvest with **skin paddle** → monitor flap
- Fibula inserted

Proximal Humerus Prosthesis Ask AI

- Resection → **neurovascular bundle** visible; biceps attached
- Prosthesis + **Dacron graft**
  - Sometimes omitted → infection
  - Muscles sutured to it
- X-ray → **proximal humerus prosthesis**

Distal Humerus Ask AI

- **Distal humerus** → rare site
- Neurovascular bundle dissected completely away
  - **Radial nerve**
  - **Median nerve**
  - Vessels
- Mass appears
- Resection → **specimen**
- **Modular elbow prosthesis** → prosthetic reconstruction

Proximal Tibia Ask AI

- **Biopsy tract** → ellipsed
- **Medial approach**
- Vessels dissected first → then beneath **soleus**
- Resection
- Post-operative appearance
- Reconstruction → **modular prosthesis**

Secondaries, Pathological Fracture and Local Recurrence Ask AI

Treatment of Secondaries Ask AI

- Mainly **chemotherapy**
- Fewer than 8 → surgical resection (wedge resection) or **lobectomy**
- **Secondaries treated as the primary**
- Recurrence → **CT chest** → secondaries?
  - Present → treat: surgery or chemotherapy

    - Left untreated
        - → affects **survival**

Pathological Fracture Ask AI

- About 10% of osteosarcoma patients
    - Rare → patient seeks advice before bone breaks
- **Telangiectatic** → more aggressive → 30% fracture
- Predisposition → lysis → lesion progression
- Example → **lytic** appearance
- Atypical location → epiphyseal or diaphyseal; lytic
- **Does not affect survival** *[Q3 · As spoken; most series link pathological fracture in osteosarcoma with worse survival.]*
- Treatment: **splint** → chemotherapy
- **Internal fixation** → very bad outcome
    - No fixation until diagnosis known
- Good sign → fracture **healing** (union, callus)
- Even **malunion** with healing → good sign

> **Pitfall**
> • Pathological fracture → splint → chemotherapy
> • Internal fixation → very bad outcome
> • No fixation before diagnosis
> • Fracture does not affect survival

Local Recurrence Ask AI

- More common → **proximal humerus**, femur
- **Pelvic lesions** → rarer, but recurrence reaches 25%
- Key → **early detection** through follow-up
- On recurrence
    - **Chemotherapy** immediately → usually, not all cases
    - **CT chest** → often with metastasis
- Earlier detection → better
- CT chest + refer to **chemotherapist** → his opinion before surgical management

Prognosis Ask AI

Metastases Ask AI

Prognosis

- Depends on metastasis → **number, time, type**
- Longer **disease-free** → better survival

Factor	Usually respond	Usually do poorly
Number	Chest metastases <8	Chest metastases >8
Site	Chest	Bone, whatever the number

Factor	Usually respond	Usually do poorly
Timing	After >24 months	On or after chemotherapy; <2 years from treatment

- **Five-year survival** → 80% or 85%; some literature up to 90% *[Q4 · As spoken; standard 5-year survival for non-metastatic osteosarcoma is about 60-70%, not 80-90%.]*
- **Ewing sarcoma** → lower, 70% or less
- **Metastases** → survival falls considerably

Prognostic Factors Ask AI

- Groups: **tumour**, patient, chemotherapy

Factor	Favourable	Unfavourable
Grade	Low	—
Detection	Early	—
Size	Small	—
Subtype	Not chondroblastic	Chondroblastic, telangiectatic
Age	Younger	Older
Chemotherapy response	—	Poor
Metastases	—	Develop on treatment, or early

- **Chondroblastic** → very bad response to chemotherapy
- Factor groups
  - **Time** of metastases
  - Patient → **age**
  - Tumour → grade, size
  - **Response to chemotherapy**
- **Serum markers** → response to chemotherapy, e.g. LDH
- Others
  - **P-glycoprotein**
  - **HER2 gene**
  - **Retinoblastoma gene**
  - **MDM gene** → especially in the parosteal type
- Radiological → **PET CT** → practical idea
- Two years without metastases → **five-year survival** 85%
- Each metastasis → survival ↓ 10% to 20%
- More reassurance → more time, **younger** age, away from telangiectatic, chondroblastic types
- = prognostic factors affecting **outcome**

Exam Pearl

- Five-year survival 80–85%; up to 90% in some literature
- Ewing sarcoma → 70% or less
- Each metastasis → survival ↓ 10–20%
- Chondroblastic → very bad chemotherapy response

Lecture Summary Ask AI

- Little new → some theory; same **staging**
- Age 15, 16, 20, 21, 22 + lesion → **sarcoma until proved otherwise**
- Bone inside bone, **ill-defined** → sarcoma
  - With **blastic** new bone formation → sarcoma until proved otherwise
  - Not only blastic → may be lytic
- **Ill-defined lytic or blastic** lesion, around adolescence → osteosarcoma number one
- Mainly **metaphyseal**; diaphyseal, epiphyseal → subtypes
- **Chemotherapy → chemotherapy → chemotherapy**
- **Chemotherapy life-saving** → most important in treatment
- Diagnosis → **routine staging**
- New today → genetics, usual treatment
- Surgeon's treatment → **wide resection** or amputation
- **Wide resection = amputation** in survival

Parents' Questions Ask AI

- Parent's questions → must know the answers

Question	Answer
How long will he live?	Depends on age, chemotherapy response, tumour type; more time → more reassurance; ordinary cases → 85%
Will it come back?	Recurrence 15% to 20%
Is amputation better?	No difference from limb salvage in survival
Why not operate straight away?	Chemotherapy → chemotherapy → chemotherapy
Why such long follow-up?	Detect local recurrence early

- Early detection → better for patient, chemotherapy; late → nothing can be done
- Osteosarcoma **not a single entity** → several entities
- This lecture → basis for the **subtypes** lecture

Questions and Answers Ask AI

Follow-up Schedule Ask AI

Period	Frequency
First two years	Every 1.5 to 2 months
Third year	Every 3 months
Fourth, fifth years	Every 6 months

- Minimum **five years**
  - Checked after five years → 5-year, 10-year survival
- Tools
  - **X-ray**
  - **CT chest** → every 3 months
  - **Bone scan** → every year
- Suspected local recurrence → **MRI with contrast** or PET CT

Order of Next Lectures Ask AI

- **Subtypes** first → then reconstructive methods
- Subtypes → light lecture, finishes osteosarcoma
- Then → **resection**, margins, reconstructive measures, their differences

Amputation versus Limb Salvage Ask AI

- Question → what decides amputation vs rotationplasty vs other?
- **Amputation or limb salvage** → no difference in overall survival
- Wide margin on MRI → 3 cm above tumour vs amputating the whole limb
  - Amputation → recurrence almost 0%; **stump** recurrence possible

Feature	Amputation	Limb salvage
Overall survival	Same	Same
Recurrence	Almost 0%; stump possible	—
Function	Inferior	Better

- Margin achievable (MRI + **clinical examination**) + better function → limb salvage
- Amputation instead
  - **Large tumour**
  - Extensive posterior extent
  - **Foot drop**
  - **Marginal vessels**
  - **Subcutaneous** tumour
  - Limb salvage here → **crippled** for two to three years
    - **Spastic**, immobile knee
- Rule → **wide margins + better function** → limb salvage; else amputation
  - Never for survival → same in both

Timing of Surgery Ask AI

- Question → who decides surgery? Suspicious X-ray → work-up → osteosarcoma → **oncology**
- Osteosarcoma, **Ewing sarcoma** → no surgical decision; known protocol
  - Chemotherapy → surgery → chemotherapy
  - After **third dose** → re-evaluation of response → prepare for surgery
  - After **fifth or sixth dose** → surgery; coordinated, automatic

Recycling Methods Ask AI

Recycling

- **Liquid nitrogen** → affects bone, but least damaging
  - Best method; used worldwide
- Others → **pasteurization**, radiation, autoclaving
  - All → kill tumour cells
- Recycling lecture → to be uploaded
- All affect **biomechanical properties**
- Liquid nitrogen preserves best
  - **Mechanical integrity**
  - **Glycosaminoglycans**
  - **Bone morphogenetic proteins**
  - **Bioavailability**
- Far better than **autoclaving**, heat methods

Follow-up Imaging and MRI with Implants Ask AI

- Follow-up → **CT chest** with contrast + bone scan; not MRI *[C2 · transcript: "CT chest without contrast"]*
- **MRI with contrast** → only if local recurrence suspected
  - Problem → non-titanium joint prosthesis
- **CT chest** every three months; bone scan every year
- Some centres → MRI on low-Tesla machines (0.5 T, 1 T) → lesion shows
- "Plate not visible around on MRI" → **inaccurate**
  - MRI possible with **titanium** joint or nail
  - Significant **artefact**; feasible in some machines
- **MRI not routine** in follow-up

Prognosis after Recurrence Ask AI

- After surgery + chemotherapy → prognosis ↓ sharply
- Any **local recurrence** or metastasis → survival ↓ 10% to 15% *[Q5 · As spoken; same fall in survival given as 10-20% at a092.]*
- Disease-free two years
  - → **5-year survival** most likely 85% or 90%
- Local recurrence after two years → 5-year survival 70% or 75%
- Type of surgery → detailed lecture
  - **Resection**, margins, reconstruction, patient follow-up → right after next lecture

Adjuvant versus Neoadjuvant Ask AI

- Past → all chemotherapy **adjuvant** (after surgery)
- **Rosen** → why not before surgery?
  - Affects the tumour
  - Attacks tumour cells circulating in blood

        - Before surgery
    - Third, by chance → specimen **pathology** → almost all tumour cells dead → chemotherapy efficacy known
- Rule → 95% tumour cells dead → effective → continue
    - Otherwise → change **doses** or second-line drugs
- Neoadjuvant → before surgery
    - Control micrometastases
    - Reduce tumour
    - Assess **chemotherapy effect**
- **Adjuvant** → after surgery

Feature	Neoadjuvant	Adjuvant
Timing	Before surgery	After surgery
Introduced	Rosen	Original, historical approach
Aims	Micrometastases, tumour size, efficacy	Guided by neoadjuvant efficacy

Multiple Myeloma

(From MGUS to Management)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 143) “OPG coming from the osteoclasts” → “OPG from osteoblasts”

C2 (p. 143) “tells osteoclasts not to release OPG” → “osteoblasts stop releasing OPG”

C3 (p. 148) “thyroid disease” → “parathyroid disease”

C4 (p. 148) “thyroid disease” → “parathyroid disease”

Queried - kept as spoken, but looks wrong

Q1 (p. 141) As spoken; MGUS prevalence over age 50 is usually quoted about 3%; 1% is the yearly progression rate.

Q2 (p. 144) As spoken; myeloma amyloid (AL) is deposited immunoglobulin light-chain protein, not sugar/glucose.

Q3 (p. 145) As spoken; losing a third of body weight in ten days is implausible; the time frame is probably misheard.

Q4 (p. 149) As spoken; the classic eye finding in POEMS syndrome is papilledema, not macular changes.

Q5 (p. 153) As spoken; Velcade is the brand name of bortezomib, so it is not a third drug.

Q6 (p. 155) As spoken; denosumab is approved and used to prevent skeletal events in myeloma bone disease.

Q7 (p. 156) As spoken; no imaging combination is 100% sensitive; the chapter earlier says PET-CT is not 100% sensitive.

Multiple Myeloma — Definition and Precursors Ask AI

Lecture Aim Ask AI

- Aim: patient with this picture → understanding and management
- **Punched-out lesions** — characteristic
  - "Rat gnawing the bone" → pieces bitten out
  - Disease acting inside the bone → how to deal with it

Definition Ask AI

- Punched-out lesions → think first of **multiple myeloma**
  - Tumor originating in the **bone marrow**
  - Synonyms: multiple myeloma, **plasma cell myeloma**, plasmacytoma
  - 27% of biopsied bone tumors
  - Associated **abnormal proteins** → blood, urine
  - Not always present: marrow disease may precede **abnormal proteins**
- Definition: marrow tumor + **abnormal proteins**

Monoclonal Gammopathy of Unknown Significance Ask AI

MGUS

- Myeloma preceded by **monoclonal gammopathy** — pre-existing, unrecognized
  - 1% of all adults older than 50 years *[Q1 · As spoken; MGUS prevalence over age 50 is usually quoted about 3%; 1% is the yearly progression rate.]*
  - Also: monoclonal gammopathy of unknown origin / **unknown significance (MGUS)**
- **Progression**: 1% of MGUS patients per year → myeloma
  - MGUS → no bone lesions, no **organ affection**
  - Alteration, not yet a lesion
  - Only one gamma globulin affected → "**monoclonal** gammopathy"
- **Unknown significance** → may or may not turn malignant
  - Only 1% → myeloma
  - Remaining 99%: no lesions, no organ affection, **uncertain fate**
  - Determinant → **genetic factors**
- Plus **exposure** to certain factors
  - Evidence: large cohorts (10,000–100,000) followed by **electrophoresis** → rate of myeloma
  - Myeloma carries **genetic affections**
- **Runs in families** → knowledge gained retrospectively
  - MGUS = **origin of myeloma**

Solitary Myeloma (Plasmacytoma) Ask AI

- **Solitary myeloma** / plasmacytoma → myeloma in a single site
  - Lecturer's view: **early stage of myeloma** → becomes multiple
- Myeloma cells acquire ↑ structural and numerical **chromosomal aberrations** → solitary becomes multiple
  - Lecturer: all cases seen became multiple
  - **Literature**: one-third → multiple; two-thirds stay plasmacytoma
  - Other series: 50/50 or 60/40
- Treated mainly by **oncologists** → orthopedists must still understand it

Plasma Cell Origin and Incidence Ask AI

Plasma Cell Lineage Ask AI

- Reason to know it → **commonest malignant tumor**
- **Blood stem cell** → lymphoid stem cells + myeloblasts (other blood cells)
- Lymphoid stem cell → lymphoblast → B lymphocyte → **plasma cell**
  - Parent cells → lymphoblast, **lymphoid stem cell**
- Plasma cells → antibodies = **immunoglobulins**
- **Plasma cell dyscrasia** → intermediate stage, not yet a tumor
  - Cell gone wild; no **pleomorphism**, no malignant cells
- Sequence: B lymphocyte → plasma cell → immunoglobulins → **dyscrasia** or plasma cell myeloma
- Plasma cell: large nucleus filling most of cell → **signet ring** appearance (characteristic)

Summary So Far Ask AI

- Myeloma = **bone marrow tumor**
  - Little sister → **plasmacytoma** = solitary myeloma
- Preceded by **MGUS** → 1% of people at age 50
  - Either stays quiescent or progresses in 1%
- Progression → according to **genetic factors**
- Plasma cell gone wild → **dyscrasia** or plasma cell myeloma
  - Dyscrasia → disease of its own
- Key: marrow tumor originating from **plasma cells**

Incidence Ask AI

- More common in **males**
- More common in **blacks**
- **Most common primary malignant tumor** of bone
- Rarely reaches orthopedists → seen more by **oncologists**
- **Bone marrow tumor** → orthopedists must diagnose and manage it
- Western Europe / USA (not important)
  - **Males** → 6.6–6.9 per 100,000
  - **Females** → 4.5 per 100,000

Pathogenesis and Systemic Effects Ask AI

Immunoglobulins Ask AI

- Marrow tumor → plasma cells → **immunoglobulins**
  - Function → attack **microbes and organisms**
  - Classes: IgG, IgA, IgM, IgE, **IgD**
- Structure: 2 arms joined by **disulfide bonds**
  - Each arm → **light chains** + heavy chains
- Electron microscopy → **heavy chain**, light chain
  - Regions on both chains grab antigen → **antigen-binding sites**
  - This molecule → what goes wild

- Dyscrasia (between normal and malignant) or tumor → **excess immunoglobulins**
  - Heavy + light chains, disulfide bonds, **constant and variable regions**
  - Variable region → holds antigens → **body defense**

Bone Resorption — Osteoprotegerin and RANKL Ask AI

- Myeloma cells → ↓ **osteoprotegerin** (OPG)
- **Osteoprotegerin** → protective protein
- Acts on osteoclasts → ↓ **osteoclastic resorption**
  - Produced by **osteoblasts**
  - Targets **RANKL** on osteoblasts
- RANKL = **receptor activator of nuclear factor kappa B (ligand)**
  - Receptor blocked → cell inactive; activated → cell works
  - OPG released from osteoblast → **neutralizes** receptor → ↓ osteoclastic activity
  - Myeloma → blocks OPG release
- OPG absent → receptor active → **osteoclastic resorption**
  - → **Indirect** route
  - **Direct** route too → myeloma = potent osteoclastic activator (potent RANKL receptor activator)
  - Osteoclast carries the **RANKL** receptor
- RANKL neutralized → osteoclast inactive; activated → **excessive** osteoclast activity
  - Myeloma cells → stimulate osteoclasts **directly and indirectly**
  - Direct → release **potent osteoclastic activators**
  - Bind OPG via **transmembrane glycoprotein** → block release of osteoclast-neutralizing protein
- Net → osteoclastic **bone destruction**
  - ↓ OPG (osteoclast inhibitor) from **osteoblasts** *[C1 · transcript: "OPG coming from the osteoclasts"]*
  - **Tumor necrosis factor** from myeloma cells → osteoblasts stop releasing OPG *[C2 · transcript: "tells osteoclasts not to release OPG"]*
- **Tumor necrosis factor** → also ↑ osteoclastic resorption
- Myeloma → ↑ osteoclastic **bone resorption**

Route	Mechanism
Direct	Tumor necrosis factors, interleukins, harmful cytokines
Indirect	Binds OPG → receptor on osteoclast no longer neutralized

Sites of Involvement Ask AI

- Does myeloma affect any bone?
- Mainly bones with **red marrow** → active, cellular, producing
  - Red marrow → mostly **flat bones**
  - **No bone exempt** from involvement
  - Red marrow exhausted → moves to **long bones** (less red marrow)
  - Distal to elbow and knee → **extremely rare** (less than 10%) → like metastasis
- Sequence: **flat bones** → long bones
  - Reason → needs red marrow (more new cell formation)

- Figure: flat bones > long bones and **skull**

Effects Inside the Marrow Ask AI

- Bone destruction → marrow cells ↑ **number and size**
- First affected → **anemia**; then thrombocytopenia
- Bone destruction → **hypercalcemia**
- Produces **abnormal proteins** → each with its own effect

Recap of the Disease Process Ask AI

- Simply: **tumor in the marrow**
  - Above 50 → **MGUS** → 1% develop myeloma
  - Via the **plasma cell**
- Plasma cell from **B lymphocytes** → immunoglobulins
  - Cause → **genetic factors** + unknown factors
  - Malignant tumor → **plasma cell myeloma**
  - ↑ effect only, not number → **plasma cell dyscrasia**
  - **Dyscrasia** = dysfunction → no malignancy, no pleomorphism yet
- Plasma cell gone wild →
  - ↑ **immunoglobulins**
  - Marrow destroyed → **hematopoiesis** stops
  - ↑ **calcium** release
  - Targets **active marrow** (more cells)
- Flat bones first → hematopoiesis stops there
  - → **Metaplasia of yellow marrow into red marrow** in long bones
  - → Attacks and destroys **long bones**
  - → **Refractory anemia**: neither red nor yellow marrow spared
  - → **Abnormal globulins**

Amyloidosis Ask AI

- Any **long-standing disease or pathology**
- → Associated with **amyloidosis**
  - Body → **sugar deposition** in cell to halt process → amyloidosis *[Q2 · As spoken; myeloma amyloid (AL) is deposited immunoglobulin light-chain protein, not sugar/glucose.]*
  - Cell and marrow destruction → **end stage** = amyloidosis

Myeloma Kidney Ask AI

- Any abnormal protein passing through kidney → **toxic**
  - **Glomeruli** → designed for known urine constituents
- Abnormal passage → affects glomeruli → **glomerulonephritis**
  - Myeloma immunoglobulins → mostly **normal structure**, abnormal function
  - Unlike natural ones → **abnormal function**; many also abnormal structure
- Volume effect: 100 million vs 10,000 molecules → far more **faulty** ones
  - Proteins through kidney → **myeloma kidney**
  - **Conglomeration** → stone-like casts
- Casts → **ischemia** (likely medullary)
  - → **Nephrocalcinosis** replaces small casts
  - → **Amyloidosis**

  - Kidney affected → besides **blood and marrow**

Clinical Presentation Ask AI

Pain Ask AI

- **Pain** = hallmark of diagnosis
  - Mild, transient by day; almost always **worse at night**
  - Present in 75%
- Famous for masquerading as **sciatica**
  - Many treated as sciatica; some underwent **discectomy**
  - Pain → most important diagnostic clue
  - Classic error → missing **night pain** that wakes patient
- Elderly + back pain + **anemia** → myeloma until proved otherwise
- Elderly + back pain + weight loss + malaise + fever → **myeloma or metastatic bone disease**
  - Ask → **significant weight loss**?

Clinical Pearl

- Elderly + back pain + anemia → myeloma until proved otherwise
- Myeloma → famous for masquerading as sciatica
- Pain → worse at night, present in 75%

Clinical Examination Ask AI

- Weight loss — e.g., one third of weight in ten days → key in **history taking** *[Q3 · As spoken; losing a third of body weight in ten days is implausible; the time frame is probably misheard.]*
- Findings
  - **Sciatica-like** presentation
  - **Pathological fractures**
  - **Refractory anemia** → weight loss, pallor
  - **Eye changes** ☐ thrombocytopenia
  - **Organomegaly**, cardiomegaly ☐ anemia
  - **Hepatomegaly** ☐ immunoglobulin deposition

Laboratory Diagnosis Ask AI

Routine Labs Ask AI

- Marrow plasma cell tumor → **predictable labs**
  - **CBC** → anemia
  - Serum calcium → **hypercalcemia**
  - **Amyloidosis** → diagnosable; body deposits glucose in cell to halt it
  - **Electrophoresis** → separation of proteins
  - 24-hour **urinary proteins**
- 24-hour urine → **abnormal proteins** excreted
  - Used in **renal failure** and myeloma
  - Measures immunoglobulin excreted over 24 hours

Immune Electrophoresis Ask AI

- **Immune electrophoresis** → which immunoglobulin is increasing
  - Immunoglobulins = proteins → **heavy + light chain**
- Only one immunoglobulin rising → **monoclonal gammopathy**
  - Lab goal → monoclonal proliferation? which type? → **diagnosis**
- All immunoglobulins ↑ → certainly **not myeloma**
  - Only one ↑ → most probably **myeloma**

	Polyclonal gammopathy	Monoclonal gammopathy
Pattern	All immunoglobulins ↑	Only one ↑
Causes	Liver disease (most famous), TB, chronic infections	Very limited
Myeloma?	Certainly not	Most probably

- Normal → **albumin > globulin**
  - **Globulin > albumin** → start to suspect
  - **Reversed albumin–globulin ratio** → suspect multiple myeloma
  - IgG, IgA, IgD rise (not IgM) → **myeloma** pattern
- Normal trace → **albumin** high, rest low
  - Abnormal → **gamma globulin** higher than albumin → reverse ratio
  - Next → **which one** increased; single → most probably myeloma
- "Most probably" → **other causes** (first lecture)
- Myeloma most famous; others → **plasmacytoma**, malignant lymphoma
- All → reversed ratio + ↑ immunoglobulins; **different one** rises in each

Condition	Rising immunoglobulin
Myeloma	IgG
Waldenström macroglobulinemia, lymphomas, leukemias	IgM > IgG

- **Monoclonal hypergammaglobulinemia** → not diagnostic of myeloma alone
- Method → proteins at **very high speed** → sedimentation amount and time
  - Normal → albumin rises in a **fixed time**
  - Globulin > albumin → time changes
- ↑ immunoglobulin number → **monoclonal gammopathy** / monoclonal hypergammaglobulinemia

> **Pitfall**
> • Monoclonal gammopathy → not diagnostic of myeloma on its own
> • Plasmacytoma, malignant lymphoma → also reversed ratio
> • Waldenström, lymphomas, leukemias → IgM rises more than IgG

Urinary Proteins — Bence Jones Ask AI

- **Urinary tests** → not important
  - **Bence Jones proteins** → non-specific

  - Heat to 50 degrees → coagulates → heat to 100 degrees → **no importance**
  - Frequently excreted in myeloma → **not diagnostic**
  - Mentioned only as **minor criteria**

Major and Minor Criteria Ask AI

- Work-up: elderly + **back pain** + anemia → suspicion
  - Labs → ESR high, or electrophoresis → **monoclonal gammopathy**
- Or straight to **bone marrow biopsy** → percentage of plasma cells
- **Major criteria**
  - **Tissue biopsy** → bone biopsy showing myeloma
  - **Bone marrow biopsy**
  - **Protein electrophoresis**
- **Minor criteria**
  - At least one **lytic bone lesion**
  - **Lower serum levels** → immunoglobulin high, rest lower
  - **Urinary proteins**
- Minor serum pattern → **IgG** high; IgM, IgA, IgD reduced
- **Diagnosis** → one major + one minor, or two major
  - One minor alone → **insufficient**
  - One major alone → **insufficient**

Must-Know

- Diagnosis → one major + one minor, or two major
- Major → bone biopsy, bone marrow biopsy, protein electrophoresis
- Minor → lytic bone lesion, lower serum levels, urinary proteins
- Bence Jones proteins → non-specific, minor criterion only

Imaging Ask AI

Plain X-ray Ask AI

- Lytic lesions → **flat bones**, ribs, long bones
- All → **punched-out lesions**
- Also called **rat bites**
- Pattern 1 → fairly good **bone quality**
- **Multiple lytic lesions**
  - Most visible → **ilium**; pelvis to a large extent
- Pattern 2 → bigger, worse picture
- Could pass as **metastasis** → not a wrong answer
  - Elderly + bony pains → **metastasis, myeloma, lymphoma**
  - **Bone quality** poor here
  - **Mixed lytic–sclerotic** → not wrong
  - **Ribs** → gnawed-bone appearance
- Characteristic site → **skull** → two diseases
- Skull punched-out lesions → **age** separates them

Disease	Age group
Multiple myeloma	Adults
Eosinophilic granuloma	Young

- **Femur** → punched-out lesion, piece of bone removed
- As if bone **bitten off**
  - Or **pathological fracture** through punched-out lesion (e.g., humerus)
- All → myeloma appearances on **X-ray**

Bone Scan Ask AI

- Staging → **X-ray + bone scan + MRI**
- Hot or cold → usually **cold**
  - Excessive bone destruction → no chance for **new bone formation**
  - Some hotness ☐ **increased turnover in adjacent bone**
  - Marrow attempts bone formation around **injured bone**
- Example → only **osteophytes** (lips) hot → osteoarthritis
- Patient full of lesions → scan shows almost **nothing**
  - **Misleading** → confuses rather than helps
  - Useful → **diploë of the skull** involvement

CT and Sestamibi Scan Ask AI

- Destruction of **sacrum and ilium**
- Destruction of vertebral **pedicle and body**
- **Bone destruction** shown
- **Whole-body technetium methoxyisobutylisonitrile (sestamibi)** → not very important
  - **Characteristic for myeloma**
  - Not available in Egypt, or few do it
  - Shows myeloma well → unlike usual **bone scan**
  - Also shows **parathyroid** disease *[C3 · transcript: “thyroid disease”]*
- Figure → **parathyroid disease**, not myeloma; resembles bone scan *[C4 · transcript: “thyroid disease”]*

MRI Ask AI

- **Whole-body MRI** → very sensitive for myeloma
- Reserved for **uncertain cases**
  - Any lesion → **low T1, high T2**
- DDx of such lesion → **fibrous dysplasia**, metastasis, myeloma, sarcoma
  - **Ill-defined**, lytic → not characteristic
  - **Punched-out lesion** / rat bites → clear as the sun
- Spine → **wedge fracture**
- Old-age marrow → **fatty tissue replacement** → may mask diagnosis
  - Often indistinguishable → aged marrow vs multiple myeloma vs **myeloproliferative disease**
  - Solution → **fat suppression**
  - Sequence → **STIR**
- STIR → suppresses fat and marrow fat → **lesions stand out**
  - Normal → **subcutaneous fat** white

  - Suppressed → subcutaneous fat darker → **lesions appear**
- Darkened subcutaneous fat = **fat suppression** → aids diagnosis

PET-CT and the Place of Radiology Ask AI

- **PET-CT** → included in the work-up
  - Not 100% sensitive nor specific
  - **MRI + PET-CT** together → radiological diagnosis
- Radiology → **not among major or minor criteria**
  - Criterion → **lytic bone lesion** on ordinary X-ray
  - Extra imaging → for **uncertain patients**
- Many myeloma patients → **normal bone marrow biopsy** → diagnosis after effort
  - Not all cases obvious
  - MRI → **low T1, high T2**
- Figure → **low T1**

Exam Pearl

- Skull punched-out lesions → myeloma in adults, eosinophilic granuloma in young
- Bone scan → usually cold, misleading
- MRI → low T1, high T2; STIR for fatty old-age marrow
- Radiology → not among major or minor criteria

Sclerotic Myeloma (POEMS Syndrome) Ask AI

- **Sclerotic myeloma**
- Background → plasma cells gone wild → **dyscrasia**
  - **Intermediate stage** → abnormal cell behavior, no malignancy yet
- Names → sclerotic multiple myeloma, **POEMS**, osteosclerotic multiple myeloma
  - **Osteosclerotic lesions**
  - Variant of **dyscrasia myeloma**
  - E = **endocrine disturbances**
  - M = **monoclonal gammopathy**
- **POEMS syndrome** → cell dyscrasia rather than true neoplasm
- Features
  - **Skin nodules**
  - **Hirsutism**
  - **Joint** affection
  - Eye → **macular changes** *[Q4 · As spoken; the classic eye finding in POEMS syndrome is papilledema, not macular changes.]*
  - **Foot disturbances**
  - Hand **arthritis**
  - **Gynecomastia**
  - **Enthesiopathy**, ligamentous affection, arthritis, tendinitis
- Most important → **hirsutism** and skin affections
- Also called **sclerosing myeloma** / sclerosing myelomatosis
- X-ray → **lucent bone destruction** + surrounding sclerosis

- Treatment → completely different from myeloma
  - No myeloma drugs; sometimes **steroids**
  - Not treated by orthopedists

Pathology Ask AI

- **Malignant plasma cell growth** → lytic lesion
- **Gross appearance** → not diagnostic, not informative; confused with many things
- Diagnostic → **extended sheets of atypical plasmacytoid cells**
- Typical vs atypical
  - **Well-differentiated** → normal-appearing plasma cells with nucleoli
  - **Dense chromatin**, prominent nucleoli, strongly basophilic cytoplasm
  - Electron microscopy → prominent **Golgi apparatus**, cytoplasmic inclusion bodies
  - Cells → ↑ **number** and size
- More prominent **nucleus and nucleoli**; many Golgi apparatus and inclusion bodies
- **Dedifferentiation**
  - ↑ nucleoli, **binucleate and trinucleate cells** → more dedifferentiated
  - Graded by patient **survival**
  - ↑ **cell number**
- ↑ **Pleomorphism**, loss of shape, ↑ nucleoli, ↑ inclusion bodies → dedifferentiation → worse for patient
  - **Electron microscopy** picture → not important
- Microscopy → very many **abnormal plasma cells**
  - **High power** → same picture
  - **Blue inclusion bodies** inside cells

Differential Diagnosis Ask AI

- Elderly + **bone pain** → myeloma, lymphoma, metastasis
  - Myeloma vs **lymphoma** (previous lecture)
  - **Primary sarcoma**
  - **Metabolic** conditions
- e.g., **hyperparathyroidism**
  - **Paget's** → usually sclerotic, not lytic
  - However **early stage** → may pass
  - Metastasis, myeloma, lymphoma + **other round cell lesions**
- **Round cell lesions** → next lecture
- Round cell lesions / **round cell tumors** → include myeloma
  - Myeloma → most famous **round cell tumor** ☐ most common primary malignant tumor

Genetics, Staging and Complications Ask AI

Genetic Factors Ask AI

- After 50 → 1% develop **MGUS**; 1% progress
- Determinants → chromosomal alterations, **translocations**, deletions

  - Site → chromosome number 13
- Present → most probably **myeloma**; absent → nothing
  - **Main factor** → who develops myeloma

Durie–Salmon PLUS Staging Ask AI

- **Durie–Salmon PLUS system**
  - Number of lesions → kidney affection → **spine involvement**
  - Stage 1A and B, **stage 2A and B**
  - More **oncology** domain
- By **bone lesion count**

Stage	Bone lesions
1A	None
1B	Less than 5
2	5 to 20

  - **Renal involvement**: none → stage 1; mild → stage 2
  - A vs B → **kidney affection**
  - Matters for **oncologists'** treatment

Complications Ask AI

- **Anemia**
- **Pathological fractures**
- **Hypercalcemia**
- **Kidney** impairment
- **Hyperviscosity** ☐ ↑ immunoglobulins and proteins
  - **Rouleaux formation** → cells stick together
  - Cause → abnormal cells, **abnormal immunoglobulins**, abnormal proteins
  - → Risk of **thrombosis** (DVT)

Treatment Ask AI

Principles Ask AI

- Mostly **oncology** → orthopedists must diagnose and know treatment
  - Treatment → **oncologists**, not orthopedics
  - Surgeon → decides need for surgery; sends patient for **treatment first**
  - **Patient must be under treatment**
- Wrong sequence (e.g., painful lesion in humeral head)
- Surgery paperwork → bone breaks → fixation → wound **infection**
  - Treatment delayed 2–3 months → **solitary becomes multiple**
- Correct → **start treatment** → **then surgery**
- Medical treatment = **the basis**, not orthopedic
- Drugs (not orthopedic domain) → **doxorubicin**, melphalan, corticosteroids

Role of the Surgeon Ask AI

- Surgery → at the very end, **not the basis**
  - Catch patient before fracture → **Mirels' score**
  - **Prophylactic fixation**
  - Fixation of **already fractured bones**
  - Hip → **arthroplasty**; maybe **modular tumor hip**
  - Rare → **palliative resection** → very large lesion, curettage impossible
- Reconstruction impossible → **resection**
  - Spine → **spine fixation** / palliative spine surgery
- Large lesion → **two-thirds of humerus** or a third
- Closed **nail** alone → holds nothing
  - **Nail + cementing** over the nail
  - Aim → **stable limb** the patient can move
  - Sarcoma → resect; metastasis → similar approach; **each its own treatment**
- Preferred → **intramedullary fixation**
  - Humerus → **interlocking humerus nail**
  - Femur → **second-generation nailing**

Arthroplasty and Tailored Treatment Ask AI

- Large lesion in head and neck reaching **acetabulum** → modular prosthesis (long-stem)
- Or ordinary **arthroplasty** → depends on
  - **Head and neck** affection
  - Patient **survival**
  - **Thrombocytopenia**
  - **General condition**
  - The **lesion**
- Like metastasis → **tailored treatment** per patient condition

Patient	Approach
Early myeloma, age 48–50, small hip lesion, responsive to treatment	Fixation with nail
Age 70, bedridden a month, fracture in bed	Minimal procedure to allow moving and nursing

- Each patient → per **stage of disease**
- **Surgery is not the treatment** → oncologists treat

> **High Yield**
> - Treatment of myeloma → medical, by oncologists
> - Start medical treatment first → surgery afterwards
> - Surgery → prophylactic fixation, fracture fixation, arthroplasty, rare resection
> - Preferred fixation → intramedullary nail ± cement

Remission and Relapse Ask AI

- **Relapsing** disease → stays and recurs
  - Longer **remission** (disease-free) → lower risk

- Case → remission 2 years → considered cured
  - Remission → advise **stem cell / bone marrow transplantation**
  - Refused → **relapse**
  - Relapse → **targeted therapy**: bortezomib, lenalidomide, Velcade *[Q5 · As spoken; Velcade is the brand name of bortezomib, so it is not a third drug.]*
  - Details → oncologists

Prognosis Ask AI

Predictors Ask AI

- 5-year **survival** → 35%
- Good vs **bad predictors**

Good predictors	Bad predictors
Young age	Old age
Single lesion	Multiple lesions; more than 7 lesions
Responsive to treatment	Growth despite treatment
	Relapses
	Respiratory manifestations, dyspnea ☐ rib affection

Plasmacytoma Outcome Ask AI

- **Plasmacytoma** → 5% of myeloma
  - Marrow levels normal
  - Diagnosis → **bone biopsy**
- Progression to myeloma → **estimates vary**
  - One-third, 40%, 50%, two-thirds
  - Lecturer → 55% most accurate
- = **myeloma in one place** → plasmacytoma / solitary plasma cell myeloma

Take-Home Messages and Surgical Safety Ask AI

Take-Home Messages Ask AI

- Elderly over 50 + back pain + anemia → **myeloma**
- **Anemia that does not resolve** → think myeloma
- Old age → myeloma = **commonest bone tumor**
- **Surgery** → not the mainstay of treatment
- Start medical treatment first → unless **bone broken**

Timing of Surgery Ask AI

- **Fracture neck of femur** → no "go take treatment first"
  - Accidental **femoral shaft** lesion, mild pain, walking, untreated → may wait
  - **Humerus** → waiting more acceptable; femur → cannot wait

  - Fractured **neck of femur** or spine → cannot wait
  - Small lesion → humerus, forearm, **tibia**
- **Distal** lesions → rare but occur
  - Distal femur, around the knee → **treatment first**, surgery later
- **Treatment is medical** → depends on disease stage
- Younger, responsive, fewer and smaller lesions → **major surgery** justified
- Older, more lesions,
- **Not responding to treatment** → avoid big procedures

Extreme Vascularity of Myeloma Lesions Ask AI

- Case: humeral nailing → stuck **wire** → opened for retrieval
  - Blood loss → 3.5 liters in ten minutes
  - Outcome → **supramaximal** inotropic support
  - Lesson → **never open a myeloma lesion** as far as possible
- **Extremely vascular**
- Case: patient off **melphalan** (unavailable) → untreated
  - **Huge femoral lesion** → planned prophylactic fixation; fractured leaving car
  - **Closed nailing** → bleeding from interlocking nail entry
- Or reconstruction nail entry → 2 liters; **hemoglobin** 10 → 7–8
- **Myeloma is disastrous** → rules
  - First → never operate **without blood**
  - Second → **ICU backup**
  - Avoid opening lesion, **curettage**, cement
  - Debate → **worldwide**
- Lecturer's school → lesions **vascular** → no need to open
- Opposing view → open + **curette** + cement
  - ↓ **Tumor overload**
  - Better **radiotherapy** effect
  - Early **weight bearing**
- Drawback → **very morbid**; heavy bleeding
- Open only **when necessary**
- Have about six units of blood + **ICU backup**

Causes of Bleeding in Myeloma Ask AI

- **Several causes** of bleeding
  - **Thrombocytopenia**
  - **Platelet dysfunction**
  - **Fibrinogen degradation**
  - **Hyperviscosity**

Pitfall

- Myeloma lesions → extremely vascular
- Never operate without blood; ICU backup
- Avoid opening, curettage and cement unless necessary
- Bleeding ☐ thrombocytopenia, platelet dysfunction, fibrinogen degradation, hyperviscosity

Questions and Answers Ask AI

Pathological Fracture Ask AI

- Q → **pathological fracture** in multiple myeloma?
- A → **same management** as pathological fracture from metastatic bone disease
  - The two → siblings
- Q → role of **denosumab** (Xgeva)?
  - A → **no role** in myeloma; no response *[Q6 · As spoken; denosumab is approved and used to prevent skeletal events in myeloma bone disease.]*
  - Pathological fracture → as in previous lecture
- **Pathological fracture algorithm** → whatever the cause
  - Bone sarcoma fracture → one line
  - Metastasis or myeloma fracture → **completely different line**

Remission vs Relapse Ask AI

- Q → difference between **remission and relapse**?
- **Relapse** → treated myeloma,
- Normal for at least a year → **myeloma returns** after treatment
- **Remission** → patient responds to treatment
  - Monitored via **bone marrow** plasma cell count
  - Count reaches zero → **cured**
- Zero → rises to 40–50% → **relapse**
- Relapse after zero → **common** in a large proportion

Term	Meaning
Remission	Response to treatment; marrow plasma cells → zero
Relapse	Return after cure (zero count, normal at least a year)

Myeloma vs Hyperparathyroidism Ask AI

- Q → multiple myeloma vs **hyperparathyroidism**?
  - Myeloma → **elderly man**
  - Discriminators → labs, **electrophoresis**, parathormone, bone scan
- All of these → **differentiate**

Feature	Myeloma	Hyperparathyroidism
Parathormone	—	High
Electrophoresis	Monoclonal gammopathy	Normal
Anemia	Present	Absent
Bone scan	Usually negative, not diagnostic	Superscan

- **Quality of bone** → also differs
- Myeloma destroys bone heavily → **bone scan usually negative**

PET-CT vs MRI Ask AI

- Q → **PET-CT or MRI** for confirmation?
  - A → **both together** → 100% sensitivity *[Q7 · As spoken; no imaging combination is 100% sensitive; the chapter earlier says PET-CT is not 100% sensitive.]*
  - Not first steps → **major and minor criteria** first
- Lytic lesion + positive electrophoresis + **positive bone marrow** + urinary protein
- → **myeloma confirmed**; usual scenario
  - None, or only one major → **bone biopsy**
  - Bone biopsy negative → **MRI, PET-CT**
  - Only **MRI with PET-CT** together → 100% sensitive

Dyspnea in Myeloma Ask AI

- Q → **dyspnea** in these patients, cause?
- Seen → **rib pain** → dyspnea
  - Lesions invading, encroaching on **chest cavity**
  - Unclear cause → refer to **oncologists**
- **Orthopedic surgeon's role** → understand, diagnose, know appearance
  - Know when to **refer to oncologist** vs when to operate

Preoperative Embolization Ask AI

- Q → **embolization** before resection?
  - A → used, but **not very effective** in myeloma
  - Bleeding tendency ☐ mainly **platelet dysfunction** and fibrinogen
- **Degradation** of fibrinogen + few platelets → embolization does little
  - Sometimes tried → **no good results**

Lesion Size vs Metastasis Ask AI

- Q → size of **punched-out lesions** separates myeloma from metastasis?
  - A → **not at all**
- Differentiation → **electrophoresis** and bone biopsy
  - Electrophoresis normal → **bone biopsy**
  - **Bone marrow biopsy** → sternum or iliac bone → plasma cells?
  - **Bone biopsy** → from long bones
  - Both needle-based → seeking **plasma cells**
- Upcoming → **bone biopsy** lecture; round cell tumors collection
  - **Types of biopsies** and the idea of each
- Topic-specific questions
- Remaining questions → **coming lectures**

Limb Salvage Surgery

(Resection Margins and Reconstruction)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 162) “vessels inside tumour: absolute indication” → “vessels inside tumour: absolute contraindication”

C2 (p. 162) “neurovascular involvement: absolute indication” → “neurovascular involvement: absolute contraindication”

Queried - kept as spoken, but looks wrong

Q1 (p. 169) As spoken; extracorporeal irradiation of tumour bone usually uses 50–300 Gy, so 20–30 Gy looks too low.

Limb Salvage: Concept and Indications Ask AI

Aim of the Lecture Ask AI

- After bone tumour resection → compensate for what was removed → **reconstruction**
- Many bone tumours aggressive — mainly the malignant ones
- **Survival rate** over time
  - 1950s–1960s → 15–20%
  - Currently → 70–85%, sometimes up to 90% (around 80–85%)
  - Improvement → advances in treatment and imaging techniques

Definition Ask AI

- **Limb salvage**: management of malignant or aggressive extremity tumour
  - Upper or lower limb
  - Resected with a safety margin
  - Limb function preserved
- Most important element → **clear and safe margin**
  - Not achievable → limb salvage not achieved → amputation better
- **Clear and safe margin** = acceptably low recurrence rate
  - = the adequate (safe) margin
  - 2–3 cm around the lesion; not less
- Two conditions fulfilled → limb salvage performed
  - **Clear safety margins**
  - Adequate function
- Upper limb salvage → **much more precious** than lower limb
  - Nothing replaces the hand → always oriented towards limb salvage
  - Lower limb → entirely different

Requirements Ask AI

- After resection
  - **Satisfactory function**
  - Acceptable cosmesis — acceptable appearance
- Result **must be better than amputation** — function and appearance
  - Otherwise → limb salvage not achieved
- Differs somewhat with **rotationplasty** — discussed later
- Good safety margin + good function + good **cosmesis** → limb salvage

Contraindications Ask AI

- Not possible at any time, in any site, in any limb
  - **Poorly located biopsies** → difficult
  - **Major neurovascular involvement** — nerves and vessels encased in tumour → very difficult
  - **Skin coverage problems** → very difficult
  - **Pathological fracture** → usually difficult
    - Heals with chemotherapy → limb salvage possible

    - **Ewing sarcoma** / osteosarcoma child with fracture → splint → chemotherapy → union → limb salvage
- Vessels inside tumour → **absolute contraindication**; rare in reality *[C1 · transcript: "vessels inside tumour: absolute indication"]*

Contraindication	Strength
Major neurovascular involvement	Absolute
Pathological fracture	Relative — salvage if it heals with chemotherapy
Poorly located biopsy	Makes salvage difficult
Skin coverage problems	Makes salvage very difficult

Must-Know

- Adequate margin → 2–3 cm around the lesion
- Major neurovascular involvement → absolute contraindication *[C2 · transcript: "neurovascular involvement: absolute indication"]*
- Pathological fracture → relative contraindication
- Result must beat amputation in function and cosmesis

Types of Resection Ask AI

- Limb salvage = **resection + reconstruction**
- Resection types → required for exam, practice, and in general

By Margin Ask AI

- Three types → marginal, wide, radical; plus intralesional
- **Pseudocapsule**: zone between tumour and host tissue; always present
- Through the pseudocapsule → **marginal resection**
- 2–3 cm beyond pseudocapsule, above and below → **wide resection**
- Whole bone or whole compartment → **radical resection**
- Inside the tumour, within the pseudocapsule (from inside) → **intralesional excision**
- Planes and uses

Resection	Plane	Use
Intralesional	Inside tumour, within pseudocapsule	Curettage
Marginal	Through pseudocapsule	Benign encapsulated tumours — lipoma
Wide	Wide zone of normal tissue, 2–3 cm beyond pseudocapsule	Sarcomas — soft-tissue and bone alike
Radical	Whole bone / whole compartment	Extensive or irresectable sarcomas → whole limb removed

By Joint Ask AI

- Second classification → intra-articular, extra-articular, intercalary
- Definitions on a simple drawing

Resection	Joint	Examples
Intra-articular	Spared — still works; metaphysis removed	Distal femur (part of joint, not whole); proximal femur without acetabulum; proximal humerus alone or glenoid alone
Extra-articular	Included — part or whole joint	Proximal humerus + glenoid together
Intercalary	Not involved	Piece of diaphysis removed — diaphyseal resection

- Specimen shown → **intra-articular resection**
- Proximal humerus + glenoid together → **extra-articular**

High Yield
- Intra-articular = joint-sparing
- Extra-articular = joint-including
- Intercalary = diaphyseal resection
- Marginal → through pseudocapsule; intralesional → inside it
- Wide → 2–3 cm beyond pseudocapsule; radical → whole compartment

Options of Reconstruction — Overview Ask AI

- Two **major categories** → endoprosthetic (metal) / biological (body's own tissues)
- **Endoprosthetic**
  - **Megaprostheses** → adults
  - **Expandable prostheses** → children
  - **Intercalary prostheses** → non-joint, diaphyseal defects
- **Biological** — no metal inserted
  - **Autograft** → vascularized or non-vascularized fibula
  - Induced membrane osteogenesis → **Masquelet technique**, used in bone non-unions
  - **Distraction osteogenesis** → external fixator or Ilizarov frame
  - Allografts
  - Rotationplasty
  - Arthrodesis
- **Bone cement spacer** → separate option; neither biological nor prosthetic

Endoprosthetic Reconstruction Ask AI

Modular Prostheses Ask AI

Prosthesis

- After resection → depends on remaining **bone stock**
- Several endoprosthetic options

- Began in 1970
- Types
  - **Modular** pieces
  - **Custom-made** → tailored to patient
- Currently → essentially all modular
- **Total femoral** and total humeral prostheses
- Articulating joint → **hinge**
- **Fully constrained prosthesis** → very old classic design
  - Old classic design → lengthened from one side
- Modular design → two ends + **middle piece**
- Modular system → also proximal humerus; all **long-stem**
- **Modular elbow** also available

Expandable Prostheses Ask AI

Expandable

- Expandable = **lengthens** — invasively or non-invasively
  - **Invasive** → child's leg opened → prosthesis lengthened surgically
  - **Non-invasive** → e.g., magnetic; other methods exist
- Much debate → **survival rate not high**
- Not among the best options in children

Intercalary Prosthesis Ask AI

Prosthesis

- Humerus piece removed → custom implant, **stem above and stem below**
- **Survival rate** not very high; not used much; must be known

Advantages and Disadvantages Ask AI

Prosthesis

Advantages	Disadvantages
Immediate weight bearing	Many complications
Superior, immediate function	Age limitations
	Long-term survival not high

- **Complications** → as with an ordinary total hip
  - Loosening
  - Infection
  - Periprosthetic fracture
  - **Stress shielding**
  - Mechanical problems
- **Age limitations**
  - Child aged 7, 8, or 9 → prosthesis very difficult, even expandable
  - Opinion on **expandable prostheses** → less encouraging than before; many problems
- **Long-term survival**
  - About 60–70% of patients at 10 years, worldwide
  - 30–40% → **revision**

Fibular Autograft Ask AI

- **Biological reconstruction** → the body's own, natural tissues
- **Autograft** → graft from the same person

Vascularized Fibula Ask AI

Vasc fibula

- Most famous autograft → **vascularized fibula**
- Advantages
  - **Hypertrophies** with weight bearing
  - Combinable → e.g., with allograft
- Disadvantages
  - **Feeding vessel** injury (femoral, brachial) → limb at risk
  - Compartment syndrome
  - **Venous gangrene**
  - Many vascular complications
  - **Specialized surgeons** → complications far less frequent
- Indication → **long defect**, more than 6 cm
- Uses
  - **Condyle-sparing**: distal femur resected, condyles spared → mobile joint; fibula inserted
    - Pedicle → **superficial femoral artery**
  - **Epiphyseal transfer** → fibula placed with its epiphysis → grows
    - e.g., proximal humerus in children
  - **Fusion** (arthrodesis) → unites above and below

Non-Vascularized Fibula Ask AI

Non-vasc

- **Simple procedure** → anyone can perform
- Usually → defects less than 6 cm
- Best site → **distal radius**
- **Less morbid** technique
- **Ain Shams University** → non-vascularized fibula for defects more than 6 cm; excellent results
- **Classic teaching** (Kasr Al-Ainy, literature) → defect more than 6 cm → vascularized fibula
  - Not only unites → also hypertrophies
- Old example → wide resection → **double-barrel** non-vascularized fibula
  - Defect actually more than 6 cm → united
- Vascularized fibula + **non-vascularized fibula** beside it → union; very good result

	Vascularized	Non-vascularized
Defect length	More than 6 cm	Usually less than 6 cm
Technique	Pedicled; specialized surgeons	Simple; anyone can perform
Biology	Unites and hypertrophies with weight bearing	Unites
Morbidity	Vascular complications — compartment syndrome, venous gangrene	Less morbid

	Vascularized	Non-vascularized
Typical use	Distal femur (condyle-sparing), proximal humerus (epiphyseal transfer), fusion	Distal radius; double barrel

Induced Membrane and Distraction Osteogenesis Ask AI

Induced Membrane Osteogenesis Ask AI

Masquelet

- Third biological type → same technique as **bone non-union**
  - Well known to Ilizarov and non-union surgeons
- Tumours → **no long-term results**; mainly fracture non-union
- Very useful in
  - Children
  - **Intercalary defects**, especially large
  - Arthrodesis
- **Not good for joint reconstruction**
- Principle
  - **Cement spacer** placed → membrane forms around it
  - Membrane → **bone morphogenetic proteins** + growth factors
  - Bone formation → from within the membrane
- Spacer over a **nail** or cement alone → same technique
  - First spacer → later removed → grafting
- Tumour problem → patient tied to **chemotherapy** schedule
  - Staged spacer → removal → graft → not feasible
- Therefore → **time-consuming**, staged, unsuitable with chemotherapy

Distraction Osteogenesis Ask AI

Distraction

- Concept → **growth plate** = natural barrier for sarcomas
- Indication → intercalary defect, intact growth plate, joint preserved
  - Especially **young children**
  - Ilizarov or even Orthofix
- Like membrane osteogenesis → **staged**, time-consuming, unsuitable with chemotherapy
- Example → **diaphyseal lesion**
  - Physis working; joints above and below working → wide resection
- Picture from a paper, not own cases
- Egypt / Kasr Al-Ainy
  - **Membrane osteogenesis** → now being worked on with microsurgery colleagues
  - Neither used much
- Literature
  - **Distraction osteogenesis** → long-term results; much published, especially abroad
  - Membrane osteogenesis → little; no long-term results, few patients
  - **Skeletally immature** + diaphyseal defect → distraction instead of fibula
- Depends on **regenerate** → needs time → conflicts with chemotherapy
  - Patients suffer before weight bearing

- **Time-consuming** → not favoured in tumour surgery
  - Aim → walk quickly → morale + restore bone
- **Not used much in Egypt** → despite literature solving many problems

	Induced membrane	Distraction
Principle	Cement spacer → membrane (bone morphogenetic proteins, growth factors) → bone	Growth plate barrier; regenerate
Best for	Children, large intercalary defects, arthrodesis	Intercalary defect, intact physis, skeletally immature
Long-term tumour results	None; few patients	Available; much published
Drawbacks	Staged, time-consuming, unsuitable with chemotherapy	Same; regenerate needs time
Egypt	Being started — Kasr Al-Ainy + microsurgery	Not used much

Allografts Ask AI

- Second biological type, after autografts → **allografts**
- Source → **amputee** (above-knee or below-knee)
  - Remaining tibia and femur → processed → bone stock for others
- → Very good **bone stock**
- **Dead bone** → provides structural material in the region
- Acts as **scaffold** → healing bone creeps over it
- Combinable with other options
- Example → defect from joint upwards → **distal femoral allograft**
  - Native bone + allograft → fixed above and below
  - **Refreshing the edges** + bone graft → union
  - Another → distal femoral allograft in the middle → nail + screws
- Now **almost not used** → no law regulating them in Egypt
  - X-rays → old patients from bone bank
- Problems
  - Infection
  - **Non-union** → dead bone
  - Weak immunogenic reaction
  - **Disease transmission** → donors tested for hepatitis B, hepatitis C, AIDS

Rotationplasty Ask AI

History Ask AI

Rotation

- **Halfway** between amputation and limb salvage
- Originally not for tumours → **TB**
- **Van Nes** → proximal femoral focal deficiency

- **Salzer**, 1974 → malignant bone tumours

Principle Ask AI

Rotation

- **Distal femoral resection** → tibia turned up → rotated 180° external rotation (clockwise)
  - **Ankle** → behaves as knee
  - **Normal sole sensation** → as if walking on ground
  - **No phantom pain** → limb not removed entirely

Advantages and Problems Ask AI

Rotation

- Best functional **MSTS score** (Musculoskeletal Tumor Society) — excellent

Advantages	Problems
Walk, ride, practice sports	Cosmesis not accepted by everyone
Low cost	Many neurovascular complications
Less pain	Needs proper family counselling to accept

Case Ask AI

Rotation

- Small piece of femur removed → tibia rotated **anticlockwise** → ankle intact, functions as knee
- Patient wears **prosthesis**
- Leg kept longer than the other → child still growing
- Walks normally; **heel sensate** → feels the ground
- Function at that time → excellent; problem → **cosmesis**
- X-ray → femur above, rotated tibia, **plate** with three screws above + three below
  - **Discrepancy** → thin tibia vs large femur

Exam Pearl

- Rotationplasty → halfway between amputation and limb salvage
- Salzer, 1974 → malignant bone tumours; originally for TB
- Tibia rotated 180° → ankle functions as knee
- Sensate sole, no phantom pain
- Best functional MSTS score; problem → cosmesis

Arthrodesis Ask AI

- Used when → **joint removed**
- Limitations → cannot sit well; drivers can no longer drive; squatting likely not possible
- **Palliative option**
  - **Metastatic osteosarcoma**, poor survival → tumour removed + arthrodesis instead of a joint (or prosthesis unavailable)
    - Walks **without pain** for remaining life
  - **Infected joint (prosthesis)** removed → arthrodesis until healed

Recycling of Tumour Bone Ask AI

Concept and Steps Ask AI

Recycling

- Fifth option → **recycling** — extracorporeal treatment of tumour bone
- Tumour-bearing bone removed → treated → **re-implanted** in place
- Sarcoma of **distal femur** → distal femoral resection → sarcoma removed from medullary canal → re-implanted
- **Complete tumour destruction** mandatory → otherwise → recurrence
- Aim → devitalize tumour without devitalizing **mother bone** — the core problem
- Steps
  - **Wide resection** → soft-tissue mass + intramedullary extent removed
  - Bone **treated** → not mere curettage → heating or freezing
  - Goal → tumour cells die; **biomechanical properties** preserved

Heating Methods Ask AI

Heating

- Measures
  - **Autoclaving**
  - Microwaving
  - Boiling → water or saline
  - **Extracorporeal irradiation** → radiotherapy device
  - Specific techniques and settings → not needed in detail
- Effects of heating
  - **Denaturation** + cell dehydration
  - **Collagen** denaturation → loss of structural integrity and strength
  - → Loss of bone strength, **bone morphogenetic proteins**, osteoconductivity
- Specific methods
  - Boiling above 100°C → **decreased mechanical strength**
  - **Autoclaving** → destroys bone biology
  - Autoclaving and boiling → **worst**
  - Recycling → very rarely, only when forced; autoclaving no longer done
- Best two heating measures
  - **Pasteurization** → heating at different temperatures, as with milk
  - **Irradiated bone** → 20–30 Gy, radiotherapy department *[Q1 · As spoken; extracorporeal irradiation of tumour bone usually uses 50–300 Gy, so 20–30 Gy looks too low.]*
- Most heating measures → loss of bone strength and **osteoconductivity**
- Heating → **not favoured**
  - Unavoidable → pasteurization or irradiation → least loss of strength and biological activity

Pitfall

- Autoclaving and boiling → worst heating methods
- Boiling above 100°C → decreased mechanical strength
- Autoclaving → destroys bone biology; no longer done
- Heating unavoidable → pasteurization or irradiated bone (20–30 Gy)

Freezing — Liquid Nitrogen Ask AI

Freezing

- **Freezing** → best method
  - Kills tumour cells
  - Maintains bone strength, cell components, proteins
- Two methods
  - **Free** → bone fully removed → container
  - **Pedicled** → bone attached to host at one side → placed in freezing container
- Advantages
  - **Bone morphogenetic protein** activity maintained
  - Adequate bone stock maintained
  - Simple method
  - **Optimal fitness** — exact fit
  - Low cost
  - Durable
- Freezing much better than heating → preserves **structural integrity**
- Problems
  - **Avascular** bone
  - **No specimen** for pathology → bone returned
  - Always need **immobilization** → cast until union
  - **Liable to fracture** → frozen or heated bone loses integrity
  - **Non-union** → slow, like allografts; both devitalized bone
- Freezing → preserves **most vitality** of bone

	Heating	Freezing
Tumour cells	Killed	Killed
Bone strength	Lost — collagen denaturation	Maintained
Bone morphogenetic proteins	Lost	Maintained
Osteoconductivity	Lost	—
Verdict	Not favoured; pasteurization / irradiation least harmful	Best method

Technique Ask AI

Freezing

- Segment to be removed → leg in place, bone rotated out → **liquid nitrogen** → returned → pedicled freezing
- **Free freezing** → bone taken out completely → container
- Bone attached to limb → pedicled; fully removed → free
- Case → **distal femur** tumour
  - Soft-tissue mass cleaned; tumour cleaned inside (holes visible) and outside
  - Out of liquid nitrogen → bone **white or grey**
  - Returned → fixed with **plate** or nail

Cement Spacer Ask AI

Indications Ask AI

Spacer

- Last option → may be **permanent**, not only temporary
- **Limited survival** → e.g., multifocal malignant tumour → palliative resection
- **Prosthesis not available** at operation → very common here
- Developed countries too → **aggressive malignant tumour** or terminal cancer → spacer instead of joint
- **Salvage procedure** → infected modular prosthesis removed → spacer

Cases Ask AI

Spacer

- **Giant cell tumour**, proximal femur → prosthesis → recurrence + infection
  - Proximal femur removed → spacer until infection clears and pathology known
  - Built over **Rush pins** → not very strong
  - Stronger → over **Küntscher nail**
- **Chondrosarcoma** + second chondrosarcoma in pelvis + chest metastases (lung nodules)
  - Aim → **palliative**, not curative; very painful lesion
  - Wide resection + spacer → pain-free for short remaining life
  - Proximal **modular prosthesis** unavailable; no other solutions
  - Küntscher-based → **more rigid** than Rush pin-based
- Role → **palliative** or temporary measure

Choosing the Reconstruction Ask AI

Summary by Resection Type Ask AI

Resection	Best usual option
Extra-articular	Arthrodesis — whole joint removed
Intercalary	Autograft — vascularized fibula

- Intercalary → **recycling** also possible → needs good bone stock
  - **Good bone stock** = bone not extensively destroyed
  - Half the bone eaten by tumour → re-implanted → fails; already dead / devitalized
  - → Recycling not always used; **vascularized fibula** always used
- **Allografts** → not available in Egypt
  - Available abroad; **no regulating law** here
  - Past use → agreement with amputees' families → stopped → **legal liability**
- **Non-vascularized fibula** → rarely used; mainly distal radius
- Above → applies to **adults**

Children Ask AI

Children

- Age-based choice

Age	Option
Less than 10 to 10–12 years	Temporary option until the child grows older
Above 12 years	Endoprosthesis — boys only

Biological vs Prosthetic Ask AI

- Trade-off → **function** vs durability

	Biological	Prostheses
Physiology	More physiological	—
Durability	More durable	Inferior long-term results; revision often needed; high failure rate
Function	Questionable — not immediate, not as good	Better; walks quickly, immediately

- **Limb salvage does not affect survival** → same as amputation
- Always oriented → **towards limb salvage**
  - Disseminated disease, goal → walk and move without pain → salvage, not amputation
  - Amputation → remaining time with **phantom limb pain**
- Limb salvage → possible on **palliative** basis
- Key determinants of the reconstructive measure
  - **Age** → less than 10, 10–12, more than 12 years
  - **Site** of resection → intra-articular, extra-articular, intercalary
  - Not patient or emotional acceptance

Amputation Still Has a Role Ask AI

Amputation

- **Amputation** → always there; was primary treatment of all bone tumours
- Indications
  - **Neurovascular bundle** involvement
  - Cannot afford **function and cosmesis** better than amputation
- Example → **huge tumour** occupying whole quadriceps
  - **Anteriorly placed open biopsy** → e.g., 15 cm anterior or posterior biopsy tract
  - MRI → vessels inside tumour
  - **Mal-treated tumour**, mal-positioned biopsy, all muscles involved
  - Quadriceps + skin removed → limb not better than amputation
  - **Morbid surgery** → about 10–15 units of blood, 8 or 10 hours
  - Result → sitting, probably in cast, **immobilized** 3, 4, or 5 months
  - Short remaining life → neither walks nor moves
- Rule → better function possible → **limb salvage**; otherwise → amputation
- Both → **do not affect survival**
- Decision → factors in **patient**, tumour, resources

Must-Know

- Limb salvage does not affect survival → same as amputation
- Amputation → neurovascular bundle involvement
- Amputation → function and cosmesis cannot beat amputation
- Reconstruction determinants → age + resection site
- Prostheses → better function; biological → more durable

Questions and Answers Ask AI

Shaping the Cement Spacer Ask AI

Spacer

- Küntscher nail + cement → mould used?
  - **Shaped by hand**; moulds exist, not here
  - Sometimes **50 ml syringe** → e.g., humerus → better shape
  - Shape irrelevant → **rigidity** matters
  - Küntscher-based → rigid; **Rush pin**-based → not rigid at all
  - Purpose → **weight bearing** on the spacer
    - Avoids **disuse atrophy** and sympathetic dystrophy

Synovial Sarcoma Around the Knee Ask AI

- Limb salvage → possible in anything, per agreed conditions
  - **Wide resection** + function better than amputation → do it
- Huge mass → whole **quadriceps**, wraps around into joint
  - Sciatic nerve at margin, vessels inside → **amputation** better

Recycling: Complications and Limits Ask AI

Recycling

- Question → complication rate; recycled bone better or worse?
- Experience
  - **Kasr Al-Ainy** → still building experience
  - **Alexandria** → worked on it earlier
  - **South-East Asia** → extensive use, religious beliefs
- Problems so far, own experience
  - **Non-union**
  - Infection
  - Young children (under 10 years) → very high **immunogenic reaction** → graft disappearance
    - Twice → entire graft resorbed → violent **giant-cell reaction**
    - **Immune system** very strong in children
- **Liquid nitrogen** source → gas company, Nasr City; very cheap; right behind the hospital
- Recycling in all malignant / **osteolytic tumours**?
  - No bone stock — whole bone eaten → never
  - Required → at least **3 cortices intact**; only part of fourth eroded
  - **Lytic tumour** in half the bone — medial, anterior, or posteromedial cortex over 10–12 cm → never, even with cement augmentation
  - After cleaning → **adequate bone stock** → recycle; inadequate → do not
  - Assessed → **MRI** + intraoperatively

Permanent Cement Spacer Ask AI

Spacer

- Can be left permanently → yes, especially **humerus**
- **Dr. Sherif** + colleagues in Tanta → prostheses vs spacers, proximal humerus
  - **No difference in function**
  - Survival → prostheses slightly higher → cement may **break**
- Built over a nail → patient can live with it normally; some do

Clinical Pearl

- Spacer → rigidity matters, not shape
- Küntscher nail → rigid; Rush pins → not rigid
- Recycling → needs at least 3 intact cortices
- Children under 10 → recycled graft may resorb completely
- Proximal humerus spacer → same function as prosthesis

Remaining Lectures in the Series Ask AI

- No further questions
- Next week → **chondrosarcoma**
- Two lectures remain
  - Chondrosarcoma
  - **Soft-tissue sarcomas** → heavy; one or two lectures per audience preference
- Material provided → still requires **studying**; broad topic
- Then **Dr. Hesham Galal** → spinal metastases; completes the series
- Closing thanks from the audience

Pathological Fractures

(Diagnosis and Management in Bone Tumors)

Prof Dr Wisam Abosena

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 189) “children 95-95% not a disaster” → “children about 95% not a disaster”

Queried - kept as spoken, but looks wrong

Q1 (p. 189) As spoken; malignant cause given as 5–10% just above, so benign share would be 90–95%, not 95%.

Treating Before Diagnosing Ask AI

- Pathological fracture → **nail** or plate? Decision depends on age
  - Lesion: diaphyseal, lytic, crack (fractured)
  - Diaphyseal lytic lesion in a young patient → very wide differential
  - Aim: know the work-up and treatment for such a patient
- Disaster example: **fibrosarcoma**
- Long nail inserted inside the sarcoma
  - Everything along the nail → **contaminated**
  - Result → whole limb removed, from a high level
  - Goal: treat the previous case correctly → never reach this outcome

Pitfall

- Never nail or plate a pathological fracture before diagnosis
- Nail inside a sarcoma → whole nail tract contaminated → limb lost

Pathological Fracture in Adults: The Algorithm Ask AI

- Follow the **algorithm** → avoids the nail-in-sarcoma disaster
  - Mostly covered in earlier lectures
  - Applies to adults; children handled separately

History Ask AI

Adult

- Routine staging starts with **history**
  - Pain at the site before the fall?
  - Pain related to walking and movement?
  - Worse at night or morning? Wakes him from sleep?
  - Known **cancer patient**?

Examination and Investigations Ask AI

Adult

- **General examination**
  - Female → breast
  - Ribs and abdomen
  - Refer to surgeon → **PR** (per rectal) examination
- Ordinary staging
  - X-rays
  - Local **MRI with contrast**
  - CT chest
  - Routine **bone scan**
- Remaining tests still done
  - Labs, **electrophoresis**, tumour markers
  - CT abdomen, CT chest
- CT chest: much debate → with or without contrast
  - Lung metastases → contrast irrelevant
  - Contrast still given → shows lesions outside the **parenchyma** (e.g. mediastinum)

  - Also shows enhancing lesions elsewhere, not otherwise seen

Solitary Lesion → Biopsy Ask AI

Adult

- Bone scan **solitary** + not known cancer → biopsy
  - Solitary on MRI and bone scan, no known cancer → primary tumour or **solitary metastasis**
  - Solitary bone scan → does not make things easy
- Never forget: **solitary lesion might be primary**
- Example: **skeletally immature** patient, fracture through the lesion → looks like osteosarcoma
  - History, general examination, bone scan → **pathological fracture** in a young person
  - Bone scan solitary, not known cancer → biopsy → **osteosarcoma**

Fractured Primary Sarcoma Ask AI

Sarcoma

- Osteosarcoma with vs without fracture → **same treatment**
- Sarcoma + fracture → treat as if **not fractured**
  - Splint → **chemotherapy**
  - Chondrosarcoma or fibrosarcoma → splint → plan **surgery** (known types)
  - Principle: primary sarcoma treated accordingly → **splint**, nothing else
- No intervention until the **biopsy** result
  - **Chemo-sensitive** → chemotherapy
  - Not chemo-sensitive → surgery
  - Osteosarcoma, fractured or not → **splint and chemotherapy**
- Sequence: elderly patient, pathological fracture → general examination → usual **staging** (first lecture)
  - Bone scan solitary → may be **primary** → biopsy
  - Complete all investigations anyway → tumour markers, **electrophoresis**
  - Any single test may lead to the diagnosis
  - Studies + biopsy → **sarcoma**
    - **Cast** or splint → identify sarcoma type
    - Chemo-sensitive → chemotherapy
    - Not chemo-sensitive → surgery

Known Cancer Patient: Timing Ask AI

Adult

- Known cancer (e.g. **prostate cancer**, breast cancer) → ask
  - Treatment received? Still on treatment?
  - Old bone scan?
- Lesion < 1 year after diagnosis → most likely **metastatic**
- Disease-free 2–3 years → new **solitary lesion** → worry
  - Treat as **primary**
- Key in adult pathological fracture → **bone scan** + timing of the lesion
  - Solitary → worry
  - **Multiple** → solved: myeloma, lymphoma or metastasis → easy
- Solitary → **sarcoma** or solitary metastasis
  - Depends on known-cancer status
  - Not known cancer + all investigations **negative** → most likely sarcoma
  - Known cancer, treated < 1 year ago → **metastasis**
  - Disease-free 2–3 years + solitary lesion → most likely metastasis, but **may be sarcoma**
- Bone scan = key → sets the direction

Bone scan	Cancer status	Likely diagnosis	Action
Multiple	Any	Myeloma, lymphoma, metastasis	Reassured
Solitary	Not known cancer, work-up negative	Sarcoma	Biopsy
Solitary	Known cancer, < 1 year	Metastasis	—
Solitary	Known cancer, disease-free 2–3 years	Metastasis or sarcoma	Biopsy; treat as primary

Indications for Biopsy Ask AI

Adult

- **Biopsy indications** in pathological fracture
  - Not known cancer, **solitary bone lesion**, everything else normal, never had cancer
  - Known cancer, all else normal, 2–3 years later → precautionary biopsy → **second cancer** or second sarcoma
- Metastatic disease → later **sarcoma** → rare but possible
- Second cancers
  - Breast → **colon cancer**
  - Breast + thyroid
  - Breast + sarcoma → rare, happens
- Bone scan multiple → reassured; **solitary** → must worry

High Yield

- Bone scan = key of adult staging
- Multiple → myeloma, lymphoma or metastasis
- Solitary + no known cancer → biopsy
- Disease-free 2–3 years + new solitary lesion → biopsy, treat as primary

Surgery After Chemotherapy Ask AI

Sarcoma

- Aggressive lesion + fracture → **never touch** without pathology
  - Must know what pathological fractures have (question from last time)
  - Otherwise → nail-in-sarcoma disaster; procedure not done properly
- Fractured sarcoma → chemotherapy → reassess **union**
  - United → operate accordingly
  - Not united → **resection** or amputation
- Example: fractured osteosarcoma, fully **staged** → splint → chemotherapy
  - Likewise **Ewing sarcoma** → chemotherapy
  - After 6 cycles: fracture **united**, soft tissue mass ↓, fracture haematoma gone → resection
  - Another after 3–6 cycles: tumour growing → **fungating** through the biopsy site
  - Soft tissue mass → no resection → **amputation**

Pathological Fracture in Adults: Differential and Fixation Ask AI

Disaster Until Proved Otherwise Ask AI

Adult

- Adult pathological fracture → most likely a **disaster**
  - No innocent lesion at 40–50 years; unlike children
  - Older age fracture → most likely a disaster
  - Severe → **sarcoma**
  - Lesser → metastatic disease, myeloma, lymphoma
- Must differentiate → each has a **different treatment**
  - 50 years, pathological fracture femur due to **fibrosarcoma** → one treatment
  - 50 years, metastatic prostate cancer, pathological fracture femur → completely different
  - 20 years, pathological fracture due to osteosarcoma → third, different
- **Pathology** mandatory → clear treatment direction
  - "Plate + biopsy" or "nail + biopsy" → **never**
- **Staging**
  - General examination, routine labs, routine X-rays
  - MRI, CT, **PET-CT**
  - Biopsy when indicated → solitary lesion, nothing else found

Fixing Metastasis, Myeloma, Lymphoma Ask AI

Adult

- Proven metastasis, myeloma or lymphoma → **intramedullary fixation**
  - Cement as needed; not routine
- Solitary metastasis (e.g. breast cancer → **hip**)
  - Large, occupying the whole hip → resect + **joint replacement**

Renal Cell Carcinoma Metastasis Ask AI

RCC

- **Renal cell carcinoma** → radio-resistant + chemo-resistant
  - Femoral metastasis + **radiotherapy** → mass will not shrink
  - Options → curettage or **wide resection**
  - Must **attack** renal cell metastasis
  - Chemo- and radio-resistant → relies on **local treatment**

Primary	Adjuvant response	Implication
Renal cell	Radio- and chemo-resistant	Local treatment: curettage or wide resection
Thyroid	Radioactive iodine	Operate + rely on adjuvant
Breast, prostate	Hormonal treatment	Operate + rely on adjuvant

Adult Summary Ask AI

Adult

- Older patient (> 40, or 35) + pathological fracture
  - Primary sarcoma, myeloma, lymphoma or **metastasis**
- Adult in his twenties → **osteosarcoma**
- Differentiate → examination, investigations, **routine staging**
- Bone scan: multiple → reassured; **solitary** → great worry

- Example: 34-year-old woman, **subtrochanteric** pathological fracture, not known cancer
  - Investigations, tumour markers, whole-body imaging → negative
  - Fix as "most likely metastatic"? → no → **biopsy first**
  - Biopsy metastatic → fix
  - Biopsy sarcoma → is it **chemo-sensitive**?
- Adult pathological fracture → **black disaster**
- Differentiation decides: nail in 1.5–2 hours vs **operation** of 5–6 hours

Pathological Fracture in Children: Approach Ask AI

- Children → completely **different story**
  - Adults (20s, 40s, 50s) → most likely (not always) black disaster
  - Children → most likely **benign tumour**
- Pathological fracture = **fracture of abnormal bone**; any lesion may cause it

History and Examination Ask AI

Child

- Same history
  - Pain before? Appearance?
  - **Fever**?
  - Previous fracture?
  - Young child → ask the parents
- **General and local** examination, as agreed
- Treatment path varies
  - One child → depends on **fracture appearance**
  - Another → independent of the fracture
  - Both may need treatment → completely different routes
- Example: **non-ossifying fibroma**
  - Well-defined lytic lesion, surrounded by sclerosis, eccentric, crack
  - → pathological fracture in NOF → treatment completely different

Causes Ask AI

Child

- **Metabolic** diseases, e.g. rickets
- Infection
- Benign or **malignant tumour**
- **Irradiation** → e.g. for Wilms tumour, rhabdomyosarcoma
- **Age-specific** approach
  - Birth–5 years → infection, eosinophilic granuloma, Ewing sarcoma, metastatic **neuroblastoma**
  - 5–10 years → **bone cysts**
    - Less common → osteosarcoma, enchondroma, osteochondroma
  - > 10 years → bone cysts, **fibrous dysplasia**, lymphoma

High Yield
- ≥ 5 years → think bone cysts
- < 5 years → worry: infection, Ewing sarcoma, eosinophilic granuloma
- Eosinophilic granuloma = younger sibling of Ewing sarcoma
- > 10 years → bone cysts or fibrous dysplasia (mostly above ten)

Work-up Ask AI

Child

- **X-rays** → mostly sufficient for diagnosis
  - Most lesions benign; unlike adults
- **MRI with contrast**, as agreed
- **Bone scan** → not routine, unlike adults
  - Avoid exposing children unless indicated
  - Indication → suspected **multiple bone lesions**
- **Biopsy** last, if needed → not routine, unlike adults
- Adults → disaster; young → usually **benign**
  - Young → clear-cut, not **tricky** as in adults
- Usual work-up, nothing added or removed
  - Except → **bone scan not routine**
  - Bone scan not the key here

Aggressive Lesion in a Child Ask AI

Child

- **Aggressive lesion** in a young child
  - **New bone formation**
  - Crack → must worry
  - Contrast: the other lesion → looks like a simple crack
- Diagnosis by X-ray? → no → **MRI**
- MRI: extensive involvement, clearly visible → certainly **not benign**

Infection Versus Ewing Sarcoma Ask AI

- Infection or not? → **biopsy**
- Imaging example: **PET-CT**, bone scan uptake → disaster-like
  - Skeletally immature → treated as a child
- Key problem in children → infection vs **Ewing sarcoma**
  - Clinical picture, history, **ESR**, CRP
  - 4-year-old, pain + fever, poor examination → **exploration biopsy**

Biopsy Technique Ask AI

Ewing

- **CT-guided core biopsy** if available; otherwise exploration biopsy
- Child < 5 years + pain + fever ≠ always Ewing → but **high index of suspicion**
- Exploration biopsy → small incision **down to the bone**
  - Same **rules of biopsy**
- **Pus** pours out → clean out → pathology + culture
- **Culture what you biopsy and biopsy what you culture**
  - Any debridement = biopsy + culture

- Tissue not reassuring → **no debridement** → biopsy + culture → await pathology
  - Ewing sarcoma → nothing lost → chemotherapy
  - Infection → 2–3 days later → formal opening, drainage, **debridement**
- Summary: child < 5, fever + pain, equivocal X-ray and MRI
  - → **biopsy**
  - No core biopsy → exploration biopsy, small incision down to bone

Obvious Infection Ask AI

Ewing

- **Infection mimics Ewing sarcoma**
- Tonsillitis → painful leg → swelling → wound opens
  - **Subperiosteal abscess** + subcutaneous abscess
  - Draining sinuses + abscess → certainly **infection**, not Ewing
- Hidden cases, esp. **young children** → great worry
- Investigations → covered before

Exam Pearl

- Infection mimics Ewing sarcoma
- Culture what you biopsy, biopsy what you culture
- No core biopsy → exploration biopsy, small incision down to bone

Simple Bone Cyst Ask AI

- Imaging appearance → covered before
- Fractured → **conservative** unless otherwise indicated

Humerus and Non-Femoral Sites Ask AI

SBC

- Metaphyseal or diaphyseal SBC + **undisplaced fracture** → splint only
  - **Arm-to-chest** bandage or neck sling (collar and cuff)
  - Review after 3–4 weeks
- Fracture → **serous fluid** drains → heals
- Humerus, **active** cyst
  - Extends toward the epiphysis (**physis**) → observation for a while

Proximal Femur Ask AI

SBC

- Proximal femur, fractured (cracked or not) → **attack**
- Common idea: cyst → **curettage** → wrong
  - Fractured cyst → no curettage
  - **Attack the fracture**; cyst heals on its own
- **Proximal humerus crack** → nothing → review after 3–4 weeks
- Proximal femoral **cystic lesion**
  - Large, like fibrous dysplasia or SBC → **never leave** → will fracture
  - Not leaving = **fixing**; several methods
  - **Screws** → if room for screws
  - No room → smooth **K-wires**, wires at two sites
  - **Curettage** → described in a paper; not routine, may be skipped
  - Above → age < **10 years**

  - > 10 years → no **physis** concern → screws directly → must fix

Fallen Leaf Sign Ask AI

SBC

- Humeral SBC → same management
- **Fallen leaf sign** → fragment broke off and fell down
- Not femur (tibia, humerus) → **cast** or splint
- Femur → intervene → **fix**
- Opening + curettage → not wrong; applies to **non-fractured** cyst
- Some (incl. lecturer) open the cyst → **biopsy** in a worried patient
- Others (our professor, many others) → never open
- Fractured cyst → already **opened itself** → no curettage → unites

Must-Know

- SBC → attack the fracture, not the cyst
- Not femur → cast or splint
- Proximal femur → never leave → fix
- < 10 years → K-wires if no room for screws; > 10 years → screws directly

Non-Ossifying Fibroma Ask AI

- Very common → **the most common**
- Usually **before twenty** (about 15 up to 20 years)
- Usually **asymptomatic**; appearance covered before
- Size variable → small or large
- Treatment → exactly like **simple bone cyst**
- X-ray → **cortical lesion**, eccentric, well-defined, surrounded by sclerosis

Fracture Management Ask AI

NOF

- Unstable fracture: spiral, **lower third** of tibia
  - Site famous for **non-union** → treat the fracture
- Open and curette? → no
  - Fix as **distal tibial fracture**; leave the NOF alone
- SBC → same → **attack the fracture**
  - Treat like casualty treats an ordinary fracture of the distal tibia
- **Displaced unstable** fracture of distal tibia → fix
- NOF + fracture → **stable or unstable**?
  - Crack or **in place** (first lecture) → cast
  - Displaced, spiral, long oblique or **unstable** → treat as ordinary fracture; ignore NOF
- Femur, **completely off-ended**
  - Large NOF occupying almost the whole femur, displaced fracture → fix
  - Curettage → not needed
  - Fix only → heals by itself
- Large NOF + displaced femoral fracture → **plating**
- NOF on **CT**: small or cracked → cast, nothing else
- NOF → **conservative**, unless displaced or in the proximal femur
  - Bone scan → no

  - Biopsy → no
- Intervention timing → after a **week to ten days** (fracture becomes sticky)
  - Some → immediately; no big difference
- Lesions so far → only **fixation** needed when indicated
- Meanwhile → cast, treat **conservatively**

Aneurysmal Bone Cyst Ask AI

- **More aggressive**; fortunately only 2%
- **Expansile**, lytic, eccentric
- Sites → almost everywhere: femur, tibia, humerus
- MRI → characteristic **fluid level**
- **May be secondary**
- **Biopsy mandatory**
  - Children otherwise → biopsy only if **infection vs tumour vs ABC** unclear
- 36% **fracture** → large proportion
- Treatment → **surgery**; no conservative treatment
  - Must be **curetted**, and curetted well
- Any lesion in **proximal femur** → never leave
  - NOF, SBC, anything
- Sometimes needed → **large** lesion

Imaging Reasoning Ask AI

ABC

- Example (old ABC picture): **epiphyseo-metaphyseal**
- Effect on bone → **expansile lytic**, some internal septation-like lines
- **Periosteal reaction** → none
- **Matrix** → none
- Old-way differential
  - Cartilage? No **calcifications**
    - Expansile cartilage lesion → **chondromyxoid fibroma**; site does not fit
    - This one → **epiphyseal**, reaching subchondral bone
  - Fibrous? → no
  - Cystic? → yes, **ABC** fits
  - Sarcoma? → fits too
- Differentiate → **staging**; ABC diagnosed only after biopsy
  - **ABC biopsy is mandatory**
- **Multiple fluid levels**, expansile, young patient → ABC or sarcoma → biopsy decides
- Characteristic **fluid levels** → fluid + air
  - Air and fluid in a closed space → separate into layers
- Expansile lesion, **proximal humerus** → simple or aneurysmal?
  - Differentiate → MRI + biopsy

Proximal Femur Ask AI

ABC

- Proximal femur → **fix**
  - Cyst **away from the physis** → ordinary plate
- **Biopsy is mandatory; attack the cyst and the fracture**

- Window → curettage → **bone graft substitute**
  - Or **cement** → if away from the physis
- Not fix-only as in SBC or fibrous lesions → **biopsy + curettage**
- **Bone graft** → some use it (not wrong); not us

ABC Versus the Well-Behaved Lesions Ask AI

ABC

- Three lesions in children → SBC, NOF, **ABC**

Lesion	Behaviour	Management
Simple bone cyst	Benign, well-behaved	Fix if fracture
Non-ossifying fibroma	Benign, well-behaved	Fix if fracture
ABC	Aggressive	Curette well + fix well

- SBC and NOF → alike in everything
- All very important → fracture → fix, as in **distal tibia** or anywhere else

Fibrous Dysplasia Ask AI

- Covered before; incidence 5–7% in children
- **Monostotic** or polyostotic
- **Fractures** in 50% of children
- High-risk in the **femur**
- Tibia → cast, **conservative**
- Deformity or femur → **intervene**
- Femur with **deformity** → osteotomy + fixation
- Femur picture
- MRI → **dark** → fibrous tissue

Diagnosis and Fixation Ask AI

FD

- Near **skeletal maturity** (still a child) + large lytic lesion
- Diagnosis → MRI, staging, **biopsy**
  - Worrying → could be an ABC
  - Assess case → decide whether **biopsy** needed
- Painful leg + large lesion → **fix**
- **Curettage** → not needed; nail or plate only
- **Nail** preferred
  - Difficult in children / at this age → some use a plate
- Principle: long implant → splints most of the bone → **bypasses the lesion**
- Only **ABC** → open + curette very well; others → no curettage, no opening
- Fractured case fixed → starting to **unite**
- Fracture → **deformity** → correct the deformity
- **Attack the fracture** or attack the deformity
- **Plate** → one method; some prefer it, not us
- Preferred → **nail** (possibly second-generation)

Benign Lesions in Children: Summary Ask AI

- Children → easy
  - **Infection** → number one; differentiate from Ewing sarcoma
  - Two **cysts** → SBC, ABC
  - Two **fibrous lesions** → NOF, fibrous dysplasia
  - Worrying one → **ABC**
    - Biopsy mandatory
    - **Bone scan** preferable
    - Very thorough **curettage**
- Rest → **conservative**, unless
  - Proximal femur
  - Displaced fracture
  - **Deformity** (fibrous dysplasia)

Lesion	Biopsy	Treatment
Simple bone cyst (UBC)	No	Attack the fracture; mostly conservative
Non-ossifying fibroma	No	Attack the fracture; mostly conservative
ABC	Mandatory	Surgery: curettage + fixation
Fibrous dysplasia	If worrying	Attack the fracture + deformity

Malignant Tumours in Children Ask AI

- **Malignant cause** → only 5–10% of children's pathological fractures
- Adults → 90–95% disaster; children → about 95% **not a disaster** *[C1 · transcript: "children 95-95% not a disaster"] [Q1 · As spoken; malignant cause given as 5–10% just above, so benign share would be 90–95%, not 95%.]*
- **Osteosarcoma** and Ewing sarcoma → fracture rates 5–13%
- Child with osteosarcoma → exactly **as in adults**
  - Sarcoma fracture, adult or child → don't touch → **splint** → chemotherapy → decide after
- Pathological fracture → does **not affect overall survival**
- Example: **skeletally immature**, diaphyseal lesion
  - Aggressive, lytic, ill-defined
  - **Periosteal reaction**
  - **Cortical breakdown**, small crack
  - Work-up → bone scan, MRI, X-ray, **biopsy**, splint
  - **Ewing sarcoma** → splint or brace → chemotherapy → follow-up
- Another: young patient → **diaphyseal** lesion
  - Medullary involvement + large **soft tissue mass** → most likely disaster
  - Biopsy → sarcoma → **chemotherapy**
  - After chemotherapy → **unites** → resection + joint replacement
  - Fracture → **splint** or cast → chemotherapy

  - After chemotherapy: united + **responded** → treat as never fractured
  - Not united → most likely more **radical surgery**

Overall Summary Ask AI

- **Do not treat before being sure**
- Being sure → **X-ray** + MRI
- **Bone scan** → adults mandatory; children only if indicated
- **Biopsy** → two situations
  - Suspected **ABC** or sarcoma
  - Young child → **Ewing or infection** unclear
- Young child → **infection** (number one) or sarcoma
  - Most famous sarcomas → **Ewing sarcoma**, osteosarcoma
- **Benign lesions** → cysts or fibrous lesions
  - **UBC** (unicameral bone cyst) → well-behaved → attack the fracture, ignore cyst
  - **ABC** → attack both; biopsy; aggressive
    - May **recur**, esp. in children
    - Biopsy required → **ABC**; entirely surgical
    - No biopsy, mostly **conservative** → UBC
  - **Non-ossifying fibroma** → completely benign → leave alone
  - **Fibrous dysplasia** → leave alone unless deformity
- Child + pathological fracture in **proximal femur** → never leave
  - Attack → ideally **before** it fractures
- Most children's fractures (good proportion, not all) → **conservative**
  - Reassure the parents
  - Surgery only if → aggressive lesion (**ABC**)
  - Lesion of the **proximal femur**
  - **Displaced fracture** → e.g. distal tibia, unstable spiral humerus
  - **Multiple fractures** (repeated, humerus or tibia) → rare
- Treat as **ordinary fracture**
  - **TENS nails** (elastic nails)
  - Plate in distal tibia
- ABC or suspected **malignancy** → completely different story
  - ABC → **extended curettage** + high-speed burr
  - Cement → if **away from the physis**; important
  - Malignant tumour → splint → **chemotherapy**
  - Primary sarcomas in children → **chemosensitive** (all except very rare cases)
  - After chemotherapy → definitive **surgery**
- Children → mainly **conservative**, unless
  - **Proximal femur** → critical site
  - **Displaced fracture** → fracture pattern
  - **ABC**
  - **Primary sarcoma** → between brackets → chemotherapy first

Feature	Adult	Child
Default	Disaster until proved otherwise	Benign until proved otherwise
Disaster proportion	90–95%	About 5%
X-rays	Not sufficient alone	Mostly sufficient
Bone scan	Mandatory, the key	Only if indicated
Biopsy	Solitary lesion	ABC, sarcoma, Ewing vs infection
Fractured sarcoma	Splint → chemotherapy	Splint → chemotherapy

- Adult → **disaster until proved otherwise**
- Young child → hopefully benign **until proved otherwise**
- **Bear in mind** infection and sarcoma
  - Not every child with a hot leg → sarcoma
- Fever after **tonsillitis** + painful leg → hopefully infection, not 100%
  - Staging, **MRI with contrast**, examination
  - Any worry (usually) → **biopsy**
  - No biopsy service → **exploration biopsy**, small incision
- Per lesion
  - UBC, NOF → **attack the fracture**
  - ABC → **curette the lesion**
  - Fibrous dysplasia → attack the fracture + **attack the deformity**
  - Malignancy → **splint** → chemotherapy
- Pathological fractures → very important; **everyone** will see them

High Yield
- Adult pathological fracture → disaster until proved otherwise
- Child pathological fracture → benign until proved otherwise
- Children: conservative unless proximal femur, displaced, ABC, sarcoma
- Pathological fracture does not affect overall survival in sarcoma
- Primary sarcomas in children → chemosensitive

Questions and Answers Ask AI

Who Takes the Biopsy? Ask AI

- Q: biopsy ideally by treating surgeon, same centre → take it or refer to a specialist?
  - Specialist better → refer to **oncology surgeon**
  - Stuck, referral hard, time running out for the child → **do it yourself**
  - Ideally → surgeon doing the **definitive surgery** takes the biopsy
- Next (short) lecture → 2–3 slides: **how to take a biopsy** (basics)

Mirels Score Ask AI

- **Mirels score** → applies to benign and malignant lesions

Bone Grafting in Fibrous Dysplasia Ask AI FD

- Q: graft in FD of **proximal femur**, with fixation?
  - Some still convinced of grafting
  - **Fibulas** shown in one case (done abroad)
  - Graft or fibula → acceptable, but must be **splinted** (fixed)
  - Graft alone → **woven bone** collapses
- **Kasr Al Ainy** school → **fixation only**: no graft, no cement

Cement in Children Ask AI

- **Not routine** → only recurrent lesions
- Exception → large **ABC**, recurrent 2–3 times
  - After **embolization** and everything; no other option
- When forced → **layer of sponge** or iliac crest graft on the physis → cement above
  - Or fill only the **metaphyseal part** of the lesion; leave the epiphysis
- Rare

ABC Is Never Splinted Ask AI ABC

- Q: proximal humerus fracture on an ABC → splint? → no → **attack**
- **ABC in children** → much worse than adults
  - Never leave; use everything available
- Attack **from the start** → no waiting (e.g. for union)
  - Waiting → very old practice (ten years ago, simple studies)

Biopsy of ABC Ask AI ABC

- Must biopsy ABC? → **yes**
  - Kasr Al Ainy view, shared by many worldwide
- Biopsy → nothing lost; at least **medico-legal** benefit
- Curettage of apparent ABC → turns out **sarcoma** → limb lost, likely life too
- Kasr Al Ainy → **ABC routine for biopsy**
  - Some respected tumour surgeons → do not
  - Many in the **literature** → support biopsy

Solitary Lesion in a Known Cancer Patient Ask AI

- Q: every **solitary lesion** → biopsy, known or not known primary?
- **Duration makes the difference**
  - Breast cancer, **disease-free** 2–3 years → solitary lesion → worry
  - Disease-free **> 1 year** → solitary lesion → must worry
  - Breast or prostate cancer 2–3 months ago → lesion → most likely **metastatic**
- Worry if
  - Disease-free > 1 year, previous **bone scan** free → sudden lesion
  - No cancer history at all → **solitary lesion**
- Known **breast cancer** on treatment 2–3 months + solitary lesion → not presumed sarcoma
- Worry → **no history**, or cancer long ago + normal work-up
  - Long ago = **> 1 year**

Fracture Neck of Femur in Hyperparathyroidism Ask AI

- Q: fracture neck femur secondary to **hyperparathyroidism**
  - Very poor bone stock; young adults and adults
- Secondary hyperparathyroidism → first identify the **cause**
  - **Parathyroid adenoma** → remove adenoma first, before the operation
  - Not adenoma → **long stem prosthesis**; anything may be added
- Among the **most difficult** cases
- Must know whether an **adenoma** exists
  - Untreated + **joint replacement** → often fractures around the prosthesis within a month